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MEMORANDUM

DATE: August 13, 2026

TO: Park County Citizens

FROM: Brian Edwards, P.E. – County Engineer

RE: Park County Road & Bridge Standards – Draft #1

Enclosed herewith you will find the first draft of the proposed revisions to the *Park County Road & Bridge Standards (RBS)*. The RBS date back more than 30 years and have evolved to consider current technology and modern engineering, construction, and maintenance standards. Although the RBS has historically been a standalone document, it is currently included as *Chapter V* in the *Park County Development Standards and Regulations (DSR)*. With the current proposed revisions to the DSR, the RBS is being physically removed from the DSR but is included by reference to eliminate the potential for duplication and discrepancies. Pending input from the professional engineering/surveying community, developers, and the public at large, the Board of County Commissioners (BOCC) will consider adopting the RBS at a later date to coincide with the timing of the DSR.

The RBS is intended to act as a reference and directive for managing, planning, designing, constructing, and maintaining the Park County Road & Bridge System (PCRBS) including consideration for site development. As such, the RBS includes information, guidelines and requirements for a wide variety of topics such as planning, addressing, mailboxes, road naming, snow plowing, road maintenance, and many other related elements pertaining to site, road, and bridge related infrastructure.

Although substantial changes, additions, and deletions are proposed, it is not this department's intent to make things more difficult or restrictive for development. In fact, several things have been incorporated or modified to improve clarity while removing unnecessary obstacles and costs that developers face in complying with Park County regulations and standards. Our intentions and focus are on uniformity, consistency, and clarity with emphasis on those standards that protect the greater public good without compromising individual property rights.

When reviewing the enclosed document, please remember and consider that this is a first draft that has not been vetted technically by engineers, surveyors, or developers. Over the next few weeks, substantive changes will likely continue to be made based on input received from all interests including the public at large.

We purposely left the changes to the current document (included in Chapter V of 2015 DSR) in this first draft in the interest of transparency. For reference, the following color-coding system was used as it relates to font color in the draft document:

- **Black font:**
 - **Wording from the current RBS (included as Chapter V in 2015 DSR)**
- **Blue font:**
 - **Administrative language that was brought over from the DSR that addresses processes for amendments, appeals, public hearings, etc.**
- **Red font:**
 - **Wording from the current RBS that is proposed to be deleted.**
- **Green font:**
 - **Wording that is proposed to be added to the RBS.**

Substantive Additions or Changes Worth Noting:

- Includes an overview and summary of historical development of the County Road & Bridge system (Section 1)
- Includes County and Wyoming State statutory requirements and processes for public input, public hearings, appeals, and amendments (Section 2).
- Provisions have been incorporated to move away from the administrative relief process currently in place regarding deviations from the RBS. The current process has the County Engineer (not an elected position) reviewing and approving variances. The proposed RBS replaces the “administrative relief process” to better align with the statutory process for variances which requires review and approval by the BOCC. The County Engineer and staff will continue to provide technical assistance and guidance to the Commissioners to assist them in making informed decisions as it relates to variances from these RBS.
- Incorporates Figures 1 and 2 to assist developers in identifying which sections of the RBS apply for residential, industrial, and commercial development.
- Includes consideration for overall county wide transportation system planning (Section 3) to integrate parallel efforts in developing a *County Transportation Safety Plan* consistent with Federal Grant funding under the *Safe Roads and Streets for All* program (plan development in progress).
- Incorporates the statutory requirements for creation, alteration, and vacation of public road rights-of-way (Figure 3).

- Updates functional classifications of various types of roadways that comprise the CRBS to better align with universal classifications adopted by Federal Highway Administration (FHWA) and WYDOT. Additional classifications for lower volume roads below the local access road standard have been incorporated to ensure design and construction standards are reasonable for the level of use (ie, average daily traffic).
- Includes an allowance whereby roads associated with development do not need to be built to County design standards when the estimated traffic will not exceed 60 vehicles per day. New residential development subdivisions that involve less than 6 residential lots (minor subdivisions in the proposed DSR) will **NOT** need to meet County design and construction standards.
- Includes a road classification (Low Volume Rural Road) that allows a 40-foot right-of-way. All current road classifications for residential, commercial, and industrial development require a 60-foot-wide right-of-way;
- Softens the requirement for when paving is required for roads serving development. Currently roads associated with development where the traffic (ADT) exceeds 100 vehicles per day must be paved per current RBS. We are proposing to move this to 200 vehicles per day, which is more consistent with industry standards for rural development.
- Consolidates and clarifies permitting requirements applicable to the County Road & Bridge System (Section 4).
- Organizes all design and construction requirements into one section for quick reference (Section 5). **Requirements for design and construction pertaining to site development as cross referenced from the DSR is included in Section 5.9.**
- Organizes all maintenance requirements into one section for quick reference (Section 6).

Obviously, there are numerous other edits and additions that may or may not be of interest. There are also circumstances and topics where more work, information, and effort is needed by staff to address ongoing problems. For example, in the past staged, or phased development that spans multiple subdivisions, owners, and even subdivisions (particularly older subdivisions) results in situations where a road is inferior and not adequately maintained (mainly private roads). Current RBS include provisions for forward planning to make sure roads are designed and constructed to consider potential future development based on zoning out to 20 years in the future. However, this can prove costly and even cost prohibitive to a smaller development. More discussion and input is needed to consider all interests and concerns in this regard before these RBS are adopted.

We welcome any and all constructive input regarding the proposed RBS. In the interest of transparency, we ask that all comments be provided in writing (email preferred). When possible, please try to list or reference specific language, sections, page numbers, etc. that you feel are problematic and need further adjustment. Comments and recommendations for possible edits shall be submitted to:

Park County Public Works Department
Attn: Brian Edwards, P.E. – County Engineer
1131 11th Street
Cody, WY 82414
(307) 527-8520
EMAIL: Brian.Edwards@parkcounty-wy.gov

On behalf of the Park County Public Works Department, we appreciate your interest and assistance in this process.

Sincerely,

Brian Edwards

Digitally signed by Brian Edwards
DN: C=US, E=Brian.Edwards@parkcounty-wy.gov,
O="Park County, Wyoming", OU=Public Works
Department, CN=Brian Edwards
Location: Cody, WY
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Brian Edwards, P.E. – County Engineer

DRAFT

ROAD & BRIDGE STANDARDS

PARK COUNTY, WYOMING

Updated 08/13/2026



RESOLUTION NO. 2026 - ____

CERTIFICATE OF ADOPTION
ROAD AND BRIDGE STANDARDS
PARK COUNTY, WYOMING

WHEREAS the Board of County Commissioners for Park County, State of Wyoming (BOCC) has been granted administrative powers to manage the business and concerns of the County per *Wyoming State Statutes (W.S.) § 18-2-102*; AND

WHEREAS the BOCC has the authority to regulate land use in unincorporated areas to promote public health, safety, and general welfare per *W.S. 18-5-201 through 208* and establish requirements for the design and construction of roads within private subdivisions in accordance with *W.S. § 18-5-306*; AND

WHEREAS *W.S. § 24-1-104* authorizes the BOCC to administer the County Road and Bridge System (CRBS) including, but not limited to, maintenance, layout, establishment, alterations, vacations, property acquisition, and traffic regulations; AND

WHEREAS the BOCC recognizes the need for and importance of establishing and maintaining a safe and reliable county road and bridge network to serve the needs of the people of Park County and those that visit this area; AND

WHEREAS the BOCC recognizes the need for and importance of establishing uniform administration, planning, permitting, design, construction, and maintenance standards for the CRBS including roads and related infrastructure in support of development for the greater public good.

NOW, THEREFORE, IT IS HEREBY RESOLVED that based on the foregoing, the BOCC on its own motion hereby establishes *County Road and Bridge Standards* to be used as a reference and guide for administering the CRBS. The *County Road and Bridge Standards* shall supersede and replace any previous road and bridge related policies and standards previously adopted by the BOCC; AND

IT IS HEREBY FURTHER RESOLVED that the BOCC hereby grants the authority to the Park County Engineer to oversee and administer the *County Road and Bridge Standards*.

APPROVED this ____ day of ____, 2026

**BOARD OF COUNTY COMMISSIONERS
PARK COUNTY, WYOMING**

Scott Steward, Chair

Lloyd Thiel, Vice Chair

Scott Mangold, Commissioner

Kelly Simone, Commissioner

Dossie Overfield, Commissioner

(SEAL :)
(ATTEST :)

Colleen Renner, Park County Clerk

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ACRONYMS USED

AADT	Average Annual Daily Traffic
AAN	American Association of Nurserymen
AASHTO	American Association of State Highway and Transportation Officials
ACI	American Concrete Institute
ACPA	American Concrete Pipe Association
ADT	Average Daily Traffic
AI	Asphalt Institute
AISC	American Institute of Steel Construction
ANSI	American National Standards Institute
APWA	American Public Works Association
ASA	American Standards Association
ASCE	American Society of Civil Engineers
ASLA	American Society of Landscape Architects
ASTM	American Society for Testing and Materials
ATTSA	American Traffic Safety Services Association
AWWA	American Water Works Association
AWSE	American Welding Society Code
BOCC	Board of County Commissioners (Park County)
BOR	United States Bureau of Reclamation
BROS	Bridge Off-system Program
CCID	Cody Canal Irrigation District
CMAQ	Congestion Mitigation and Air Quality Program
CMP	Corrugated Metal Pipe
CRBS	County Road & Bridge System (Park County)
CRCMF	County Road Construction & Maintenance Fund
CUHP	Colorado Urban Hydrograph Procedure
DIP	Ductile Iron Pipe
DSR	Development Standards and Regulations (Park County)
EPA	Environmental Protection Agency
FEMA	Federal Emergency Management Agency
FHWA	Federal Highway Administration
GP&L	Garland Power & Light
GVID	Greybull Valley Irrigation District
HMID	Heart Mountain Irrigation District
HOA	Home Owners Association
IMSA	International Municipal Signal Association
ISO	Insurance Services Office
ITE	Institute of Transportation Engineers
LID	Lakeview Irrigation District
LTAP	Local Transportation Assistance Program
LUP	Lands Use Plan (Park County)
MDU	Montana Dakota Utilities
MSHA	Mine Safety and Health Act
MUTCD	Manual on Uniform Traffic Control Devices
NBIS	National Bridge Inspection Standards
NEC	National Electrical Code
NEMA	National Electrical Manufacturers Association

NEPA	National Environmental Protection Act
NFIP	National Flood Insurance Program
NHPA	National Historic Preservation Act
NOAA	National Oceanic and Atmosphere Administration
NRWD	Northwest Rural Water District
OSHA	Occupational Safety and Health Act
PCPWD	Park County Public Works Department
P&Z	Planning and Zoning Department (Park County)
P&ZC	Planning and Zoning Commission (Park County)
R&B	Road & Bridge Division of the Park County Public Works Department
RAP	Recycled Asphalt Pavement
RBS	Park County Road & Bridge Standards
RCP	Reinforced Concrete Pipe
RIMD	Road Improvement and Maintenance District
RNAC	Road Naming and Addressing Committee (Park County)
SCS	Soil Conservation Service
SHPO	Wyoming State Historic Preservation Office
SID	Shoshone Irrigation District
SMP	Shoshone Municipal Pipeline
SS4A	Safe Roads and Streets For All Program
SW	Solid Waste Division of the Park County Public Works Department
TAP	Transportation Alternatives Program
TSP	County Transportation and Safety Plan (Park County)
UL	Underwriters Laboratories, Inc.
USACE	United States Army Corps of Engineers
USDA	United States Department of Agriculture
UD&FCD	Urban Drainage and Flood Control District
USFS	United States Forest Service
USGS	United States Geological Survey
WDEQ	Wyoming Department of Environmental Quality
WG&F	Wyoming Game & Fish Department
WID	Willwood Irrigation District
WPWSS	Wyoming Public Works Standard Specifications
WYDOT	Wyoming Department of Transportation

1.0 INTRODUCTION

1.1 PURPOSE AND INTENT

These *Park County Road and Bridge Standards*, hereinafter referred to as “RBS”, establish a uniform road and bridge development, design, construction, and maintenance policy for the unincorporated areas of ~~throughout~~ Park County. The RBS are not intended to address roads and bridges in incorporated areas (ie, City of Cody, City of Powell, Town of Meeteetse). Further, the RBS do not address roads and bridges developed and maintained by the Federal Government on public lands (i.e., Bureau of Land Management, Bureau of Reclamation, etc.) nor State Government on State Lands (i.e., Wyoming Department of Transportation, Wyoming Office of State Lands, etc.).

The purpose and intent of the *RBS Standards* is to provide consistency and uniformity throughout the County Road and Bridge System (CRBS), safe and attractive travel corridors, efficient traffic flow, effective maintenance, and to protect the public investment in the infrastructure. Procedures are outlined in the *RBS Standards* for the administration, planning, design, construction, and maintenance of roads and bridges within unincorporated Park County (not including State or Federal lands).

Except as otherwise provided for in the *Park County Development Standards and Regulations (DSR)*, the RBS apply to all roads within unincorporated Park County that are properly established County roads, roads constructed or to be constructed within established County rights-of-way and those roads accessing any division of land created under the *Wyoming Real Estate Subdivision Act, W.S. §18-5-301 through 318*. In addition, the RBS apply to private and public roads serving commercial and/or industrial development. Refer to Section 2.1.2 for specific application of the RBS as it relates to land development. The *RBS Standards* are considered minimum and are not intended to replace or to conflict with ~~this nor~~ any other federal or state law or regulation. Should there be a conflict, the more stringent law, regulation or statute shall apply.

1.2 AUTHORITY

The County Engineer ~~and the Road & Bridge Foreman have~~ *has* been delegated authority by the Board of Park County Commissioners (BOCC) to administer and enforce the *RBS se Standards*, ~~issue Right-of-Way permits, hereinafter referred to as “ROW permit(s)”, review plans and conduct inspections.~~ This authority includes the following responsibilities:

- Administer and oversee the Park County Public Works Department (PCPWD) which includes the Road & Bridge (R&B), Solid Waste (SW), and Engineering Divisions.
- Planning and evaluation of CRBS needs.
- Administration and tracking of PCPWD budgets as approved by the BOCC for each fiscal year.
- Seek, secure, and administer supplemental funding via local, State, and Federal sources for County initiated road and bridge system projects.
- Administer and oversee the design, construction, and maintenance of CRBS infrastructure.
- Promote interagency coordination and cooperation while ensuring compliance with applicable local, State, and Federal regulatory and permitting requirements.
- Review and approve construction plans and related documents associated with applicable road and bridge infrastructure serving land development (when applicable).
- Review and approve site, drainage, floodplain, and access plans serving new development in coordination with the Park County Planning & Zoning Department when required by the DSR.
- Administer the Road Naming and Addressing Policy including the issuance of permits for new road names and new addresses.

- Review and approve construction/installation by others of improvements, repairs, and/or maintenance within dedicated County Road rights of way through a standardized permitting process (i.e., Right of Way Permits).
- Administer and oversee County traffic safety and sign programs to ensure uniformity and consistency throughout the CRBS while emphasizing the protection of public health and safety.
- Administer and oversee procurement of County equipment and materials utilized by the PCPWD for County infrastructure improvements and maintenance.
- Administer and oversee the County's gravel road and pavement management programs.
- Review and consider CRBS related service requests and/or complaints from the public.
- Administer the creation, alteration, or vacation of public roads in accordance with Wyoming State statutory requirements.
- Conduct traffic studies to assist in identifying traffic safety trends and needs to assist in planning CRBS infrastructure projects.
- Maintain public records consistent with Wyoming State statutory requirements pertaining to CRBS infrastructure.
- Coordinate with the BOCC in ensuring that the RBS is updated when necessary to ensure adherence to the goals and objectives outlined in any BOCC approved County Land Use Plan (LUP) and/or Transportation and Safety Plan (TSP).

The County Engineer ~~and/or Road & Bridge Foreman~~ may designate ~~a designee as appropriate to enforce~~ qualified PCPWD personnel (i.e., Road & Bridge Foremen, Project Manager, or Operations Manager) as appropriate to assist in enforcing the ~~RBS~~ **RBS Standards**. In the County Engineer's absence, the BOCC shall designate a qualified individual to oversee the PCPWD including administration and enforcement of the ~~RBS~~.

Park County is in the process of developing a *County Transportation Safety Plan (TSP)* through a federal grant (*Safe Roads and Streets for All Program*). Ideally and ultimately, the *RBS* shall consider and be consistent with the pillars, goals, and objectives identified and developed within the framework of the *TSP*. The *RBS* supersedes and replaces all previous County Road & Bridge-related policies and standards. The *RBS* shall be updated when deemed necessary by the County Engineer and the BOCC as usage, needs, engineering practices, and construction methods evolve. In addition, this document shall be amended when deemed necessary by the County Engineer and the BOCC, depending on fundamental changes to the *DSR* and/or the *TSP* when applicable. The process for amending these *RBS* are outlined in *Section 2.2.4*.

1.3 HISTORICAL DEVELOPMENT

Covering 6,943 square miles (*United States Census Bureau, 2012*), Park County is known for having diverse terrain including mountains and streams, open range, and productive agricultural lands. The CRBS mirrors this diversity with a variety of different road and bridge types, characteristics, and challenges.

The framework for the current CRBS dates to before the founding of the City of Cody in 1896 (*Houze, 2014*) and long before Park County was legally established as a subdivision of the State of Wyoming in 1911 (*Houze, 2014*). Early road development throughout the County involved transforming stagecoach routes as well as establishing and improving a route to Yellowstone National Park (established as a national park in 1872). Much of the original road construction in this area was managed by the United States Army Corps of Engineers between 1872 and 1905 (*Williamson, 2016*). Subsequent road improvements were initiated to improve connectivity between communities including the City of Cody, City of Powell, Town of Meeteetse, and Frannie, as well as the Garland, Clark, and Ralston townsites.

Early development in Park County was bolstered by agricultural interests with the construction of the Buffalo Bill Dam and associated irrigation projects. The road system at the time was also improved and expanded to serve emerging oil and gas projects throughout the County.

With the invention and introduction of the automobile, the need for better, safer, and more reliable roads became increasingly important in Park County by the early 1920s. Large infrastructure projects in the 1960s such as the construction of the tunnels through Shoshone Canyon on State Highway 14/16/20 further improved connectivity throughout the County. Eventually, the State of Wyoming took over the primary connecting routes and developed more reliable roads as part of the State Highway System. As people moved to the area and developed rural lands in Park County, the CRBS continued to evolve with a network of “farm to market”, residential, and collector roads, several of which have been improved and are now maintained by Park County.

Road and bridge related policies have been enacted by the BOCC over many years to keep up with the growing public infrastructure needs and demands. The need to establish uniform and effective design, construction and maintenance requirements as it relates to the CRBS is essential in the interest of preserving public safety and reliable access while supporting economic growth and prosperity throughout Park County. The *RBS* has evolved from the policies adopted by the BOCC and reflect the standard operating procedures currently in practice by the PCPWD as it relates to the CRBS.

2.0 ADMINISTRATION

2.1 APPLICABILITY

2.1.1 Applicable State Statutes

The *RBS se-Standards* are based on the following *Wyoming State Statutes*:

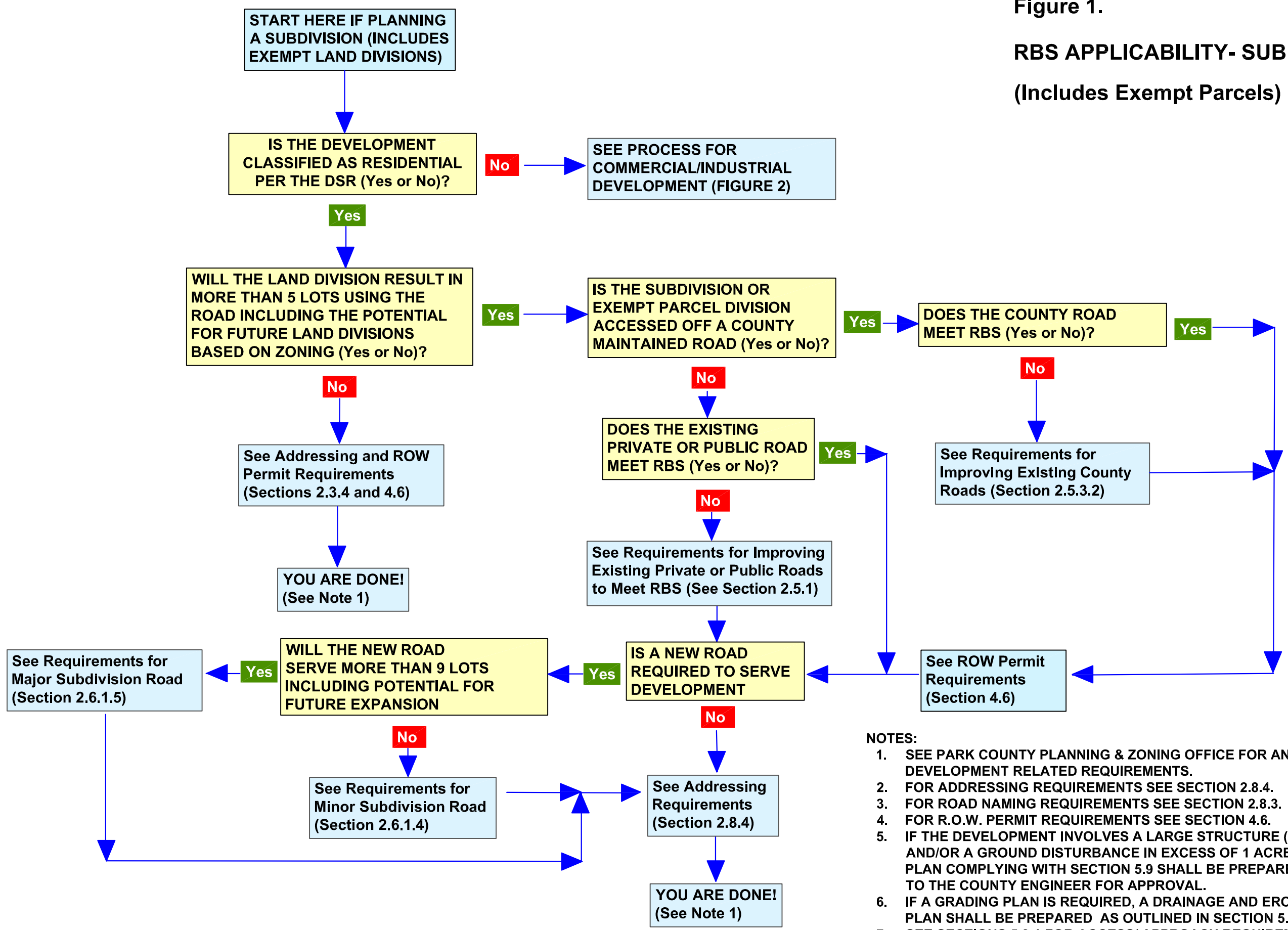
- *W.S. §§ 1-1-120*, limits civil liability when providing public assistance in the event of an emergency (Good Samaritan Law).
- *W.S. §§ 6-6-301 through 307*, prohibits unlawful conduct within governmental facilities, including land under the control of a governmental body.
- *W.S. §§11-28-105*, BOCC authority regarding cattleguards and fences placed within public road rights-of-way.
- *W.S. §16-3-106*, establishes the public process for promulgation, amendment, or appeal of a rule or regulation;
- *W.S. §16-9-101 through 108*, establishes the emergency notification system including 911;
- *W.S. §18-2-101*, “*General Powers*”, authorizes the Board of County Commissioners to manage the business and concerns of the County in the exercise of its corporate or administrative powers.
- *W.S. §§ 18-3-504*, outlines general powers and duties of the Board of County Commissioners.
- *W.S. § 18-5-201 through 208*, et seq., provides for a county’s authority to regulate land use in unincorporated areas to promote public health, safety, and general welfare.

- W.S. § 18-5-306, outlines a county's authority regarding the design and construction requirements for roads within private subdivisions. ~~(a)(vii) provides for a county's authority over subdivision roads.~~
- W.S. § 18-12-103, outlines the process for which residents may petition the BOCC to establish a Road Improvement and Maintenance District (RIMD).
- W.S. § 18-12-112, summarizes the powers and authority of the RIMD.
- W. S. §24-1-104, authorizes the BOCC to administer the ~~CRBS County road system~~ including, but not limited to, maintenance, layout, establishment, alterations, vacations, property acquisition and traffic regulations.
- W. S. §24-1-105, establishes width restrictions for County Roads.
- W. S. §24-1-106 through 109, grants county commissioners and authorized officials the authority to close or restrict the use of county roads for safety, construction, or maintenance.
- W. S. §24-2-110, dictates state cooperation and funding for county road construction, and tasks the Wyoming county roads standards committee with furnishing construction and maintenance standards (establishes the County Road Construction & Maintenance Fund program).
- W. S. §24-3-101, et seq. outlines the authority and procedures to be followed in the establishment, vacation or alteration of County highways.
- W. S. §24-3-108, authorizes BOCC the ability to establish a County public road with consent of all impacted landowners.
- W. S. §24-3-203, allows counties to adopt resolutions and maps to legally identify, clarify, and establish the locations of existing county roads.
- W.S. §§ 24-6-101 through 111, provides counties with additional authority over access issues related to public highways.
- W.S. § 35-10-401 provides that public highways, including county road rights-of-way, shall not be obstructed, and provides for criminal penalties.
- W.S. § 37-12-301 through 305, *Wyoming Underground Facilities Notification Act* and associated requirements.

2.1.2 Applicability to Land Development

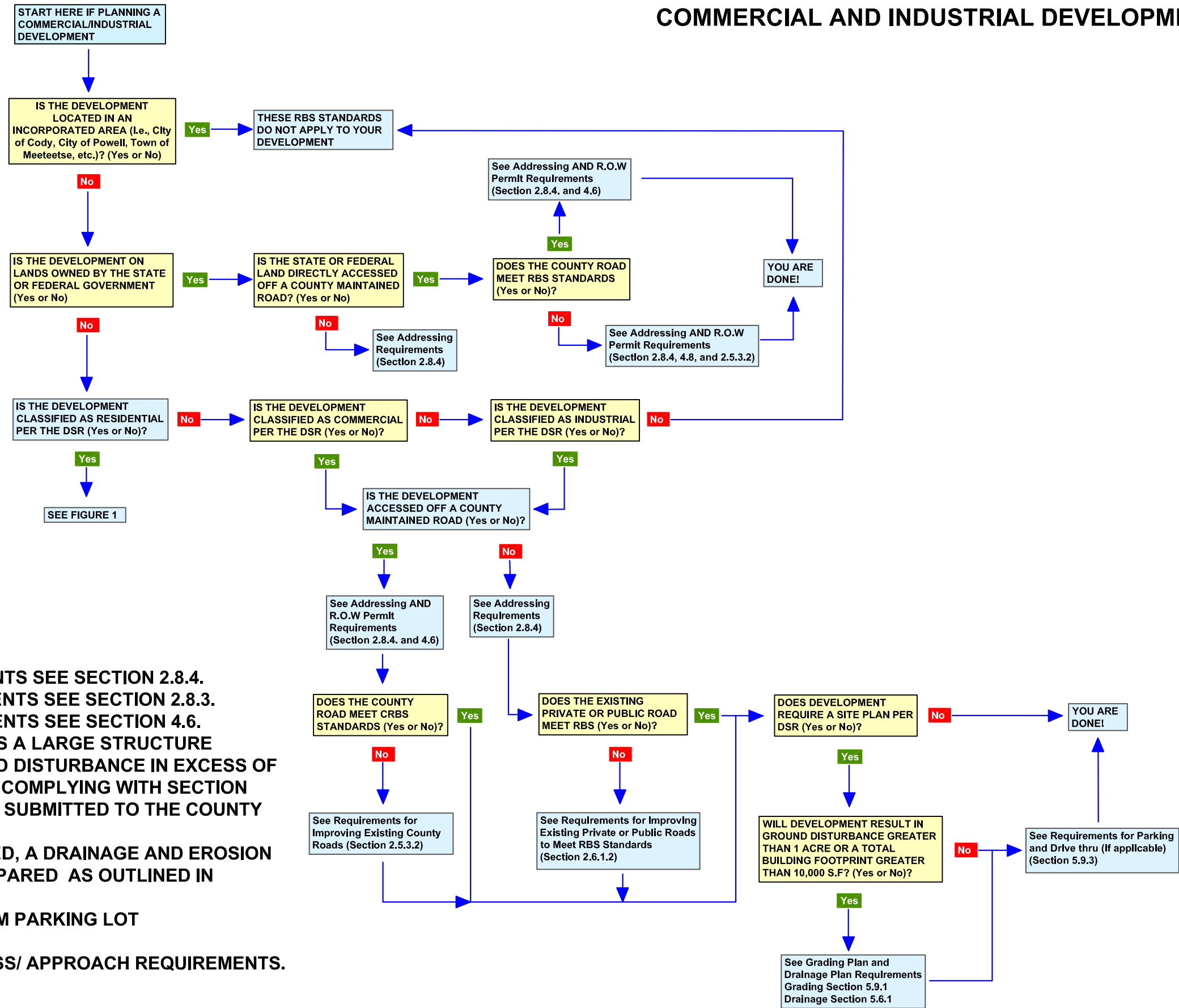
The BOCC have adopted *Development Standards and Regulations (DSR)* to address land development in the unincorporated areas of Park County. Questions regarding the DSR and applicability to specific types of development shall be directed to the Park County Planning and Zoning Office. As it relates to development, certain sections of the *RBS* may be applicable. **FIGURE 1** addresses applicability to residential development and provides a flow chart guide to specific sections of the *RBS* based on the type, size, scope, and extent of residential development. **FIGURE 2** addresses the applicability of the *RBS* to commercial and industrial development in unincorporated Park County. Questions regarding the applicability or interpretation of the *RBS* shall be directed to the PCPWD (307-527-8520 or publicworks@parkcounty-wy.gov).

Figure 1.
RBS APPLICABILITY- SUBDIVISIONS
(Includes Exempt Parcels)



- NOTES:
1. SEE PARK COUNTY PLANNING & ZONING OFFICE FOR ANY ADDITIONAL DEVELOPMENT RELATED REQUIREMENTS.
 2. FOR ADDRESSING REQUIREMENTS SEE SECTION 2.8.4.
 3. FOR ROAD NAMING REQUIREMENTS SEE SECTION 2.8.3.
 4. FOR R.O.W. PERMIT REQUIREMENTS SEE SECTION 4.6.
 5. IF THE DEVELOPMENT INVOLVES A LARGE STRUCTURE (≥10,000 S.F.) AND/OR A GROUND DISTURBANCE IN EXCESS OF 1 ACRE THEN A GRADING PLAN COMPLYING WITH SECTION 5.9 SHALL BE PREPARED AND SUBMITTED TO THE COUNTY ENGINEER FOR APPROVAL.
 6. IF A GRADING PLAN IS REQUIRED, A DRAINAGE AND EROSION CONTROL PLAN SHALL BE PREPARED AS OUTLINED IN SECTION 5.6.
 7. SEE SECTIONS 5.9.1 FOR ACCESS/ APPROACH REQUIREMENTS.

RBS STANDARDS APPLICABILITY-
COMMERCIAL AND INDUSTRIAL DEVELOPMENT



NOTES:

1. FOR ADDRESSING REQUIREMENTS SEE SECTION 2.8.4.
2. FOR ROAD NAMING REQUIREMENTS SEE SECTION 2.8.3.
3. FOR R.O.W. PERMIT REQUIREMENTS SEE SECTION 4.6.
4. IF THE DEVELOPMENT INVOLVES A LARGE STRUCTURE (≥10,000 S.F.) AND/OR A GROUND DISTURBANCE IN EXCESS OF 1 ACRE THEN A GRADING PLAN COMPLYING WITH SECTION 5.9.1 SHALL BE PREPARED AND SUBMITTED TO THE COUNTY ENGINEER FOR APPROVAL.
5. IF A GRADING PLAN IS REQUIRED, A DRAINAGE AND EROSION CONTROL PLAN SHALL BE PREPARED AS OUTLINED IN SECTION 5.6.1.
6. SEE SECTION 5.9.3 FOR MINIMUM PARKING LOT REQUIREMENTS.
7. SEE SECTIONS 5.9.2 FOR ACCESS/ APPROACH REQUIREMENTS.

REV.	DATE	BY



DRAWN BY:	B. Edwards
DESIGNED BY:	B. Edwards
APPROV. BY:	BOCC
SCALE:	Not to Scale
DATE:	08/02/2026
FOR NO.	

RBS STANDARDS APPLICABILITY- COMMERCIAL/ INDUSTRIAL	Wyoming
PARK COUNTY R&B STANDARDS	Park County

2.2 APPLICATION OF STANDARDS

Application of Standards: All applicable new road and bridge construction, and any applicable upgrading of ~~the~~ existing roads or bridges (refer to Section 2.1 for applicability), commencing after the effective date of these *RBS Standards* shall adhere to these *Standards* unless a variance is requested and approved by the BOCC. ~~an Administrative Relief is granted in accordance with Chapter V, Section 7, et seq. of these Standards.~~

2.2.1 Enforcement of Standards

2.2.1.1 Organization and enforcement

The County Engineer is authorized and directed to enforce all provisions of the *RBS* ~~ese Standards~~. The County Engineer may appoint the Road & Bridge Foreman, Project Manager, Operations Manager, ~~construction inspector or other related technical officer or inspector~~ or other PCPWD employee to act as an authorized representative.

Whenever any work is being done contrary to the provisions of the *RBS*, ~~se Standards~~, the County Engineer or his designated agent may order the work stopped by a written notice which shall be served on any persons engaged in the doing or causing of such work to be done, and any such persons shall forthwith stop such work until authorized by the County Engineer to proceed.

2.2.1.2 Liability

Park County, or its authorized representative, charged with the enforcement of the *RBS* ~~se Standards~~, acting in good faith and without malice in the discharge of duties set forth herein shall not be rendered liable for any damage that may accrue to persons or property as a result of any act or by reason of any act or omission in the discharge of such duties. *Nothing in the RBS as stated or otherwise interpreted shall hinder, negate, replace or otherwise impact Park County's status as a governmental entity regarding governmental immunity as outlined in Wyoming State Statutes.*

2.2.1.3 Violations

It shall be unlawful for any person, firm or corporation to construct, enlarge, alter, repair, relocate, improve, remove, excavate, convert or demolish any public improvements or common facilities or permit the same to be done in violation of the *RBS* ~~se Standards~~. Utility Service Providers, Contractors, or any entity/person who violates these *RBS Standards* shall be subject to denial of future ROW Permits ~~and may be subject to fines or penalties.~~

2.2.2 Variances

2.2.2.1 Purpose

The variance procedure provides authorization to deviate from the literal ~~standards and~~ requirements of the *RBS* ~~se regulations~~ due to special or extraordinary circumstances ~~of the land~~ where strict application and enforcement of the *RBS* ~~se regulations~~ would result in exceptional practical difficulty or undue,

unique, and unnecessary hardship preventing the development, use, and access of ~~the land, building, or structure as otherwise permitted by these regulations.~~

2.2.2.2 Applicability

~~(2) A property owner, developer, contractor, or governmental authority may seek relief from these Standards regulations when the strict application of these Standards regulations would result in undue, unique, and unnecessary hardship, impacting depriving the owner of the reasonable use of the property. The BOCC Board may vary or adjust the strict application of any of the requirements of the RBS se regulations if it would adversely impact or restrict deprive the owner of the reasonable use of the land (including, buildings, or structures if applicable) involved. The BOCC Board may only vary substantive requirements of the RBS se regulations, such as design or maintenance standards, plat standards, application requirements, etc. The BOCC shall seek input from the County Engineer or other Wyoming licensed professional engineer when making decisions regarding variances in design standards where engineering judgement is needed to ensure protection of public health and safety. The BOCC Board and County Engineer may not waive or vary any procedural or statutory requirements in administering the RBS. for subdivisions set forth in W.S. §18-5-301 through 18-5-317.~~

2.2.2.3 Application Submittal

Those seeking a variance from the RBS shall complete and submit a **complete Variance Request Application** (See **APPENDIX D**) ~~shall be submitted~~ to the PCPWD ~~P&Z Department~~, including all applicable supplementary materials and payment.

2.2.2.4 BOCC Consideration

~~(1) (2) Public Hearing: Once a Variance Request Application is deemed complete by the County Engineer, the P&Z Director, a public hearing shall be scheduled during a regular meeting of the BOCC P&Z Commission to consider the request at the earliest available time. The hearing shall be noticed at least fourteen (14) days prior to the date of the hearing, pursuant to the public hearing requirements of the RBS se regulations (APPENDIX EA). All costs associated with noticing any public hearing related to a variance request shall be paid by the applicant.~~

Recommendation: Within forty-five (45) days after the closure of the public hearing, the BOCC ~~P&Z Commission~~ shall resolve to recommend approval, approval with conditions, or denial of the variance request. ~~The P&Z Department shall forward the recommendation to the Board for their consideration. e. Board Decision: (1) (2) f. Public Hearing: Following receipt of the P&Z Commission's recommendation, the Board shall hold a public hearing to consider the Variance Request Application at the earliest available time during a regular meeting. The hearing shall be noticed pursuant to the public hearing requirements of these regulations. All costs associated with noticing any public hearing related to a variance request shall be paid by the applicant. Decision: Within forty-five (45)~~

~~days after the close of the public hearing, the Board shall resolve to approve, approve with conditions, or deny the variance request. Variance Approval Criteria:~~

The applicant shall have the burden of demonstrating that the application satisfies the applicable review criteria. Variances from ~~the 1-9 Park County 2025 DSR—Chapter I—ADMINISTRATION requirements of the RBS~~ ~~se regulations~~ shall be considered an extraordinary remedy. If the application does not meet the criteria for a variance, an amendment to the regulations, pursuant to Chapter 1, Section 3, may be sought. When considering, reviewing and deciding whether to approve a variance application, the ~~P&Z Commission and BOCC Board~~ must find that a majority of the following criteria have been met:

1. The practical difficulties arising from application of these ~~RBS regulations~~ are significant or create exceptional and undue hardship upon an applicant, provided the conditions of ~~Section 7.d. 4.~~ below are met.
2. If an individual is proposing to construct a ~~low volume or local access~~ road and the application of the ~~RBS se Standards~~ would result, in the opinion of the ~~BOCC County Engineer~~, in excessive cut and fill slopes, visual scarring or other environmental damage.; A variance to the ~~Administrative Relief in~~ road design standards (~~Section 5~~) may be ~~approved~~ granted if ~~granting the Administrative Relief the action~~ will result in lessened environmental damage, and the conditions in ~~Chapter V, Section 7.4. below of these Standards~~ are met.
3. If documentation is provided including, but not limited to, technical references from recognized professional organizations, or significant changes to standards adopted by reference in ~~APPENDIX B24~~ offer options to the ~~RBS, se Standards; a variance Relief~~ may be granted by the ~~BOCC County Engineer~~. Further, if it is determined at the sole discretion of the ~~BOCC County Engineer~~ the option is an improvement or viable alternative which maintains the same level of care and safety intended by the ~~RBS, se Standards; Administrative Relief~~ a variance may be granted.
4. A variance providing relief ~~Administrative Relief~~ from the difficulties or hardships described in ~~Chapter V, Sections 7.a. and b. 1. and 2. above of these Standards~~ may be granted provided variance Relief will not result in substantial detriment to public health, safety and welfare, or substantial impairment of the road design and construction standards. Prior to acting, the ~~BOCC and County Engineer and the Road and Bridge Foremen~~ shall review the request for a variance ~~an Administrative Relief~~ and, if necessary, refer any variance request for ~~Administrative Relief~~ to the appropriate fire district, Sheriff's Department and other interested agencies for comment. The ~~BOCC County Engineer~~ shall decide on whether a variance ~~an Administrative Relief request~~ should be

granted based on input from the County Engineer and applicable agencies.

In reviewing such requests, the BOCC ~~County Engineer~~ shall, at a minimum, consider the following:

- A. The effect of using a lesser standard of public health and safety including the ability of emergency vehicles to gain access using roads built to a lesser standard.
- B. The severity of the terrain crossed by the road alignment.
- C. The availability of alternative alignments where the same or more stringent road standards could be met with the same or less environmental damage.
- D. The length of road segments which will be built to a lesser standard; and
- E. Future maintenance requirements, including, but not limited to, snow removal.

Costs may be considered included in the review of a variance ~~an Administrative Relief~~ request, but shall not supersede public health, safety and welfare or substantial impairment of the road design and construction standards. If costs are considered, the BOCC ~~County Engineer~~ shall consider the initial and long-term costs. The applicant shall provide all cost data requested by the BOCC ~~County Engineer~~, as required, to provide a complete cost analysis.

~~(1) (2) (3) (4) (5) 1. There are unusual physical circumstances or exceptional conditions, including, without limitation, irregularity, narrowness or shallowness of the lot, exceptional topography or other physical conditions peculiar to the affected property; The special circumstances and unique physical conditions are not the result of actions of the applicant taken after the adoption of these regulations; The variance sought is the minimum variance necessary to afford the requested relief and would be the least modification of the applicable provisions of these regulations; Granting of the variance is consistent with the general purposes and intent of these regulations; and Granting of the variance will not: (a) be injurious to or impair the reasonable use and enjoyment or development of adjacent properties; (b) (c) alter the character of the community; or otherwise be detrimental to the public health, safety and welfare.~~

In approving a variance, the P&Z Commission and Board the BOCC may impose reasonable conditions and restrictions on such approval, and the premises to be developed or used pursuant to such approval, it determines necessary to prevent or minimize adverse effects from the proposed variance on other properties in the vicinity, and on the general health, safety and welfare of the public. Such conditions imposed upon any variance shall be set forth in the resolution granting such variances, shall be limited to direct impacts of the proposed use or development, and shall be proportional to the impacts. h. Expiration: A variance

~~shall expire and become void one year following the date of approval, unless otherwise specified by the Board, unless construction or development is commenced prior to the expiration date and diligently pursued to completion. The Board may grant an extension for up to one additional year for good cause, including if construction or development is commenced prior to the first year's expiration date and diligently pursued to completion. Extension requests shall be submitted to the Board prior to the expiration date. i. Denial: If a variance application is denied by the Board, another application which is substantially the same cannot be submitted within one (1) year of the denial. If the variance is denied a second time, another application that is substantially the same cannot be submitted within three (3) years of the second denial.~~

2.2.3 Administrative Relief

~~Whenever there are practical difficulties involved in carrying out the provisions of these Standards, the County Engineer may approve Administrative Relief. Relief from the design criteria and construction specifications contained in these Standards may be granted by the County Engineer under the following circumstances.~~

2.2.3 Process for Appeals

2.2.3.1 Purpose

The purpose of the appeals procedure is to define the circumstances by which a person may appeal a final decision made by the ~~P&Z Commission, P&Z Director,~~ ~~or~~ County Engineer or his/her assigns.

2.2.3.2 Applicability

~~(1) (2) (3)~~ An aggrieved or adversely affected person may appeal any final determination, interpretation, decision or similar action made by ~~P&Z Commission, the P&Z Director, or~~ the County Engineer or his/her assigns pursuant to the ~~RBS se regulations~~. The person filing the appeal is the "appellant". Final decisions made by the ~~BOCC Board~~ pursuant to the ~~RBS se regulations~~ may be appealed to the courts in accordance with state law.

2.2.3.3 Initiation of Appeal

~~(1) (2)~~ Appeals shall be made in the form of a written notice of appeal letter delivered or postmarked to the ~~BOCC's Board's~~ Administrative Assistant within twenty (20) days of the date the written determination, interpretation or decision was made. Such notice shall identify the date and nature of the written determination, interpretation or decision, and set forth in plain and concise language the facts and reasons for the appeal, including any relevant citation to any rule, regulation, or resolution section relied upon. Such notice shall also include a copy of the written determination, interpretation or decision being appealed. The burden of proof on appeal shall rest with the appellant. ~~17 Park County 2025 DSR Chapter I ADMINISTRATION~~

2.2.3.4 Scheduling of Appeal

Upon receipt of the notice of appeal, the BOCC's ~~Board's~~ Administrative Assistant shall schedule a public hearing to consider the appeal during a regular or special meeting of the BOCC ~~Board~~. The public hearing shall occur as soon as time is reasonably available on the BOCC's ~~Board's~~ schedule, but no later than forty-five (45) days after the filing of the notice of appeal.

2.2.3.5 Notice of Public Hearing

The BOCC's ~~Board's~~ Administrative Assistant shall send written notice to the appellant of the time, date, and location of the appeal hearing by certified mail, at the expense of the appellant, not less than fifteen (15) days prior to the appeal hearing. The BOCC's ~~Board's~~ Administrative Assistant shall notice the public hearing at least fourteen (14) days prior to the hearing date, pursuant to the procedures for public hearings in **APPENDIX E A** of the ~~RBS~~ **se regulations**. Said notice shall include the following information: a brief description of the original proposal or project that is the subject of the appeal, the location of the subject property, appellant's name, hearing date, time and location, and how additional information may be obtained.

2.2.3.6 Board Hearing on Appeals

~~(2) (3) (4) (5) Evidence:~~ The BOCC ~~Board~~ shall decide what evidence is material to the appeal. ~~Testimony:~~ The BOCC ~~Board~~ shall allow an opportunity during the hearing for the appellant, staff, and any member of the public to offer either written or oral testimony regarding the appeal under consideration.

2.2.3.7 Official Record of Appeal Hearing

The official record of an appeal hearing shall consist of materials submitted by the appellant, correspondence received concerning the appeal, testimony offered at the hearing, any written documents or exhibits presented at the hearing, any applicable resolutions, official actions, written determinations, interpretations or decisions relevant to the appeal, and the official file maintained by the ~~P&Z Department or Public Works Department~~ PCPWD on the action from which the appellant has appealed.

2.2.3.8 Basis of Decision on Appeal

Review of the decision being appealed shall be limited to the record established before and relied upon by the BOCC ~~Board~~. The Board shall not have the authority to override provisions of the ~~RBS~~ **se regulations**. Any decision shall include a basis for the decision and cite specific sections of the ~~RBS~~ **se regulations**.

2.2.3.9 Decision on Appeal

The ~~Board~~ BOCC shall hear all relevant evidence and in no more than thirty (30) days after the close of the public hearing shall render its decision. The BOCC ~~Board~~ may reverse or affirm, wholly or in part, the written determination,

interpretation, decision or action, but in so doing shall comply with applicable law. All decisions on an appeal shall be adopted by resolution, containing a concise listing of facts and reasoning supporting same, and shall be promptly mailed to the appellant and sent to the ~~P&Z Director or~~ County Engineer. The resolution shall be recorded in the Clerk's Office.

~~If an applicant does not agree with the determination made by the BOCC County Engineer, the applicant may appeal the decision using the procedures described herein. to the Board of County Commissioners.~~

2.2.4 Process for Amendment of Standards

2.2.4.1 Authority

Amending the ~~RBS se regulations~~ is a matter committed to the legislative discretion of the ~~Board~~ BOCC. Pursuant to W.S. §16-3-106 (*Petition for promulgation, amendment or repeal of rules*), any interested person may petition the BOCC ~~Board~~ requesting the promulgation of a new rule, or an amendment or repeal of all or part of the ~~RBS se regulations~~. Nothing in the ~~RBS se regulations~~ shall be construed to limit the BOCC's ~~Board's~~ inherent authority to initiate rulemaking proceedings.

2.2.4.2 Procedure for Submission of Petition

Any petition to amend the ~~RBS~~ ~~authorized by Subsection a of this Section~~ shall be submitted to the BOCC's ~~Board's~~ Executive Assistant using the ~~Text Amendment Application~~, a form provided to the public by the PCPWD (See **APPENDIX F**) ~~P&Z Department~~, which shall include the following components:

1. ~~(1)(2)(3)(4)(5)~~ Nature of the request (e.g., promulgation, amendment or repeal).
2. Proposed action(s), including the exact language to be promulgated, amended, and/or repealed.
3. Statement justifying the proposed action(s), which may include relevant data, views and arguments.
4. Statement describing the consistency of the proposed action(s) with other adopted County Regulations, the *Park County Land Use Plan (LUP)*, *DSR*, *TSP*, and the intent of these ~~RBS regulations~~.
5. and a ~~A~~ fee tendered in the correct amount according to the fee schedule adopted by separate resolution (if applicable). ~~I-4 Park County 2025 DSR — Chapter I — ADMINISTRATION~~

2.2.4.3 Consideration of Petition

Upon submission of a petition, the BOCC's ~~Board's~~ Executive Assistant shall schedule the matter for consideration at the next available time during a regular meeting of the BOCC ~~Board~~. As soon as practicable, but in no event later than sixty (60) days after the date of submission of the completed petition, the BOCC ~~Board~~ shall resolve to either 1) deny the petition or 2) initiate rulemaking proceedings in accordance with W.S. §16-3-103 (*Adoption, amendment and repeal of rules*). The action of the BOCC ~~Board~~ in denying a petition is final, shall be in writing (stating its reasons for the denial), and is not subject to review/appeal.

The resolution on the petition shall be recorded in the Clerk's Office and a copy mailed to the petitioner. Where the BOCC Board resolves to initiate rulemaking proceedings, the process shall continue as described in 2.2.4.4. ~~under subsection d of this Section.~~

2.2.4.4 Public Participation

The BOCC Board should consider reasonable means by which it may encourage citizen participation and the collection of information throughout the RBS regulation amendment process. The methods by which this is to be accomplished may include but are not limited to: public gatherings (e.g., open houses and planning area/community meetings); citizen advisory committees; technical working groups for participants from relevant agencies (e.g., utilities, special districts, government agencies); citizen surveys; studies (e.g., demographics, economics, growth projections, water availability, etc.). The BOCC Board may choose to procure professional services to assist with meeting facilitation, drafting documents, information gathering or related tasks. The amount of public engagement shall be determined by the BOCC Board but shall meet the minimum requirements of W.S. § 16-3-103.

2.2.4.5 Review Procedure for Amendments:

Any proposal for amendments to the RBS ~~se regulations~~ shall be considered in accordance with W.S. §16-3-103 and the process that follows. ~~(1) (2) (3) Staff Review:~~ Within fourteen (14) days of the BOCC's Board's decision to initiate rulemaking proceedings, the PCPWD P&Z Department staff shall schedule a public hearing during the next available time at a regular BOCC P&Z Commission meeting. Notice of the public hearing shall be given at least ~~forty-five (45)~~ ~~thirty (30)~~ days in advance of the hearing, according to the public hearing notice procedure in these regulations. Further, the notice shall include a summary of the proposed amendments and a link to the County website where the full request submittal and staff recommendations may be viewed.

~~P&Z Department staff, in conjunction with Public Works-PCPWD staff for changes relating to Chapter V, shall prepare a staff report to be provided to the BOCC P&Z Commission at least five (5) days prior to the hearing. P&Z Commission Review:~~ In the public hearing, the BOCC P&Z Commission shall review the amendment request and consider the recommendations of the requestor and staff as presented in the staff report, in addition to receiving and considering public testimony. ~~P&Z Commission Recommendation: (a) Within forty five (45) days after the close of the public hearing, the P&Z Commission shall, by resolution, recommend that the Board approve, approve with modifications, or deny the requested amendments to the regulations. The resolution shall be carried by the affirmative votes of not less than a majority of the entire Commission, and be recorded in the Clerk's Office with the associated meeting minutes. 1-5 Park County 2025 DSR Chapter I ADMINISTRATION (b) The P&Z Department and the Board's Executive Assistant shall schedule a public hearing during the next available time at a regular meeting of the Board. Notice of the public hearing shall be given at least forty five (45) days in advance of the hearing, according to the public hearing notice procedure in these regulations. Further, the notice shall~~

~~include a summary of the proposed amendments and a link to the County website where the full request submittal may be viewed, in addition to the P&Z Commission's resolution. A complete staff report packet, including the Commission's recommendations, shall be prepared by staff and provided to the Board at least five (5) days prior to their hearing. (4) (5) f. Board Review:~~

In the public hearing, the BOCC Board shall review the amendment request and consider the recommendations of the requestor and staff ~~and the P&Z Commission~~ (as presented in a PCPWD P&Z Department staff report), in addition to receiving and considering public testimony. ~~Board Decision:~~ Within forty-five (45) days after the closure of the public hearing, the BOCC Board shall resolve to adopt, adopt with modifications, or deny the requested amendments to the RBS regulations. ~~In the event the Board adopts the Commission's resolution with modifications, the Board need not resubmit any such measure to the Commission but may rely on its own inherent authority to make such modifications.~~ Whatever course the BOCC Board resolves upon, it must be in full compliance with W.S. §16-3-103. The resolution shall be carried by the affirmative votes of not less than a majority of the entire BOCC Board, and be recorded in the Clerk's Office. Failure of a department to submit a staff report shall not be grounds for invalidating the BOCC's Board's action.

2.2.4.6 Publication and Filing

Within thirty (30) calendar days following the date amendments are adopted by the BOCC Board, the PCPWD P&Z Department shall revise the text and/or graphics of the RBS ~~se Standards regulations~~ to reflect the adopted amendments and make this text available to the public. The PCPWD P&Z Department shall file a copy of the adopted revised text in the Clerk's Office, PCPWD, the P&Z Department, and on the County's website. ~~g. Validity of Regulations:~~ The amended RBS regulations shall take effect on the date specified by the Board BOCC in their resolution of approval.

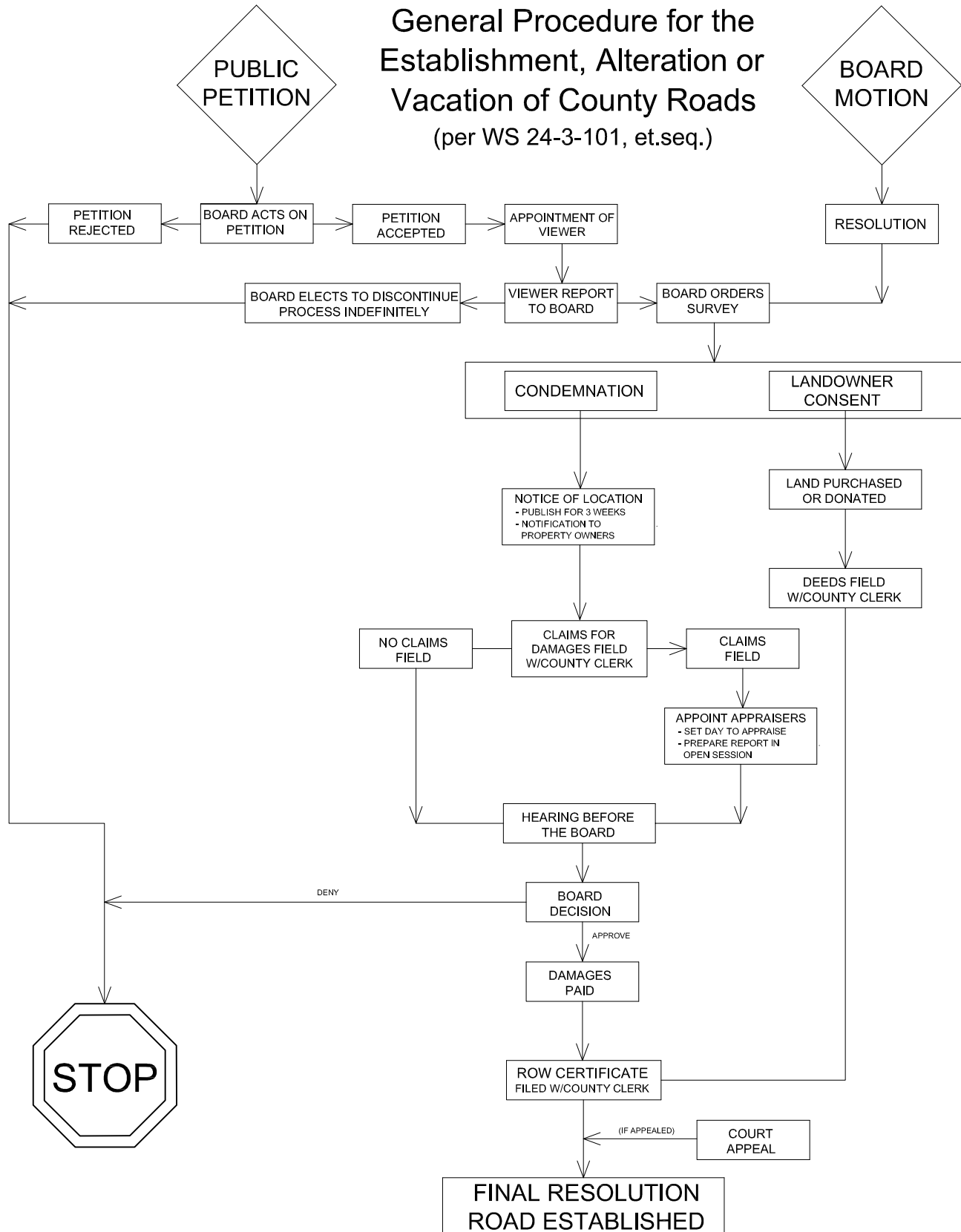
2.3 CREATION, ALTERATION, OR VACATION OF PUBLIC ROAD RIGHT-OF-WAYS

~~Establishment, vacation or alteration of County roads:~~ Any party wishing to initiate the ~~(establishment, vacation or alteration)~~ of a dedicated public road or easement must conform to the statutory requirements of W.S. §24-3-101, et seq. A flow chart illustrating the process for creation, alteration, or vacation of a public road right-of-way is provided in **FIGURE 3**.

2.4 ROAD IMPROVEMENT AND MAINTENANCE DISTRICTS

Road Improvement and Maintenance Districts (RIMD) are local, citizen-initiated special taxing districts created to construct, improve, or maintain roads in specific areas. As outlined in W.S. § Title 18, Chapter 12, RIMD are funded by property assessments levied on residents within the boundaries and are governed by elected or appointed local boards. In subdivisions where a Homeowners Association (HOA) has been established, the RIMD may be part of and administered through the HOA.

General Procedure for the Establishment, Alteration or Vacation of County Roads (per WS 24-3-101, et.seq.)



Residents outside incorporated city limits, such as in rural subdivisions, may petition the BOCC to create a RIMD as outlined in *W.S. §18-12-103*, et seq. The petition (See **APPENDIX G**) shall outline the road limits and extents, planned improvements (if any), and provisions for maintenance. The petition shall also include an estimate of the cost of the improvements and ongoing maintenance estimates as well as a mechanism for collecting assessments from the impacted landowners that rely on and utilize the road(s) to be covered by the RIMD. The petition shall include signatures of all the landowners who utilize and rely on the road.

The petition to establish a RIMD shall be submitted to the BOCC's Administrative Assistant in writing. The BOCC's Administrative Assistant shall send written notice via certified mail to the petitioner and all impacted landowners of the time, date, and location of the public hearing, at the expense of the petitioner, not less than fifteen (15) days prior to the public hearing. The BOCC's Administrative Assistant shall notice the public hearing at least fourteen (14) days prior to the hearing date, pursuant to the procedures for public hearings in **APPENDIX E** of these *RBS*. Said notice shall include the following information: a brief description of the proposed RIMD proposal that is the subject of the public hearing, the location of the said road or roads, petitioner's name, hearing date, time and location, and how additional information may be obtained.

At the public hearing, the BOCC shall hear from the petitioner(s) and shall consider the petition to establish a RIMD. The BOCC may then choose to approve the RIMD, authorize the approval of the RIMD with conditions, or deny the request. If a RIMD is approved or conditionally approved by the BOCC, a separate legal entity shall be created at the petitioner's expense with authorization to collect special taxes or assessments as allowed per *Wyoming State Statutes*. The RIMD shall include a Board of representatives consisting of interested landowners appointed by the BOCC or the HOA if one exists. Funds assessed and collected by the RIMD shall be maintained in a dedicated improvement and maintenance fund account and shall be used only for the purpose of the designated road improvements and maintenance as approved by the RIMD.

2.5 UPGRADING EXISTING ROADS AND BRIDGES

2.5.1 Upgrades to Existing Non-County Roads to Accommodate New Development

When a proposed development will utilize an existing private road or public road not maintained by the County, the developer shall provide an estimate of the amount of traffic the proposed development is expected to add to the existing roadway. If the development will result in a substantial increase in traffic that will impact the classification of the roadway (see *Section 3.3.2*), the existing roadway shall be improved to bring the road up to County *Standards* for the given classification. If the existing roadway is sub-standard and does not meet County *Standards* for the given classification, the roadway shall be brought up to current design and construction standards regardless of whether the development will result in a change in classification. Refer to **FIGURE 1** (Residential) and **FIGURE 2** (Industrial/Commercial) to determine which design and construction standards are applicable based on the type, size, and scope of the planned development. Once the functional classification is known (*Section 3.3.2*), refer to *Section 5* for the design requirements applicable to a given functional classification.

2.5.2 Upgrades to Existing Subdivision Roads Requested by Landowners

~~Upgrading requested by property owners:~~ Upgrading existing substandard roads within a subdivision ~~or exempt land division~~ shall be at the expense of the property owners served by such roads. Upon request of the property owners ~~that utilize the road~~, the County may assist in the formation of a ~~Local Improvement District~~ RIMD as provided by W.S. §18-12-101 et seq (See Section 2.4).

Completion of the improvements ~~and bringing the road up to the Standards described herein~~, does not in itself constitute acceptance by the County for maintenance. Only the BOCC can authorize acceptance of a road for maintenance by the County.

2.5.3 Upgrades to Existing County Roads & Bridges

2.5.3.1 Upgrades by County to Consider Growth

It shall be the goal of the PCPWD to improve and maintain all dedicated County Roads and Bridges to the minimum standards outlined herein to the extent possible and practical. However, depending on budget constraints and revenue allocations approved by the BOCC, necessary upgrades to existing County Roads may be delayed until sufficient funding is available. PCPWD shall prioritize improvements to County Roads based on the following levels of importance:

- *Priority 1 - Improvements necessary to remedy unsafe infrastructure.*
- *Priority 2 - Improvements associated with critical arterial or collector routes.*
- *Priority 3 – Improvements that serve the greatest number of people (i.e., traffic use).*
- *Priority 4 - Improvements that will result in reduced long-term maintenance costs.*
- *Priority 5 – Improvements to serve future growth and development.*
- *Priority 6 – Improvements to low volume rural roads.*

2.5.3.2 Upgrades to Existing County Roads to Accommodate New Development

~~Upgrading needed to accommodate new development:~~ Where new development is proposed along existing County roads, the developer's proposal, ~~if required by the County Engineer~~, shall include an analysis of the projected traffic volumes and impacts, along with information on existing road widths, curves, intersections and drainage. This information shall be reviewed by the County Engineer, and recommendations shall be made as to what improvements are necessary to accommodate the ultimate traffic to be generated by the new development. If requested by the County Engineer, a traffic impact study in accordance with accepted traffic engineering standards shall be prepared by a Wyoming licensed professional engineer and shall be submitted to the County Engineer for review and approval. ~~f submitted by the developer.~~

~~Serving new developments:~~

~~Requirements for upgrading:~~ Existing County roads serving a new development or an area proposed for either platting or re-platting after the effective date of these ~~RBS Standards~~ shall be upgraded to the *Section 5.0 Design Standards* when one (1) or more of the following conditions occur:

1. Existing ~~County R~~oads which do not meet these Standards for the classification are improved or modified.
2. Existing ~~County R~~oads meet ~~local-access Rural Road Standards~~, but the projected ADT will exceed ~~one two~~ hundred (~~24~~00), thus requiring paving of the roads; and /or
3. Existing ~~County R~~oads meet ~~local-access Rural Road Standards~~, but the projected ADT exceeds the maximum for ~~Rural Roads local-access roads~~, thus requiring improvement to a ~~Local Access Road secondary road~~ status.
4. The County Engineer determines that a threat to public safety and welfare exists due to the increase in traffic volumes caused by the new development.

Existing County Roads that must be improved due to impacts from development shall be classified per Section 3.3.2 and improved to the corresponding design and construction standards outlined in Section 5.

2.5.4 Financial Responsibility

~~Payment of costs:~~ Park County shall be responsible (as funds are available) for the costs to improve and maintain dedicated County Roads and bridges and those existing public roads that are maintained by PCPWD if called for in existing agreements. However, costs to improve existing County maintained roads and existing privately maintained roads resulting from impacts attributed to a development shall be the responsibility of the developer. When upgrades to any road are deemed necessary resulting from a development to meet current standards, the developer shall be responsible for all costs incurred unless approved otherwise by the BOCC through a variance request. The BOCC may authorize a proportional cost sharing of the improvements if other development(s) in the area contributes to a change in road classification and need for improvements.

~~Payment of costs:~~ The developer may be responsible for a portion of costs incurred to upgrade existing County roads. The portion of costs to be paid by the developer shall be determined using a formula of proportionality as agreed upon by the County Attorney and the Board of County Commissioners. Administrative Relief from this requirement MAY be granted in accordance with Section 7 of these standards.

2.6 NEW ROADS AND BRIDGES

2.6.1 Serving Development- Based on Land Use

~~New roads to be built by developers:~~ Where new roads are proposed to be built by a developer to serve a major subdivision (6 or more residential lots), an exempt parcel division serving 6 or more lots, or a commercial/industrial development with an expected Average Daily Traffic (ADT) of 60 vehicles per day or more, the developer's proposal shall include an analysis of the projected traffic volumes, information on topography, drainage and extent of cuts and fills, along with construction plans and technical specifications (See Section 5). The road design plans and construction specifications shall include the minimum information listed in **APPENDIX H** and shall be prepared

under the direct supervision of a Wyoming Licensed professional engineer. The construction plans and technical specifications shall be reviewed and approved by the County Engineer in conjunction with the preliminary plat, or if no plat is required, prior to commencement of construction. The total cost of roads required to serve new development shall be borne by the developer. It shall be the developer's obligation to obtain all necessary rights-of-way, permits, agreements and easements prior to approval of construction plans.

2.6.1.1 Agricultural Development

New private roads developed and constructed to serve general agricultural interests are exempt from the design and construction standards outlined in *Section 5* unless a development or commercial business is planned that will result in an ADT of 60 or more vehicles per day using the proposed roadway. If traffic on a private agricultural roadway intermittently or occasionally exceeds 60 or more vehicles per day (i.e., during harvest season), the road does not have to meet the *Section 5* standards provided that the ADT during normal usage periods does not exceed an ADT of 60 or more vehicles per day. The burden is on the developer to show that the estimated ADT is representative of the planned and intended use.

If a private roadway to serve an agriculture based commercial business is planned with an ADT of 60 or more vehicles per day, see *Section 2.6.1.2*.

2.6.1.2 Commercial and Industrial Development

New private roads developed and constructed to serve small commercial or industrial developments that will result in an ADT less than 60 vehicles per day are exempt from the design and construction standards outlined in *Section 5*. Large commercial development expected to generate an ADT of 60 or more vehicles per day shall be designed and constructed to the standards outlined below. The burden is on the developer to show that the estimated ADT is representative of the planned and intended use.

- ADT of 0 to 59 vehicles per day – Exempt from *Section 5* Standards*
- ADT of 60 to 99 vehicles per day – See Low Volume Rural Road Standards
 - See *Section 3.3.2.7*;
- ADT of 100 to 199 vehicles per day – See Rural Road Standards
 - See *Section 3.3.2.6*;
- ADT of 200 to 399 vehicles per day – Local Access Road Standards
 - See *Section 3.3.2.5*;
- ADT of 400 to 999 vehicles per day – Minor Collector Standards
 - See *Section 3.3.2.4*;
- ADT of 1000 to 2,599 vehicles per day – Major Collector Standards
 - See *Section 3.3.2.3*;
- ADT of 2,600+ – Arterial Standards (State Highways & Interstates)
 - See *Sections 3.3.2.1 and 3.3.2.2*

**Even though the planned road is considered exempt from the design and construction standards provided in Section 5, all roads and driveways must be*

built and maintained to ensure access by emergency services vehicles (ambulance, fire trucks, etc.) to the extent possible and practical.

2.6.1.3 Recreational Development

New private or public roads developed and constructed to serve recreational areas or interests shall be constructed to the standards for a Rural Road functional classification (See *Section 3.3.2.6*) at a minimum. Special considerations may be warranted depending on the size and scope of the development to consider the potential for larger truck/trailer combinations or recreational vehicles. See *Section 2.6.1.2* for developments involving camping or RV Parks.

2.6.1.4 Residential Subdivisions (Minor)

New roads serving residential subdivisions classified as “Minor” in the DSR or exempt land divisions involving less than 6 residential lots are generally exempt from the road design and construction standards described herein (*Section 5*). However, if the size of the lots is such that the zoning will allow additional land division in the future, these *Standards will* apply (See *Section 2.6.1.5*) if the potential future division will result in more than 6 lots using the new road. Determinations as to whether subdivision road design and construction standards apply shall be based on the maximum projected design loading based on the potential for future development with zoning considered.

If the potential for future development results in design and construction of a road that will be substantially underutilized for the foreseeable future, the developer shall establish and provide a road improvement and maintenance agreement. All current and future lots shall be party to the road improvement and maintenance agreement. The developer may also elect to develop a RIMD to administer the road improvement and maintenance agreement. *Example:* A road to serve a two-lot subdivision that could potentially be split into a maximum of 10 lots would need to be designed and built initially to a *Local Access Road Standard* (*Section 5* design standards apply). Obviously, the cost to design and construct a *Local Access Road* to serve 2 lots is likely cost prohibitive. As such, in lieu of designing and constructing a road to serve 10 future lots today, the developer may establish a road improvement and maintenance agreement and/or a RIMD, the details of which would be recorded with the plat. When RIMD (See *Section 2.4*) are used, they would include a funding mechanism where current lot owners and future lot owners will contribute to a fund specifically established to improve and maintain the roadway at some time in the future.

2.6.1.5 Residential Subdivisions (Major)

New roads servicing residential subdivisions classified as “Major” in the DSR or exempt land divisions involving 6 or more residential lots must comply with the road design and construction standards described in *Section 5*. All new roads constructed to serve development shall be designed and constructed to consider the maximum design loading based on the potential for future development based on zoning.

Residential subdivision roads or exempt land division roads that are expected to receive traffic involving more than 200 vehicles per day (average daily traffic, ADT) shall be paved. In lieu of a traffic study, it is assumed that each residential lot will result in 10 vehicles per day. In other words, in absence of a traffic study, twenty (20) residential lots can be expected to produce an ADT of 200 vehicles per day, thus requiring paving if a new roadway is constructed or an existing road is to be upgraded.

2.6.1.6 Financial Responsibility

~~Payment of costs for road construction:~~

~~(a) Developer responsibility:~~ Any and all costs of new road construction in new developments are the responsibility of the developer. The developer is also responsible for the design, rights-of-way acquisition and construction of the new roads, whether public or private, according to the ~~RBS. see Standards.~~

~~(b) Payback agreements:~~ During the approval process for a proposed development, the developer may be required to construct a new road or to make improvements to an existing road which also benefits future developments. The ~~BOCC Board of County Commissioners~~ may establish a plan of compensation to the original developer whereby subsequent beneficiaries pay a fair share for the use of future benefiting developments. The ~~BOCC Board of County Commissioners~~ shall determine the equitable distribution of benefits and costs if applicable.

2.6.2 New County Roads and Bridges

The CRBS currently includes more than 600 miles of roadway that is maintained by the PCPWD. Roughly half of the County maintained roads are paved. As the County continues to grow and change with new residents and interests moving into the area, new County roads and bridges will be planned and developed as outlined in *Section 3.0*.

2.7 IMPROVEMENTS WITHIN DEDICATED RIGHT-OF-WAYS/EASEMENTS

The following information addresses the process and requirements for initiating improvements or maintenance within dedicated road easements and/or rights-of-way.

2.7.1 Improvements in Public Right of Ways (Not County Maintained)

Contractors, utility service providers, and individuals wishing to install, repair, or maintain utilities or infrastructure within a dedicated public road right-of-way that is not maintained by Park County must seek prior approval from the local Homeowner's Association (HOA) or Road Improvement and Maintenance District (RIMD) before initiation of work. If an HOA or a RIMD does not exist for the roadway, but it is dedicated for the use and benefit of the public, a Right-of-way permit application shall be submitted to the PCPWD for review and approval before initiation of work. See *Section 4.6* for ROW Permit requirements.

In issuing ROW permits for roadways dedicated to the use and benefit of the public, PCPWD's primary interest is in ensuring that safe public access and use of the public roadway is not hindered or restricted. ROW permit fees will be waived for ROW permit applications associated with public roads not maintained by Park County. The County Engineer may require a traffic control plan as part of the ROW permit process depending on the nature and extent of the work planned.

In issuing ROW permits for public roads not maintained by the County, PCPWD is authorizing work within the easement and the right for the improvements to exist within the easement. The responsibility and liability for reclaiming the area disturbed or damage to any existing infrastructure lies entirely with the permittee.

2.7.2 Improvements in Public Road Rights of Way Maintained by the County

Contractors, utility service providers, and individuals wishing to install, repair, or maintain utilities or infrastructure within a dedicated public road right-of-way that is maintained by Park County (including dedicated County Roads) must seek prior approval from the PCPWD before initiation of work. A *Right-of-way Permit Application* (See **APPENDIX I**) shall be submitted to the PCPWD for review and approval before initiation of work. See *Section 4.6* for ROW Permit requirements.

Any improvements and modifications to utilities and infrastructure located within the limits of the County maintained right-of-way shall be completed by licensed and bonded contractors. Any permanent improvements shall not restrict or interfere with PCPWD's ability to maintain the roadway.

2.7.3 Improvements in Private Road Easements

Any improvements and modifications to utilities and infrastructure located within the limits of any private road easement will require prior approval by the adjacent landowners that utilize the roadway. Any disturbances to the ground and/or associated vegetation shall be promptly reclaimed and brought back to the pre-construction conditions by the entity making the repairs/improvements to the extent possible and practical and to the satisfaction of the impacted landowners. Any damage to private property or existing infrastructure located within or adjacent to the easement shall be promptly repaired with the person or entity causing the damage fully responsible and liable for all costs including legal fees. Legal action involving work in a private road easement or damage to private property resulting from work in or adjacent to a private easement shall be resolved as a civil matter in District Court.

2.8 ROAD NAMING, NUMBERING, AND ADDRESSING

~~Rural road naming and addressing policy~~

1) ~~Statement of Purpose:~~

It is the purpose of the ~~RBS se Standards~~ to promote public health, safety and general welfare, and to provide a coordinated and uniform road naming and addressing system within Park County.:

~~a) To protect human life and health.~~

~~b) To optimize the response for emergency services, such as fire, ambulance, rescue and relief efforts undertaken at the expense of the general public.~~

- ~~e) To apply to all addressing within the jurisdiction of Park County, including but not limited to, state, county, public and private roads and easements; and~~

The Park County road naming, numbering, and addressing system has been established to provide a reliable system for locating and identifying roads and facilities throughout the County. ~~Road Numbering:~~ All County Roads, public roads (not maintained by Park County), and private roads located within unincorporated Park County are assigned a County road name or number which is used for identification purposes to help ~~reduce emergency services response time~~ ~~speed emergency access by fire and ambulance~~, and to assist in locating utilities. In addition to the road naming and numbering system, facilities are assigned a numerical address for identification purposes as described herein.

~~Designation of addressing responsibility. The Public Works Department, PCPWD under the direction of the County Engineer, shall be responsible for the implementation and enforcement of the RBS se Standards. The duties shall include, but not be limited to, implementation and enforcement of these Standards. Road name signs. It is neither the desire nor the intent of the County, through its road naming and addressing policy, to differentiate between public and private roads, nor to imply a right of ingress or egress. Generally, the following sign conventions will be followed:~~

2.8.1 Road Naming and Addressing Committee

The Park County Road Naming and Addressing Committee (RNAC) was formed several years ago for the purpose of administering the road naming and addressing policy described herein. The committee is tasked with reviewing and approving road names as well as changes to road names and addresses. The committee shall consist of representatives from the following offices/agencies at a minimum:

~~Address Review Committee. An Address Review Committee will be comprised, at a minimum, of members from the following agencies:~~

~~Park County Engineer's Office Park County Sheriff's Office~~

~~City of Cody City of Powell~~

~~Town of Meeteetse~~

~~Fire Districts, including: Clark, Cody, Powell and Meeteetse.~~

- a. Park County Engineer's Office
- b. Park County Planning & Zoning Office
- c. Park County Sheriff's Office
- d. City of Cody
- e. City of Powell
- f. Town of Meeteetse
- g. Fire Districts (Clark, Cody, Powell, and Meeteetse)

2.8.2 Road Naming and Numbering System for Designated County Roads

County Roads that have been established in accordance with *Wyoming State Statutes* are generally named based on a letter and number-based grid system (See **APPENDIX J**). These dedicated County Roads are typically marked with a shield shaped sign with gold lettering and a blue background (*MUTCD M1-6*). **FIGURE 4** shows a typical designated County Road sign used in Park County and throughout the State of Wyoming. County Road signs shall be installed and maintained in accordance with the *Manual of Uniform Traffic Control Devices (latest edition)*. County roads with established

names or numbers do not necessarily mean the roads are established as County roads or are maintained by the County.

~~(i) County Roads Formally established County roads will be marked with a County Route Marker, MUTCD designation M1-6. County road names or numbers do not necessarily mean the roads are established as County roads or are maintained by the County.~~

FIGURE 4 – TYPICAL COUNTY ROAD SIGN



2.8.3 Road Naming for Non-County Roads

~~Address location on a road requiring a road name.~~ Non-County Roads including unimproved roads or driveways ~~A road~~ shall be named in the following cases:

- (i) *New address is the third address located on a particular access.*
- (ii) *Increased development is anticipated to be located on a particular access that may result in three (3) or more addresses off ~~of~~ a known road.*
- (iii) *Any road that in the opinion of ~~any member of the~~ RNAC Address Review Committee should have a new road name.*

~~And (iv)~~ Once ~~it is determined~~ a new road name is required, the following procedures shall be followed:

- A. *If a landowner is initiating the need for a new or revised road name, an application form (See **APPENDIX K** and **Section 4.7**) shall be submitted to the PCPWD with the appropriate fee.*
- B. *PCPWD ~~The County~~ shall review the property ownership on adjacent properties to the road in question.*

- C. The property owners ~~shall be~~ **are** contacted in writing and given the opportunity to nominate names for consideration.
- D. Once submitted, the ~~RNAC Road Naming and Address Review Committee shall~~ reviews the nominations.
- a. *Any conflicts or perceived conflicts between proposed road names and existing road names, which are already established in Park County, will not be allowed. All names not disqualified through the review process will be submitted to the landowners, so they have an opportunity ~~chance~~ to vote.*
- b. *If the landowners cannot agree on a road name, the County Engineer will assign a road name.*
- E. Once a road name is accepted, all the affected landowners will be notified outlining the new road name and corresponding new addresses. A copy of the notification **shall be sent to:**
- h. Park County, Wyoming**
 - i. Board of County Commissioners
 - ii. Park County Engineer
 - iii. Park County Road & Bridge Foremen
 - iv. Park County Planner
 - v. Park County Sheriff
 - vi. Park County Assessor's Office
 - vii. Park County Clerk's Office
 - viii. Park County Homeland Security
 - ix. Park County Fire Chief
 - i. City of Cody**
 - i. Chief of Police
 - ii. Planning & Zoning Director
 - iii. Fire Chief
 - j. City of Powell**
 - i. Chief of Police
 - ii. Planning & Zoning Director
 - iii. Fire Chief
 - k. Town of Meeteetse**
 - l. Emergency Dispatchers (911)**
 - m. United States Postal Service.**
 - n. Federal Express**
 - o. UPS**
 - p. Park County School System**
 - i. Park County School District 1
 - ii. Park County School District 6
 - q. Cody Regional Health**
 - r. Utility Service Providers**
 - i. Beartooth Electric
 - ii. Black Hills Energy
 - iii. Garland Power & Light
 - iv. MDU
 - v. Northwest Rural Water District
 - vi. PacifiCorp
 - vii. TCT
- ~~a. is also sent to emergency services, utility companies, county offices, postal service, etc.; and~~

- F. In conjunction with the appropriate notifications described above, ~~the~~ **PCPWD County sign technician is notified for installation of** will install the new road sign and associated address markers if applicable. Road name signs and installation shall be in accordance with MUTCD requirements.

Non-county roads (ie, ~~(ii) Other Named Roads Other named roads~~, public, private, subdivision, etc., will be marked with a **Road Street** Name Sign, MUTCD designation D3 (See **FIGURE 5**). Requests for signs with additional wording such as “Private Road”, “Private Drive” or other private signs will be denied by the County. This does not preclude the property owner, at **owner’s his** expense, from installing these signs on private property. Such signs shall be installed outside of any County or public rights-of-way and may not be attached to the road name sign or signpost.

~~(h) Interpretation. In the interpretation of these Standards, all provisions shall be:~~

~~(i) Considered as minimum requirements;~~

~~(ii) Liberally construed in favor of the governing body; and~~

~~(iii) Deemed neither to limit nor repeal any other powers granted under state statute.~~

~~(i) Warning and disclaimer of liability. The degree of protection required by these Standards is considered reasonable for regulatory purposes. These Standards shall not create liability on the part of Park County, any officer or employee thereof for any damages that result from reliance on these Standards or any administrative decision lawfully made thereunder.~~

FIGURE 5 - TYPICAL ROAD NAME SIGN FOR NON-COUNTY ROADS

See MUTCD Type D3 Signs



2.8.4 Addressing

Park County uses a mileage-based system for addressing, which means addresses are assigned from a known point, usually the beginning of the road. For example, an address marker measurement from the start of the road measuring 0.15 miles would be assigned an address marker of "15" if located on the left side of the road (i.e., odd numbered addresses) and "16" if located on the right-hand side of the road (i.e., even numbered addresses).

All residents on a single parcel of property shall submit an application to the PCPWD for an address with the appropriate fee and shall have a County assigned address. Other routinely occupied facilities, such as barns, trailers, commercial structures and other facilities, may be required to be addressed. ~~Application (a) Application for an address.~~ Application for an address shall be made on forms furnished by PCPWD Public Works (See APPENDIX L).

General Provisions:

~~A formal written procedure follows, as outlined in Sections (4) (Application) and (5) (Process) below.~~

~~Land to which these Standards apply. These Standards shall apply to all areas within the jurisdiction of Park County.~~

~~Basis for establishing. To provide for a uniform coordinated rural addressing system for Park County.~~

Process

~~Application review.~~ Upon receipt of a *Request for an Address* form, PCPWD Public Works will review for the following:

- (i) Legal access (proof to be provided by Applicant);
~~(A) Proof of legal access may be required to be furnished by the applicant.~~
- (ii) Property ownership
- ~~(iii) If address location is on a currently named road proceed to (5)(b) below; and~~
- ~~(v) If a private road name will be required proceed to (5)(c) below.~~

~~New address location on a currently named road.~~

Once the application is complete and all appropriate fees are paid, PCPWD Public Works shall provide the applicant with an approved marker and instruct the applicant to place the marker at the location of the proposed driveway. ~~;~~ and PCPWD shall then

~~(ii) Public Works shall~~ determine an address for the location based on the location of applicant placed marker and shall place a permanent address marker as shown in **FIGURE 6.**

~~Modification of an existing address.~~ If PCPWD Public Works or a member of the RNAC Address Review Committee determines a modification of an existing address is needed, PCPWD Public Works will initiate a change following the procedures described herein. When an unauthorized address is discovered, PCPWD Public Works shall notify the property owner of the violation and initiate the process in accordance with these Standards to correct the address. ~~outlined in Sections (b) and (c) above.~~

~~Assignment of road name(s) and address(es).~~ Once the Applicant and Public Works have complied with the requirements of these Standards, Public Works will assign an address, and if required, a road name.

~~Notification of address. Once a road name and an address are determined, Public Works will notify the applicant and appropriate agencies.~~
~~Correction of self-assigned address.~~
~~until all costs have been received.~~

FIGURE 6 - TYPICAL RURAL ADDRESS MARKER

<Insert Photo>

2.9 LICENSED PROFESSIONALS

PCPWD may require 3rd party professional services including architecture, engineering, and land surveying to assist with various projects involving County infrastructure. Procurement for professional services shall be in accordance with the *Park County Procurement Policy* and shall follow *Wyoming State Statutes* for qualifications-based selection when applicable.

Preparation of construction plans and technical specifications for new road and bridge construction or improvements to existing road and bridge infrastructure applicable to the design standards in *Section 5* shall be prepared under the direction of a Wyoming licensed professional engineer. The engineer of record shall certify that construction was completed in accordance with the approved construction plans/technical specifications and these *RBS*.

3.0 ROAD & BRIDGE SYSTEM PLANNING

In conjunction with the development of the *Park County Transportation Safety Plan (TSP)*, short-term (5-10 years) and long range (20+ years) plans are to be developed by PCPWD, regularly reviewed with the BOCC, and updated as necessary. Having short- and long-range plans in place will help guide fiscally responsible planning for major transportation infrastructure projects to ensure that adequate funding is in place when needed.

The BOCC, County Engineer, and R&B Foremen will consider input from the public in developing short-term and long-range plans to ensure that public needs and trends are clearly communicated, understood, and considered when forecasting future road needs.

3.1 COUNTY TRANSPORTATION SAFETY PLAN

Park County is a large and diverse county located adjacent to and east of Yellowstone National Park in Northwest, Wyoming. With a population of roughly 29,624 people and 6,967 square miles of land mass (2020 Census), the County's demographics are bolstered largely by tourism, agriculture, and oil & gas related interests. The County encompasses millions of acres of public lands including national parks, national forests, wilderness areas, and rangeland all parts of the greater Yellowstone ecosystem. Much of the County is characterized by sloping terrain, scenic vistas, co-existing wildlife ecosystems, streams/rivers, and remote areas that can prove challenging to access during periods of inclement weather. The eastern part of the County is largely agriculturally based, which offers challenges pertaining to maintaining reliable "farm to market" agricultural routes. It is estimated that roughly 23% of land within the jurisdictional boundaries of Park County is privately owned (www.parkcounty-wy.gov, 2026).

The County maintained road and bridge system involves more than 630 miles of roadway of which roughly half are paved. The County also is responsible for more than 70 bridges, 82 box-culverts, 2,063 culverts, 175 cattleguards, and more than 6,200 signs (*Park County Public Works Department*, 2026).

Park County, Wyoming recently adopted an updated County *Land Use Plan (LUP)* in March 2024. Development of the LUP engaged Park County communities, citizens, and interests while developing guidelines and planning tools to best manage the future growth and development of the County for the greater public good. After the adoption of the LUP, Park County is currently in the process of updating *County Land Development Standards & Regulations (DSR)*. This public process administered by the Park County Planning and Zoning Department has involved multiple public meetings, working groups, and other public outreach efforts. A component of the update to the DSR involves updating the *RBS*.

The *RBS* provides a guideline for uniformity and consistency of the CRBS as well as transportation infrastructure necessary to serve development. Park County's intention is for the LUP and the TSP to serve as guiding principles for the development of the DSR and the *RBS* moving forward. In other words, the RBS are intended to reflect the will of the people with an eye toward future needs and priorities. These priorities and needs are best reflected in the LUP and the TSP.

A fundamental pillar of the *Safe Roads and Streets for All (SS4A)* program is an emphasis on public transportation safety. Specifically, the goal is to evaluate existing infrastructure while developing a transportation plan designed to decrease fatalities and serious injuries related to roadway accidents. The purpose of the TSP is to evaluate existing transportation infrastructure as well as growth/development trends to plan for current and future transportation needs of Park County. The TSP has been funded through the SS4A program (*Federal Award Number 693JJ32640293*) with local matching through the State of Wyoming.

3.2 INTERAGENCY RESPONSIBILITY AND COORDINATION

Although the TSP is focused on County maintained road and bridge infrastructure, there shall be a concentrated effort to integrate the County transportation network with other

transportation systems and modes administered/maintained by Federal, State, Local (Cities and Towns), and private road infrastructure interests (i.e., residential, agriculture, commercial, private, etc.) within Park County and surrounding areas. Ultimately, and ideally, the TSP is to provide a framework and planning tool for forecasting and prioritizing future transportation infrastructure needs.

In planning and developing the CRBS, PCPWD shall coordinate efforts with other local governmental entities such as the City of Cody, City of Powell, and Town of Meeteetse to ensure integration and safe transitions of the CRBS when entering and leaving incorporated areas. In addition, PCPWD shall coordinate with various State (WyDOT) and federal agencies (FHWA, BLM, BOR, etc.) to the extent possible and practical for the greater public good. If possible, this coordination may include planning, funding opportunities, equipment procurement, materials procurement, and shared maintenance.

Each year the BOCC allocates revenue through the budget process for administration by PCPWD for the CRBS. Historically, budget levels have been sufficient to address basic maintenance of the CRBS. Typically, revenue allocated in the local budget (Budget Category 3100) is insufficient to cover the costs of major road or bridge related capital improvement or rehabilitation projects. For this reason, the PCPWD actively and aggressively seeks funding from outside sources at the State and Federal Level to help cover necessary capital improvement projects. The following sections summarize some of the Local, State and Federal partnerships as well as funding opportunities.

3.2.1 Local Coordination

PCPWD works regularly with the City of Cody, City of Powell, and Town of Meeteetse to integrate road & bridge related planning efforts. When possible and practical, PCPWD works with the street departments at the municipal level to coordinate overlapping maintenance activities and occasionally road/street materials procurement in situations where volume-based purchases will result in lower costs to the public.

PCPWD currently maintains a shared maintenance agreement with the City of Powell to address maintenance of public roads at the outskirts of the City limits. These roads include County Road 10 (Tower Boulevard), County Lane 8, County Road 8 (Panther Boulevard) and Monroe Street. A copy of the shared maintenance agreement is included in **APPENDIX M** for reference purposes.

Although not obligated to do so, PCPWD also assists the Town of Meeteetse with occasional street chip seal projects as budgets and resources allow. Specifically, PCPWD has historically donated labor and equipment towards this effort with the Town of Meeteetse reimbursing the County for the materials used (chips and chip oil) at the County's cost.

3.2.2 State Coordination and Funding Requirements

PCPWD shall coordinate with WyDOT regarding situations where County Roads intersect with State Highways. PCPWD shall secure a ROW Permit from WyDOT when/if improvements are necessary within the State Highway ROW.

3.2.2.1 WyDOT Administered Funding for County Road & Bridge Projects

WyDOT administers multiple State and Federal funding programs which are available to counties for the purpose of improving and maintaining local road and bridge infrastructure. These programs include but are not limited to:

- Transportation Alternatives Program (TAP)
- Bridges- Off System Program (BROS)
- Congestion-Mitigation and Air Quality Program (CPAQ)
- High Risk Rural Road Program (HRRP)

3.2.2.2 County Road Construction & Maintenance Fund

The Wyoming Legislature allocates a percentage of fuel tax and severance tax revenues back to the individual counties through the County Road Construction & Maintenance Fund (CRCMF) program. The fuel tax revenues are distributed monthly, and the severance tax revenues are distributed quarterly. The funds are maintained by the County Treasurer with oversight by the County Engineer in a separate interest-bearing account.

Funds are allowed to accumulate within the account for road and bridge related capital improvement projects. The guidelines for the use and tracking of the program are outlined in *Wyoming State Statutes §24-2-110*. A select committee administers the CRCMF program and has developed a CRCMF manual for use and reference in the proper utilization of the funds consistent with the statutory requirements. Annual reports are submitted by each County to the CRCMF committee for review and approval to ensure that all requirements are met. A summary of historical CRCMF account tracking as it relates to the CRBS is included in **APPENDIX N** for reference.

3.2.3 Federal Government Coordination and Funding Requirements

Certain roads crossing federal lands are managed and maintained by various federal agencies including the Bureau of Land Management (BLM), Bureau of Reclamation (BOR), United States Forest Service (USFS) and Federal Highway Administration (FHWA).

3.2.3.1 Forest Access Shared Maintenance

Park County has historically entered into a shared maintenance agreement with the United States Department of Agriculture Forest Service for certain County Roads that access the national forest system. Currently, these roads include County Road YXD (Hunter Creek Road), County Road 7GQ (Sunlight Basin), and County Road 4DT (Wood River to Kirwin). The term of this agreement is typically for a 4-year period with an opportunity for addenda on an annual basis to consider special projects such as cattleguard or culvert replacements. The base agreement generally consists of basic periodic road grading which is often seasonal in nature (i.e., summer months).

The County's obligation under the agreement is to provide labor and equipment to routinely grade the roadway. The Forest Service typically reimburses the County \$15,000 per year for the County's efforts in keeping the forest access roads maintained at a level that ensures seasonal access to forested areas by the public. A copy of the latest agreement between Park County and Forest Service is included in **APPENDIX O** for reference.

3.2.3.2 Access Permits and Easements for Crossing Federal Lands

Designated County Roads that cross federal land (typically BOR, USFS, or BLM managed) require an authorization from the appropriate federal agency in the form of an access permit and easement. As these permits/easements expire they must be renewed to allow continued use and maintenance by the County for the benefit of the public.

3.2.3.3 Federal Funding for County Road & Bridge Projects

Some funding opportunities for local road and bridge infrastructure are administered through the Federal Highway Administration (FHWA) or other federal agencies. Most of these programs are very competitive and require local matching (typically 15 to 20% depending on the program). These funding programs include but are not limited to:

- Federal Lands Access Program (FLAP)
- Rural Transportation Alternatives Program (RTAP)
- Bridge Investment Program (BIP)
- Safe Roads and Streets for All (SS4A)
- Forest Reserve Secure Rural Schools (SRS)

Some federally funded programs span several years and require planning level studies to be completed before an application for an implementation grant can be submitted and considered. Federal programs are regularly discontinued or added based on funding limitations and priorities enacted by Congress each year.

Capital improvement or major rehabilitation projects for County managed roads or bridges that utilize federal funds require compliance with federal regulations and requirements. These requirements include special considerations for procurement of materials/labor, environmental considerations (NEPA), and reporting to ensure that federal dollars are spent consistently with the intent of the specific federal funding program.

3.3 ROAD & BRIDGE TYPES AND CLASSIFICATIONS

Our nation's roadway system is a vast network that connects places and people across jurisdictional boundaries. Planners and engineers have developed elements of this network with travel objectives in mind. These objectives range from serving long-distance passenger and freight needs to serving local commercial, agricultural, residential, and recreational travel. The road types and functional classification of roadways define the role each element of the roadway network plays in serving these travel needs. Ultimately, the coordinated and systematic maintenance of an accurate

roadway functional classification system ensures that funds are allocated where they are most needed, to enable people and goods to move fluidly and safely through the transportation system.

PCPWD has established road and bridge types and classifications for the purpose of planning and assigning minimum design, construction, and maintenance standards based on the type of use and volume of traffic (ADT). The road and bridge types and classifications outlined herein are adapted from FHWA standards for rural road systems.

3.3.1 Road Types

Types of Roads: Roads contained in the County-wide circulation system are classified based on ~~the~~ functional use and traffic volumes. Ownership of the road rights-of-way and who has responsibility for maintenance is determined in accordance with state statutes and is controlled by the BOCC. ~~Board of County Commissioners.~~ Establishment of the road rights-of-way does not guarantee the road is eligible for or maintained by the County.

3.3.1.1 County Roads

Designated County Roads are roads which the BOCC have accepted the road rights-of-way and ~~has~~ assumed responsibility for ~~the~~ road maintenance ~~-County owned and maintained roads: Under this category, Park County, through the Board of County Commissioners,~~ in accordance with State Statutes. ~~has~~

~~County road rights-of-way not maintained by County:~~ In some instances, the County has accepted road rights-of-way but does not routinely maintain the roadway. These rights-of-way are available for the use of the public or reserved for the future needs of the County.

~~County owned roads, but maintained by others:~~ In certain cases, private property owners using County owned roads for access desire a higher level of service than the County provides. In such cases, the County and property owners may reach an agreement ~~with the County to assign~~ ~~assigning~~ maintenance responsibilities to the property owners.

In other cases, ~~established~~ County ~~owned~~ roads may be maintained by an adjacent city or town. Such arrangements ~~may be~~ ~~have been~~ made when it makes more sense for ~~the~~ a city or town to maintain a portion of a County Road because of its location and its connection to town streets in exchange for the County maintaining sections of outlying town roads.

The process for creating, altering, or vacating County roadways is established in *Wyoming State Statutes* and is described in **FIGURE 3** and *Section 2.3* of these *Standards*.

3.3.1.2 Private Roads

~~Privately owned and maintained roads:~~ Private Roads are ~~This category includes~~ ~~all~~ roads where the adjacent property owners retain ownership of the road rights-

of-way either through a deed, easement or covenants. ~~Adjacent landowners and they~~ generally hold responsibility for ~~its~~ maintenance of and improvements to private roads. Private roads generally are associated with residential subdivisions or land divisions (large exempt parcels, etc.). Private roads can also be created or established to serve industrial, commercial, or agricultural interests. ~~Included are subdivision roads, whether private or available for public travel.~~

Refer to *Section 2.1.2* for applicability of these *Standards* as it relates to development. See *Section 2.5.1* for requirements when upgrading non-County roads. See *Section 2.5.2* for requirements when upgrades to existing subdivision roads are initiated by the adjacent landowners. See *Section 2.6.2* for requirements for new private roads serving development.

3.3.1.3 Public Roads Not Maintained by County (Not Including County Roads)

In some cases, developers may establish a road and will dedicate the use of the road to the public by easement and/or subdivision plat. Even though a road is dedicated for use by the public, this does not establish the road as a County Road, nor does it obligate the County regarding ongoing improvements and maintenance of the roadway. Typically, public roads not maintained by the County are improved and/or maintained through shared maintenance agreements with the adjacent landowners. These agreements are typically administered through an HOA or RIMD (See *Section 2.4*).

3.3.1.4 Public Roads Maintained by County (Not Including County Roads)

Prior to 1990, some subdivision roads in Park County were dedicated for public use and improved to County Road Standards in place at the time. These roads were subsequently accepted by the BOCC for limited maintenance by the County through signed agreements. These roads include:

- 26th Street
- Absaroka Drive
- Appaloosa Lane (north end)
- Arabian Lane
- Arrow Avenue
- Bannock Road
- Beacon Drive
- Beacon Hill Estates
- Black Street
- Cactus Drive
- Canyon Lake Drive
- Carbon Street
- Cody Landfill Road
- Country Place
- Davis Road
- Del Rio Lane
- Del Rio Road
- Emery Road
- Fremont Street
- Gerber Lane
- Hastings Horseshoe
- Heather Road
- Hitching Post Drive
- Jonathan Road
- Lewis Avenue

- Maddox Lane
- Marquette Drive
- McCullough Road
- McIntosh Road
- Milky Way Drive
- Moonlight Drive
- Morgan Lane
- Mountain Drive
- Mountain View Drive
- Musser Road
- Mustang Lane
- Natrona Street
- Nez Perce Drive
- North 41st Street
- North 44th Street
- North Ridge Drive
- North Star Drive
- Olive Road
- Panorama Lane
- Pat Ohara Mountain Drive
- Pinto Road
- Quarter Horse Lane
- Railway Street
- Range View Drive
- Riverside Avenue
- Ross Road
- Ryan Road
- Sage Drive
- Schneider Road
- Sheridan Avenue
- Sunburst Drive
- Temple Creek Avenue
- Vali Road
- Wild Horse Road

Limited maintenance by the County consists of road grading (twice per year) for gravel roads and chip sealing, patching, and crack sealing for paved roads. It also includes snow plowing and traffic sign replacement/maintenance. Limited maintenance does not include adding additional gravel road base material or pit-run to gravel roads. It also does not include drainage repairs/improvements nor improvements/repairs/maintenance to cattleguards.

Users of the limited maintenance public roads or adjacent landowners may petition the PCPWD to make improvements to the road including the road surface, associated drainage conveyances or cattleguards. These improvements require a ROW permit from the PCPWD (See *Section 4.6*) and shall be completed at the expense of the adjacent landowners. PCPWD will not participate (financially or otherwise) in the improvement of the limited maintenance of public roads without prior approval by the BOCC.

3.3.1.5 Other Public Roads (City, State, & Federal)

Other public roads throughout Park County not mentioned in *Sections 3.3.1.3* or *3.3.1.4* are maintained by other governmental entities. For example, the City of Cody, City of Powell, and Town of Meeteetse maintain the public streets within their incorporated limits.

There are currently no interstate highways within Park County. WYDOT maintains and improves various State Highways throughout Park County including:

- State Highway 114 (Deaver)
- State Highway 120 (North – Clark)
- State Highway 120 (South – Meeteetse)
- State Highway 14-16-20 (East- Greybull)
- State Highway 14-16-20 (West – North Fork)
- State Highway 14A (Powell)
- State Highway 212 (Beartooth)
- State Highway 290 (Pitchfork)
- State Highway 291 (South Fork)
- State Highway 294 (Badger Basin)
- State Highway 295 (Elk Basin)
- State Highway 296 (Chief Joseph)
- State Highway 310 (Frannie)

Roads crossing federal lands that are not designated State Highways or County Roads are usually BLM or USFS roads depending on whether they cross BLM lands or the national forest. Occasionally, the federal government will grant easements for private, commercial, or industrial use. For example, there are numerous gravel roads on BLM lands in Oregon Basin and Elk Basin that are maintained by various oil and gas companies and or mining interests. Conditions and requirements for the type of use and level of maintenance are typically spelled out in the agreement/easement with the applicable federal agency.

3.3.2 Functional Classifications

Road classification: Applicable roads within the CRBS ~~County roads~~ are classified according to functional classifications based largely on FHWA guidelines for rural roads. At the County level, functional classifications shall be established by the County Engineer. ~~and the Road and Bridge Foremen.~~ The County Engineer will determine which classification applies to any given road. Criteria used to design roadways are based on their functional classification and traffic volume.

Current road functional classifications include Principal Arterials, Minor Arterials, Major Collectors, Minor Collectors, Local Access Roads, Rural Roads, Low Volume Rural Roads, and Primitive/Unimproved Roads. Roads classified as Principal Arterial or Minor Arterial Roads are typically associated with State Highways and Interstates. The functional classifications that are applicable to these design standards and the roads that comprise the CRBS will generally range from Low Volume Rural Roads to Major Collectors.

These *Standards* establish maximum traffic volumes for certain classifications of roadways as summarized ~~stated~~ in **TABLE 1. Table 5-1.** If a proposed development will cause these maximum limits to be exceeded on the adjacent roads, the developer shall, ~~subject to “Upgrading existing roads” (h)(1)(a)(ii),~~ be responsible for the cost of improving the affected roads to a classification where the maximum limit is not exceeded as outlined in *Section 2.5* of these *Standards*. In calculating whether the maximum limits will be exceeded, the projected cumulative traffic volume based on surrounding land

uses and approved zoning at ultimate development shall be used, not just counts of existing traffic levels.

~~Refer to Section 7. Administrative Relief from Design and Construction Standards.~~

TABLE 1- DESIGN CAPACITY BY FUNCTIONAL CLASSIFICATION

CLASSIFICATION	ADT
Principal Arterial	>5000
Primary (Arterial)	>700
Minor Arterial	2,600 to 4,999
Collector (Major)	1,000 to 2,599
Collector (Minor)	400 to 999
Secondary (Collector)	100–700
Residential	≥99
Local Access	<99 200 to 399
Rural Road	100 to 199
Low Volume Rural Road	60 to 99
Primitive/Unimproved*	0 to 59

*Design and Construction Standards (Section 5.0) do not apply to Primitive/Unimproved Roads

~~Design Standards: Any and all publicly traveled roads, whether public or private, requiring upgrading or improvement shall be built in accordance~~

~~For planning purposes, Park County uses the following functional classifications:~~

3.3.2.1 Principal Arterials

Arterials prioritize mobility, carrying high traffic volumes over longer distances. These serve major centers of metropolitan activity, carry the highest traffic volumes, and serve the longest trips. Principal arterials are broken down into the following subcategories:

- FFC 1 (Interstate): The highest classification, designed for maximum mobility with full-access control and high-speed limits.
- FFC 2 (Other Freeways and Expressways): Divided highways with controlled access that are not part of the interstate system.
- FFC 3 (Other Principal Arterials): Serve major regional corridors and intra-urban travel; They offer high mobility but may still provide direct access to adjacent land uses.

Park County does not manage or maintain any roads that fit within the Principal Arterial Classification. In Wyoming and Park County, Principal Arterial Roads are managed and maintained by WyDOT. Examples in Park County of Principal Arterial Roads include:

- State Highway 114 (Deaver)
- State Highway 120 (South – Meeteetse)
- State Highway 14-16-20 (East- Greybull)
- State Highway 14-16-20 (West – North Fork)
- State Highway 14A (Powell)

- State Highway 310 (Frannie)

3.3.2.2 Minor Arterials

~~Primary roads (arterial): They provide a means of intra-county travel. Primary roads should provide for relatively high overall travel speeds with minimum interference to through movements. Typically the average daily traffic (ADT) is greater than seven hundred (700) vehicles per day (VPD).~~

Minor arterials connect and augment the principal arterial system. They carry medium-length trips, connect smaller cities and rural centers, and distribute traffic to smaller collector roads while allowing for more direct land access. Examples of Minor Arterial Roads in Park County include:

- State Highway 14A (Byron)
- State Highway 295 (Powell North of Highway 14A and South of Lane 8)
- State Highway 112 (Beartooth)
- State Highway 120 North (Clark)
- State Highway 291 (South Fork from 14-16-20 to Chugwater Drive)
- State Highway 296 (Beartooth)

Like Major Arterials, Minor Arterials in Park County are generally managed and maintained by WyDOT.

3.3.2.3 Collectors (Major)

Collector Roads are mid-level roadways designed to balance traffic mobility with property access. These roads serve as connectors in the highway network, "collecting" local traffic from residential or commercial streets and funneling it onto larger, high-speed arterial roads. Roads are classified by the level of mobility they provide versus the access they allow to adjacent properties. Collector roads sit squarely between the highest and lowest tiers of the road network. They allow for moderate traffic speeds while still providing direct access to neighborhoods and businesses.

Major Collector Roads generally carry higher traffic volumes, serve larger geographic areas, and link localized neighborhoods to regional commercial, industrial, or arterial systems. For the CRBS, major collectors will generally be designed and constructed to manage traffic volumes between 1,000 and 2,599 vehicles per day (ADT). Major Collector Roads can be either State (WyDOT) Highways or County Roads. Examples of roads in Park County that are Major Collectors include:

- Beacon Hill Road
- County Lane 8 (North of Powell between Road 10 and Road 8)
- County Road 2AB
- County Road 6WXE (Skyline to Red Lakes)
- County Road 8 (Panther Blvd)
- County Road 10 (Tower Blvd)
- State Highway 290 (Pitchfork)

- State Highway 291 (South Fork-Chugwater Dr to End of State Maint.)
- State Highway 294 (Badger Basin)
- State Highway 295 (North and South of Powell)
- State Highway 296 (Chief Joseph)
- Monroe Street (South of Powell)

~~Secondary roads (collector): Secondary roads serve as collectors of traffic from residential and recreational areas to the primary road system. Secondary roads provide a link between local roads and arterial, and allow for the movement of through traffic in neighborhoods. Secondary roads should be designed so they do not disrupt the activities and land uses they serve. Secondary roads should provide for relatively high overall travel speeds. In addition, access to secondary roads should be designed so as to minimize interruption of traffic flows. Typically the ADT is greater than one hundred (100) VPD but less than seven hundred (700) VPD.~~

3.3.2.4 Collectors (Minor)

Minor Collector Roads are more localized and collect traffic directly from neighborhood streets and distribute it to major collectors or arterials. Examples of Minor Collectors include:

- County Lane 9 (West of Road 4)
- County Road 1AB and 8VC (Clark Canyon Road)
- County Road 1XG (Elk Basin to Frannie)
- County Road 3EX (Sage Creek)
- County Road 3FK (Oregon Basin)
- County Road 3EM (Oregon Basin)
- County Road 3JU (from Powell Highway to CR 3FK in Oregon Basin)
- County Road 3LE (Lower Greybull)
- County Road 6NS (Lower Southfork)
- County Road 6WX (End of State Maintenance to end of County Road)
- County Road 6UU (Diamond Basin)
- County Road 7GQ (Sunlight Basin)

Minor Collector Roads are generally County Roads and are associated with traffic volumes ranging from 400 to 999 vehicles per day (ADT).

3.3.2.5 Local Access Roads

Local access roads (often referred to as “local roads”) are the lowest-tier classification in the Federal Highway Administration (FHA) hierarchy. Local access roads are primarily used to provide direct access to abutting properties rather than accommodating them through traffic. Local access roads generally prioritize property access over mobility and are commonly associated with lower speed limits and shorter trip distances. For the CRBS, *Local Access Roads* are generally associated with traffic volumes ranging from 200 to 399 vehicles per day (ADT).

~~Local access roads: Generally, local access roads provide access to private property such as farm, ranch or sparsely populated residential areas. Typically, the ADT is less than ninety-nine (99) VPD.~~

3.3.2.6 Rural Roads

For CRBS planning and uniformity of design standards, Park County has established additional functional classifications below the hierarchy of the FHA categories. Rural Roads can be associated with larger residential subdivisions and can provide access to multiple smaller subdivisions. These roads are generally not paved and generally are not through roads. For the CRBS and these *RBS*, Rural Roads typically have traffic volumes ranging from 100 to 199 vehicles per day.

3.3.2.7 Low Volume Rural Roads

Low Volume Rural Roads typically serve single mid-size to large subdivisions. They are typically dead-end roads that are privately improved and maintained by the adjacent landowners. These roads are generally not paved. For the CRBS and these *Standards*, Low Volume Rural Roads typically have traffic volumes ranging from 60 to 99 vehicles per day. Low Volume Rural Roads are the lowest functional classification for which the County design and construction standards outlined in *Section 5* apply.

~~Recreational roads: Recreational roads provide access to few, if any, year round residents.~~

3.3.2.8 Primitive/Unimproved Roads

Roads in the Primitive/Unimproved functional classification as it relates to these *Standards* are typically associated with low volume private roads that serve smaller residential subdivisions or small commercial/industrial developments that generate minimal traffic. These roads can either be improved (i.e., gravel road) or unimproved. Roads in this classification are improved to the satisfaction of the landowners that use the road as the design and construction standards in *Section 5* do not apply to this functional classification. The roads should be maintained in a condition that ensures access by emergency vehicles including during periods of inclement weather. Primitive/unimproved roads can also include agricultural, field access roads, or recreational roads that generate minimal traffic. Primitive/Unimproved Roads for the purpose of these *RBS* are associated with traffic volumes between 0 and 59 vehicles per day (ADT).

3.3.2.9 One Way Roads

One-way roads are not used in rural Park County. However, in certain residential, industrial, or commercial developments, one-way traffic may be desirable to achieve site specific access or operational goals. For example, one-way roads may be desirable within RV Parks or camping areas where space for backing larger vehicles or truck/trailer combinations is limited. One-way roads will be considered on a case-by-case basis as part of the Site Plan review/approval

process for situations where traffic flow can be improved with one-way traffic infrastructure.

One-way roads will not be allowed in the following situations:

- i. *Property owners at the far end of a one-way loop road tend to take short cuts and drive the wrong way to reach their properties, thus increasing the chances ~~for~~ of accidents;*
- ii. *Emergency vehicles must, in certain cases, take a more circuitous route to reach their destinations;*
- iii. *One-way roads can cause confusion for people not familiar with the area; and*
- iv. *In winter, snow plowing often reduces the driving surfaces of roads because snow accumulates along the edges. On one-way roads, this reduction may pose a serious safety problem because it hampers access for emergency vehicles and limits the area available for their operation.*

3.3.3 Bridge Program Planning

The diverse terrain throughout Park County results in numerous situations where bridges are necessary to cross natural and man-made drainage conveyances. These include structures over rivers, streams, creeks, and irrigation canals.

Bridges in incorporated Park County that are on public roads are typically owned and maintained by the County, State (WyDOT), or Federal Government (BLM, BOR, or USFS). In some cases, bridges that cross irrigation infrastructure are owned and maintained by the local irrigation districts.

3.3.3.1 Off-System Bridges

WYDOT assists local cities, towns, counties, and other agencies in regularly inspecting bridges used by the public that span distances in excess of 20 feet. Most bridges are inspected by WYDOT every 2 years. However, bridges having certain levels of deterioration are inspected more frequently. Bridges requiring specialized access to specific elements, such as nonredundant steel tension members or submerged members, receive special inspections designed to inspect these members.

WYDOT inspections are completed in accordance with the requirements set forth in the National Bridge Inspection Standards (NBIS). The inspection procedures are based on the codes, instructions, assessment criteria, and reporting requirements of the *FHWA Specifications for the National Bridge Inventory*, the *AASHTO Manual for Bridge Element Inspection*, and the *AASHTO Manual for Bridge Evaluation*.

Bridges require periodic maintenance to remain safe and serviceable, and regular inspections help to identify those elements needing repair or replacement. When a condition is found that may affect the ability of the bridge to carry the anticipated loads, a load rating is completed and when necessary, the bridge may require a load posting or other restrictions imposed until the condition can be corrected or

the bridge is replaced. The inspection results are submitted to the FHWA and the local government entity responsible for the bridge on an annual basis.

As part of the short-term and long-range planning process, PCPWD shall utilize the WyDOT inspection reports to prioritize the most pressing bridge maintenance, improvement, and maintenance needs. A sample WYDOT bridge inspection form is included in **APPENDIX P**. A summary of Park County's bridge rating data and information as of the date of these *RBS* is included in **APPENDIX Q**.

3.3.3.2 Privately Owned Bridges

There are numerous privately owned and maintained bridges located throughout the County. It is the responsibility of the landowner (bridge owner) to ensure that the bridge is safe for the people that use the bridge for access including emergency services vehicles. The Park County Fire Chief maintains a database of load ratings for private bridges so they know which bridges their equipment can cross in the event of a fire or other emergencies. Although not required by PCPWD, owners of private bridges are encouraged to have a Wyoming licensed engineer assess their private bridge and calculate a load rating. It is also recommended that the load rating be provided to the Park County Fire Department for their records to ensure that response time during a fire emergency is minimized.

3.3.3.3 Bridge Types

Bridges are broadly classified into fixed and movable structures.

Fixed Bridges:

A fixed bridge structure refers to a stationary bridge with no moving parts, designed to remain in one place. Fixed bridges span an obstacle using a combination of the foundation, substructure (piers and abutments), and superstructure. A fixed bridge is designed to stay in place to carry traffic over waterways, valleys, or other roadways without the need to open or pivot for marine traffic. Examples of fixed bridge types are shown in **FIGURE 7** and are described generally below.

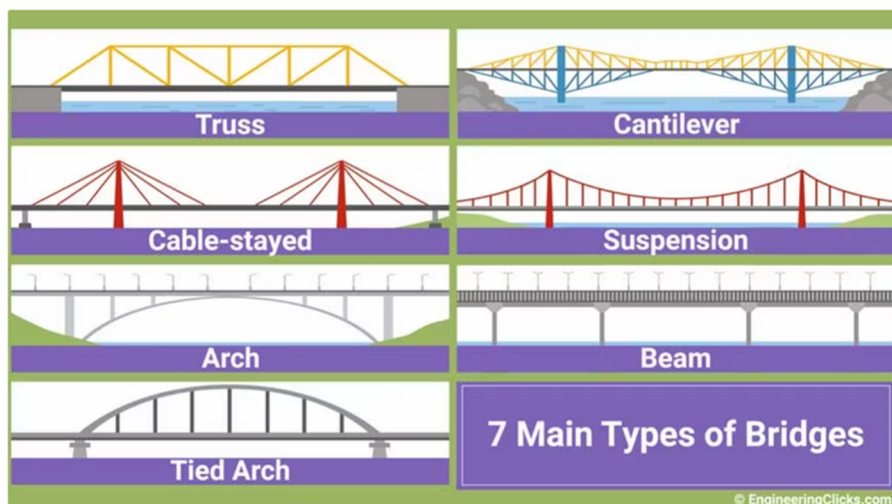
- i. **Beam Bridges:** The simplest and oldest type. They consist of a horizontal deck supported by piers at each end. Best for short distances (e.g., highway overpasses).
- ii. **Truss Bridges:** Feature a lightweight, rigid superstructure of crisscrossed triangles. This design efficiently distributes weight by handling forces in either tension or compression.
- iii. **Arch Bridges:** Curved structures that channel the load outward into heavy abutments at the ends. Naturally strong, they are often built with stone, concrete, or steel over valleys and rivers.

- iv. **Suspension Bridges:** The deck hangs from massive steel cables draped over towering pylons and anchored into the ground. Ideal for massive spans (e.g., Golden Gate Bridge).
- v. **Cable-Stayed Bridges:** Feature a deck supported by diagonal cables that run directly from the main vertical towers to the roadway.
- vi. **Cantilever Bridges:** Built using horizontal beams (cantilevers) that are supported on only one end, usually extending outward from large piers to meet in the middle.
- vii. **Tied-Arch Bridges:** Combine elements of arch and suspension designs. The arch sits above the deck and uses vertical ties (cables) to hold the roadbed, balancing forces internally rather than relying entirely on ground abutments.

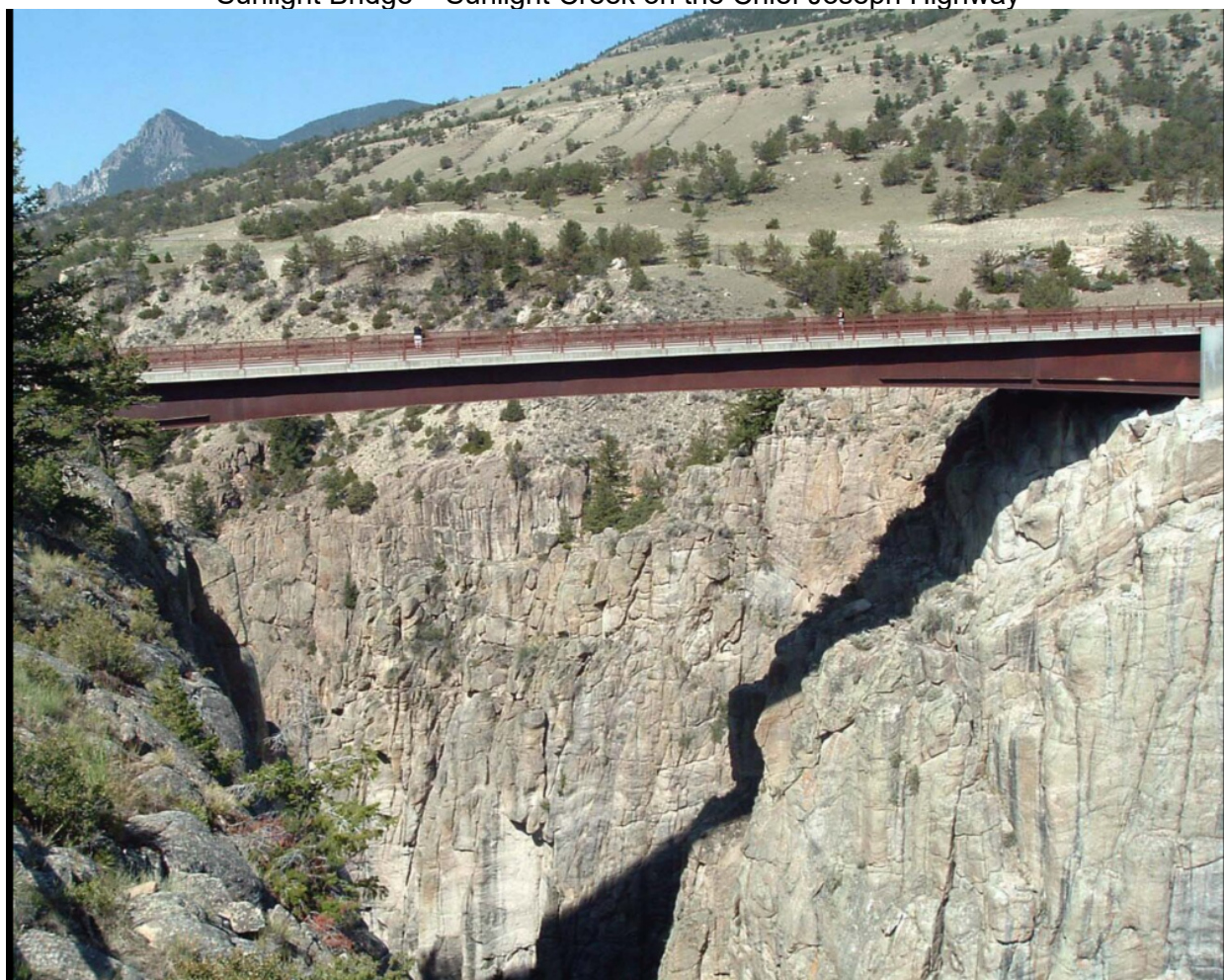
Moveable Bridges:

A moveable bridge is a structure that shifts—via rotation, lifting, or retraction—to let ships and boats pass. Moveable bridges serve a dual purpose, carrying vehicles, trains, or pedestrians when closed, and offering unlimited vertical or horizontal clearance for maritime traffic when opened. These structures are categorized by how they move and include the following types:

- i. ***Bascule Bridges:*** Also known as drawbridges, these use a counterweight to balance a hinged section of the deck as it swings upward. They are the most common moveable span.
- ii. ***Vertical-lift Bridges:*** Instead of swinging, these elevate the entire bridge deck straight up, much like a giant elevator.
- iii. ***Swing Bridges:*** These bridges rotate horizontally on a central pivot point or table, unblocking two channels at once.

FIGURE 7 - FIXED BRIDGE TYPES (TYPICAL)**FIGURE 8 - TYPICAL BEAM BRIDGE**

Sunlight Bridge – Sunlight Creek on the Chief Joseph Highway



RE: Wikipedia

FIGURE 9 - TYPICAL TRUSS BRIDGE

Bridge DDP on County Road 4DT (Wood River)

Typical Warren Truss Type (with Verticals) – Constructed in 1925 (still in service)



Photo Taken 06/03/2026 – WyDOT Bridge Inspection Report

FIGURE 10 - TYPICAL ARCH BRIDGE

Hayden Arch Bridge (Shoshone River)

Managed by the Bureau of Reclamation – Constructed in 1925 (still in service)

*Photo published in the Cody Enterprise – June 11, 2025***3.3.3.4 New Bridges**

New bridges or major rehabilitation of existing bridges that are used by the public shall meet the requirements outlined in *Section 5* and shall be designed by a Wyoming licensed professional engineer (structural engineer).

3.3.3.5 Bridge Replacement Priorities

Bridges used by the public that represent a health or safety risk shall be promptly repaired or replaced. If funds are unavailable and a bridge is considered structurally deficient with the potential for eminent failure (ie, fracture critical), the bridge shall be closed by the County Engineer (County bridges) or the entity responsible for the bridge. Replacement or major rehabilitation of private bridges is the responsibility of the bridge owner including all costs and liability.

County bridges shall be regularly inspected with the condition reviewed. A bridge inventory shall be maintained ranking the bridges in order of priority. Bridges with

the highest risk or danger to public health and safety shall be given the highest priority in regards to major rehabilitation or replacement (See **APPENDIX Q**).

4.0 PERMITTING

Development, improvement, or maintenance involving the CRBS may require specific permitting requirements depending on the location, functional classification, type, and jurisdiction of the road and bridge system. The following information is intended to summarize permitting requirements that may be applicable.

4.1 FLOODPLAINS

Roads can be constructed in floodplains, but they are strictly regulated to ensure the road does not increase flood heights, cause dangerous washouts, or adversely impact neighboring properties. Park County participates in *the National Flood Insurance Program (NFIP)*. As such, any new road construction or substantial improvement to existing roads located within a mapped floodplain may require a Floodplain Development Permit (FDP). The local floodplain administrator responsible for administering the NFIP at the County level is the P&Z Director. The P&Z Director shall be consulted to determine whether a FDP is required (Also see DSR).

Building roads in a designated floodplain typically requires a “*No-Rise Certification*” to show that the planned improvements will not result in a substantial increase in the base flood elevation. If the new road or substantial improvement of the roadway is to be constructed in a regulatory floodway (the most hazardous part of a floodplain), a hydraulic and hydrologic (H&H) engineering analysis is required to prove the road will not raise water surface elevations during a 100-year flood.

The road design must incorporate adequate culverts or bridges to allow floodwaters to cross without creating a damming effect. The roadbed and side slopes must be reinforced (often with riprap or geotextiles) to withstand scouring and prevent the road from washing out during heavy flows. The local floodplain administrator (P&Z Director) shall be consulted regarding other specific floodplain related requirements and conditions.

4.2 WETLANDS

Roads can be constructed in wetlands, but this practice is strictly regulated in most cases. In the United States, any construction that involves dredging or filling wetland areas may require a permit. The U.S. Army Corps of Engineers (USACE) should be consulted before constructing or improving a roadway within a designated wetland area to determine if any permitting is required.

For projects with minimal environmental impact (i.e., walkways), the USACE may allow construction under a pre-approved Nationwide Permit. For larger, more disruptive roads that permanently alter a wetland, an Individual Permit may be required. Individual permits require a more thorough review process and may require public notice. In some cases, additional permitting may require mitigation efforts to compensate for any destroyed or adversely impacted wetlands.

4.3 CULTURAL RESOURCES

Construction of new roads or substantial improvements of existing roads can threaten archaeological sites, historic structures, and traditional cultural places through physical damage or vibrations. While laws like *Section 106* of the *National Historic Preservation Act (NHPA)* require federal undertakings to undergo cultural resource reviews, purely private projects lacking federal funding or licensing are generally exempt from these requirements. New roads or substantial road improvements crossing federal lands may be subject to the NHPA. In addition, public road projects that utilize federal funding often require NHPA compliance.

In Wyoming, a cultural resource review is a regulatory compliance process required for any project taking place on federal lands, utilizing federal funds, or requiring federal permits. It ensures that development balances with the preservation of historic and archaeological sites under *Section 106* of the NHPA. Wyoming State Historic Preservation Office (SHPO) is the lead state agency that manages reviews, consults with federal agencies, and protects historic/archaeological properties.

Although private landowners constructing or improving private roads on private property are exempt from NHPA, developers are encouraged (but not required) to follow avoidance and preservation strategies and consult with SHPO when known or possible cultural resources exist.

4.4 ENVIRONMENTAL CONSIDERATIONS

For new county road construction, the primary environmental focus is mitigating impacts on local ecosystems, water quality, and community health. Key considerations include stormwater runoff management, habitat preservation, noise and dust pollution control.

When new roads or substantial improvements to roads are contemplated across federal lands or when federal funds are used for road construction, strict adherence to *National Environmental Protection Act (NEPA)* requirements is necessary. For most County road, public road, and private road projects, NEPA compliance is not applicable.

PCPWD does not have any separate permitting requirements addressing environmental considerations as it relates to roads serving development. However, developers and landowners are encouraged to consider the following to the extent possible and practical when contemplating new roads or substantial improvements to existing roads:

- *In some situations, road locations can represent a public safety hazard when crossing known wildlife corridors (i.e., vehicle vs. wildlife impacts). Roadways can cause habitat loss and fragmentation. New roads and substantial improvements to existing roads should be designed and constructed to minimize impacts to wildlife if possible.*
- *Mowing or maintaining the road shoulder can improve visibility for motorists and minimize the potential for vehicle vs. wildlife impacts.*
- *Bridges and culverts designed to allow aquatic passage are encouraged when applicable.*
- *The road location (alignment), construction materials, and maintenance should consider potential nuisance dust and noise impacts on neighboring properties.*

Construction of new roads, substantial improvements to existing roads, or site improvements related to development that results in a land disturbance in excess of 1 acre may require storm water quality permitting through the Water Division of the Wyoming Department of Environmental Quality (WDEQ). Disturbances that exceed 1 acre, but less than 5 acres must follow the WDEQ small construction general storm water permit guidelines. Disturbances that are larger than 5 acres must follow the WDEQ large construction site general permit requirements.

4.5 WILDLIFE CONSIDERATIONS

Governor's Executive Order EO 2019-3 addresses development in core sage grouse areas. The *Policy* requires developers to avoid, minimize, and mitigate habitat impacts in designated "Core Population Areas" ensuring new activities do not cause population declines. *Wyoming Governor's Executive Order 2019-3* (regarding Greater Sage-Grouse Core Area Protection) does not apply to private property. The land-use regulations, development stipulations, and mitigation requirements outlined in the order apply exclusively to state and federal lands. Compliance with the order on private land is entirely voluntary. Private landowners are not restricted in how they use their land as it relates to sage grouse protection, nor are they legally forced to follow state development or conservation guidelines.

If State or Federal lands, funds, or permits are required to complete a development or road improvement project, it is recommended that the Wyoming Game & Fish Commission be consulted to determine applicability pertaining to *Wyoming Governor's Executive Order 2019-3*.

4.6 RIGHT OF WAY PERMITS

Permits for road and bridge construction: Prior to the commencement of construction, improvements, repairs, or maintenance within a dedicated public road right-of-way ~~the County rights of way of any road, bridge, structure or facility,~~ the project proponent shall obtain approval of construction plans and have obtained a County ROW Permit from the PCPWD ~~County Engineer~~ as described in the following sections: ~~in accordance with Chapter V, Section 4, et seq., of these Standards.~~ The rights-of-way requiring a County permit include County Roads, County maintained public roads that are not dedicated County Roads, and public roads not maintained by the County unless there is an established HOA or RIMD. County projects are exempt from obtaining ROW Permits, but are not exempt from complying with the RBS. ~~these Standards.~~

~~b. Construction of roads:~~

~~(1) ——— Permits required for road construction: Whenever road construction within County Road rights of way results in earth disturbance, the individual responsible for the construction must obtain an approved ROW Permit from the County Engineer's office prior to commencing construction. The submittal requirements and procedure for obtaining ROW Permits are stated in Chapter V, Section 4 of these Standards.~~

4.6.1 Purpose and Intent

SECTION 4: Right of Way (ROW) Permits

~~a. Purpose and intent:~~ A ROW Permit shall be obtained whenever a developer, contractor, property owner, utility company or other individual proposes to install utility lines or culverts or do any work in ~~public County~~ road rights-of-way. ROW Permits are required to ensure the method of installation meets the specifications in these ~~RBS Standards, provide while considering for~~ the safety of the public. ROW Permits are also intended to ensure that improvements and maintenance follow generally accepted engineering practices including and adequate re-vegetation of disturbed areas within the ROW. ~~outside the roadway is done.~~ They are also intended to assure adequate reconstruction and/or repair of any damage caused to ~~public County~~ roads or road rights-of-way.

4.6.2 Road Cuts

Road cuts are discouraged by PCPWD on paved roadways. All paved ~~public~~ roadways shall be bored unless a bore creates an unsafe or hazardous condition. If, after the contractor attempts a bore, and the bore is shown to be physically impossible using standard boring techniques, the County Engineer may allow an open cut. Open cuts will only be allowed when the County Engineer determines that a bore is impossible based on a field inspection of the bore location.

If an open cut is allowed, the contractor, at his own expense, shall meet the requirements outlined in ~~Section 5.3.9~~ in addition to the following minimum criteria:

- i. ~~Submit a Traffic Control Plan for approval by PCPWD (See APPENDIX R);~~
- ii. ~~Install and maintain all traffic control devices prior to the start of and during construction;~~
- iii. ~~Work continuously to complete the work as expeditiously as possible;~~
- iv. ~~Trench backfill shall be completed made with WyDOT WDWSS Grading "W" crushed base compacted to ninety-five (95) percent of the soil maximum dry density based on a Modified Proctor Standard Proctor at ± two (2) three (3) percent of optimum moisture in accordance with ASTM D-698 D-1557 or with an approved lean concrete slurry. Compaction by flooding the trench shall not be allowed; and~~
- v. ~~Any other requirements deemed necessary by the County Engineer and/or the Road and Bridge Foremen.~~

~~To maintain a reasonable uniform road surface appearance,~~ The County Engineer may require a chip seal over the area of disturbance plus ten (10) feet on either side if deemed necessary to maintain a uniform road surface. Should this be required and the contractor is unable to chip the area due to weather or other causes, the contractor may bond the work, or a cost shall be determined by the County Engineer and invoiced to the contractor upon completion of the chip surfacing by the County.

4.6.3 Surety

Before issuing a ROW Permit, the County Engineer may require the applicant to post surety, in the form of a bond, letter of credit, cashier's check or other approved form, in an amount sufficient to complete the project or restore the construction area. The ~~surety~~

shall be based on the estimated costs ~~shall be costs~~ for the County to complete the project, including all legal and administrative costs.

Criteria to be used in determining whether a surety will be required may include, but is not limited to:

- i. *Estimated cost of the project. If the project's estimated costs are less than five thousand dollars (\$5,000) ~~seven hundred fifty dollars (\$750.00)~~, generally a surety will not be required;*
- ii. *Location of the proposed work. If the work involves cutting or disturbing a paved roadway, a surety may be required; and*
- iii. *Past experience with the contractor/developer.*

Except for emergencies, if the work and installation are not completed as stated and in accordance with these *Standards* determined by the County Engineer, ~~the County~~ PCPWD shall give written notice of the defects to the permittee at least thirty (30) days prior to the expiration date of any bond. The notice shall also state the bond will be called unless satisfactory corrective work is done within twenty (20) days of the notice. If satisfactory corrective work is not done within the required time limit, the work shall be in default, and PCPWD ~~the County~~ shall call the bond. In an emergency, the contractor shall make modifications immediately to protect the health, safety and welfare of the public.

4.6.4 Submittal Requirements for ROW Permit Applications

~~c. Submittal requirements for ROW permits:~~ Applications for ROW Permits (~~See APPENDIX I~~) shall be submitted to the PCPWD ~~County Engineer~~ for review and action. Approval shall be granted only if the proposed work meets the purpose, intent and specifications in these *Standards*, and any required fees have been paid.

Approval of a ROW Permit may be accompanied by any conditions deemed reasonable by the County Engineer to assure protection of health, safety and welfare of the public, the protection of public facilities and compliance with these *Standards*.

Applications for ROW Permits shall be submitted at least four (4) business days prior to planned commencement of construction for minor installations, which may include, but not be limited to, mailbox and turnout installations, driveways and minor utility work. ROW Permit Applications for ~~Twenty (20) business days prior for~~ major installations, which may include utility and irrigation installations, and subdivision access(es) ~~shall be submitted at least twenty (20) business days before commencement of work within the ROW. , applications for ROW Permits shall be submitted.~~ ROW permits will not be issued more than sixty (60) business days in advance of any installation. Construction shall not commence without an approved ROW Permit.

Consideration shall be given to how the proposed installation affects County road maintenance and improvement programs. Approval shall be granted only if the proposed installation meets these ~~RBS~~ *Standards*, the required fees paid and surety posted, if required.

The ROW Permit must be signed by the County Engineer ~~or Project Manager~~ and a Road and Bridge Foreman for approval.

The following information, including specific dimensions for lengths, widths and relative locations, must be submitted with any application for a ROW Permit, unless ~~specific~~ ~~waved~~ ~~are~~ waived by the County Engineer as unnecessary:

1. Fee as required by resolution of the Board of County Commissioners;
2. Completed permit form;
3. Permit submittal requirements:
 - a. For minor installations, sketch plan showing the following:
 - i. Location of any excavations using dashed lines;
 - ii. Location of road and road rights-of-way;
 - iii. Location of any driveways;
 - iv. Existing structures, if any; and
 - v. Proposed structures, including any garages;
 - b. For major installations, construction plans and specifications
 - i. All required items as outlined for minor installations;
 - ii. Evidence of legal access, easements, etc;
 - iii. Construction schedule: As part of its approval of any permit, the County Engineer shall review and approve a construction schedule. The approved schedule shall not be changed after the permit is issued without the written consent of the County Engineer; and
 - iv. Traffic Control Plan (See 4.6.14), in conformance with the MUTCD.
4. Additional permit requirements: In addition to the requirements listed above, the following may also be required in the permit application for a ROW Permit. At a minimum, a site plan showing the following information:
 - a. Well location, if any;
 - b. Septic system location, if any;
 - c. Location of property lines;
 - d. Location of required setbacks and their dimensions (refer to DSR);
 - e. Location of proposed driveways, accesses and their grades (refer to 5.9.2);
 - f. Parking areas (refer to 5.9.3), see Chapter V, Section 2.f. of these Standards;
 - g. Location and size of drainage culverts, if applicable (refer to 5.6.4);
 - h. Mailbox locations (refer to 5.9.6);
 - i. Location and size of trees and shrubs within one hundred (100) feet of the proposed point of access;
 - j. Surety covering the cost of reconstruction and/or repair of damage caused to County road or road rights-of-way;
 - k. Pre and post construction photos as required; and
 - l. Revegetation plan.

4.6.5 Supervision of Right-of-way Work

Supervision of right-of-way work: The permittee shall at all times conduct work within County public road rights-of-way so as to avoid obstruction and hazard to the traveling public and in conformance with the approved Traffic Control Plan (refer to 4.6.14). Materials and/or equipment necessary for construction shall not be stored in the public road County rights-of-way at any time unless approved in advance by the County

Engineer. The roadway and roadside area where work has been performed shall be thoroughly cleared of all debris and extraneous material and shall be restored to a condition equal to or better than the original when construction is concluded.

4.6.6 Inspection and Testing of Work

All work undertaken within the ~~public road~~ ~~County Road~~ rights-of-way shall conform to these ~~RBS Standards~~, and to approved plans and specifications. ~~e. — Inspection and testing of work:~~ Adequate inspections to assure compliance with these ~~RBS Standards~~ are required. It is the responsibility of the permittee to contact the ~~PCPWD County Engineer~~ at least three (3) business days in advance of any required inspections. In-progress inspections of all elements of work will eliminate the need for extensive post testing. At least one (1) inspection at the conclusion of construction is required. In making this inspection, the County Engineer, ~~Project Manager~~, or a Road and Bridge Foreman shall check for compliance with these ~~RBS Standards~~ and approved plans and permits. ~~The inspection shall also note any damage to public facilities and for if adequate cleanup of roadway surfaces and the rights-of-way has occurred.~~

Any work or material which does not conform to these ~~RBS Standards~~, pavement failures or broken asphalt, damaged signs or fencing and remaining debris either in the roadway or adjacent property, or improper drainage shall be brought to the attention of the permittee. Any work in which untested or unaccepted materials are used shall be ~~ordered~~ removed and replaced at the permittee's expense. Any required corrective work shall be ~~made completed~~ at the permittee's expense and shall be ~~corrected done~~ to the satisfaction of the County Engineer. If immediate corrections are not made, further construction shall be stopped.

In determining whether the work done by a permittee is acceptable, the County Engineer will consult with the Road and Bridge ~~Foreman Department~~. If a determination is made that testing is required, the number and location of the tests shall be determined by the County Engineer. If the County Engineer determines testing by an independent lab is necessary, the cost of such testing shall be paid by the permittee.

4.6.7 Responsibility for Rework

~~f. — Responsibility for rework:~~ The permittee shall be fully responsible for the maintenance and correction of any faulty construction, including unstable road cuts and chuck holes developed during the construction period and for a period of two (2) years following the final inspection of the work. All deficiencies shall be resolved to the satisfaction of the County Engineer at the property owner's and/or permittee's expense. Failure to do so could be cause to deny acceptance and denial of future permits. The permittee shall be responsible for a period of two (2) years after completion of work for any maintenance or repair necessary to keep the roadway in an acceptable condition.

4.6.8 Schedule for Work within the County Right-of-way

~~h. — Construction specifications and schedule for work within the County rights-of-way:~~ In issuing a ROW Permit, the County Engineer shall also review, and if acceptable, approve a Construction Schedule and Traffic Control Plan (~~refer to 4.6.14~~).

The approved construction plans, specifications and schedule cannot be changed without the approval of the County Engineer, except in emergency situations.

4.6.9 Emergencies

~~j.—Emergencies:~~ If a true emergency exists where time is not available to follow the procedures for obtaining a ROW Permit or for making modifications to the approved plans, specifications and schedule, a contractor may, after receipt of approval by the County Engineer, **Project Manager** or a Road and Bridge Foreman, proceed with the work. Within twenty-four (24) hours, the applicant shall submit an application for a ROW Permit.

4.6.10 Expiration of ROW Permits

~~k.—Expiration of Permits:~~ ROW Permits are for work to be completed within twelve (12) months of requested start date, or a new ROW Permit must be obtained. All work authorized by each ROW Permit issued by the County Engineer under the provisions of these **RBS Standards** shall expire if the work is not substantially begun within six (6) months from the date of the permit. If the construction of work authorized by the permit is suspended or abandoned for a period of six (6) months at any time after the work is begun, a new ROW Permit shall be required before such work can be resumed, or an extension may be allowed provided no changes in the original approved plans and specifications have been made or required by the County Engineer.

Any permittee holding an unexpired ROW Permit may apply for an extension of the time within which work may begin under the permit if the permittee is unable to begin work within the time required by this section for good cause and the cause is acceptable to the County Engineer. Permits expire when the end of the approved construction schedule is reached and must be renewed in advance to prevent the County from calling any bond or financial guarantee posted by the permittee.

4.6.11 Posting of ROW Permits

~~l.—Posting of ROW Permits:~~ ROW Permits shall be available on the job site.

4.6.12 Protection of Public Safety and Convenience

~~m.—Protection of public safety and convenience:~~ The permittee shall at all times conduct work to ensure the least possible obstruction and hazard to the traveling public. The permittee shall provide for the safety and convenience of the residents along roads where work is being done, and for the protection of people and property at all times. Adequate warning signs, barricades, lighting, flags and other devices as specified in the MUTCD, and as approved by the County Engineer, shall be provided, maintained, and paid for by the permittee. Certified flagmen shall be posted to guide the traveling public where only one (1) traffic lane remains open, or through otherwise unsafe operations.

If, in the opinion of the County Engineer, **Project Manager**, or Road and Bridge Foremen, an unsafe condition exists, or the contractor is not in conformance with the approved traffic control plan, the County Engineer may suspend **ALL** operations until the situation

is corrected. If the contractor does not remedy the situation immediately, the County Engineer may correct the problem and bill the contractor for any expenses incurred.

~~(1) Construction procedures for rights-of-way work:~~ The permittee shall plan rights-of-way work so it does not create safety hazards or maintenance problems, render portions of rights-of-way unusable for future road improvement or obstruct major floodways.

~~(2) Compliance with safety standards:~~ The permittee's operations shall conform to the applicable requirements established by the Federal Occupational Safety and Health Act (OSHA), and any other applicable laws or regulations. ~~(3) Staging of installations:~~ Staging of projects may be required by the County Engineer to produce the least disruption possible for the traveling public. A permit for any subsequent stages may not be issued until the prior stage has satisfactorily progressed or has been completed.

4.6.13 ROW Permit Work Suspension or Revocation

~~n. ROW Permit work suspension or revocation:~~ The County Engineer, **Project Manager** or Road and Bridge Foremen may suspend or revoke any **ROW** permit, in writing, issued under the provisions of these ~~RBS Standards~~ whenever the permit is issued in error or on the basis of incorrect information supplied by the applicant, or when the applicant is not in compliance with the permit conditions or a hazard is created which would pose a threat to the health, safety and welfare of the public.

Should the ROW Permit be revoked or suspended, all work shall be suspended until a new permit or work under the revoked or suspended permit is authorized by the County Engineer. The County Engineer may remove any hazard or work determined not to be in compliance with permit conditions. Any cost incurred by the County shall be billed to the **permittee applicant**.

4.6.14 Temporary Traffic Safety Plans

The County Engineer may require a *Temporary Traffic Safety Plan (TTSP)* to be submitted with the ROW Permit Application depending on the nature and location of the work to be completed. When required, the TTSP shall conform to the minimum requirements outlined in **APPENDIX R** and shall be approved by the County Engineer before work proceeds.

4.7 ROAD NAMING PERMIT

As outlined in *Section 2.8*, Park County has established a uniform road naming, numbering, and addressing policy for public safety purposes. When it has been determined that a new road name is required, the procedures in *Section 2.8.3* should be carefully followed. A *New Road Name Permit Application* (See **APPENDIX K**) shall be submitted to the PCPWD along with the appropriate fee (See *Section 4.10*). Questions regarding the road naming and/or permit requirements should be directed to the PCPWD offices located at 1131 11th Street or by calling (307) 527-8520.

4.8 ADDRESS PERMIT

When it has been determined that a new address is required, the procedures in *Section 2.8.4* should be carefully followed. A *New Address Permit Application* (See **APPENDIX L**) shall be submitted to the PCPWD along with the appropriate fee (See *Section 4.10*). Questions regarding addressing and/or permit requirements should be directed to the PCPWD offices located at 1131 11th Street or by calling (307) 527-8520.

4.9 ROAD CLOSURE PERMIT

~~i. Road Closure Procedure:~~ Road closures are not permitted unless justified based on the basis of overall benefit to the ~~general~~ public. Requests for road closures shall be specified on the permit form (See **APPENDIX S**) submitted by the applicant, and no road closures shall be undertaken unless approved as part of ~~the a~~ ROW Permit issued by the County Engineer. All requests for ANY road closure shall be submitted in written format (Temporary Road Closure Application) as outlined by event type further described below. Road closures are only permitted between the hours of 8:00 a.m. and 5:00 p.m. unless otherwise authorized by the County Engineer. At a minimum, the following information shall be submitted with the request for closure:

- a. Simple Event Closure (~~Closure for non-complicated event~~—less than eight (8) hours in a single day):
 - Temporary Road Closure Application to be submitted a minimum of ten (10) business days prior to the anticipated need for the closure.
 - A completed ROW Permit Application.
 - Traffic control plan conforming to the requirements of the MUTCD (*Latest Edition, refer to Section 4.6.13*), to include a detour plan (if necessary) or plan to maintain access for local residents, school buses, postal delivery vehicles and all emergency services (including but not limited to Sheriff's Department, local Police Department(s), Fire Districts, Search & Rescue and Highway Patrol).
 - Reason and time frame of the anticipated closure, including a sketch plan/diagram of work to be completed.
 - List of names and phone numbers of responsible people (at least two (2)); and
 - Proof of Liability Insurance in the amount of one million dollars (\$1,000,000.00). A copy of liability insurance is to be filed in Public Works Office.
- b. Complex Event Closure (Closure for ~~complicated~~ event – more than 8 hours and/or multiple days):
 - Temporary Road Closure Application to be submitted a minimum of twenty (20) business days prior to the anticipated need for the closure.
 - All items as outlined in "Simple Event Closure"; and
 - Detailed schedule of closure times and locations including when closure will be suspended each day during construction period (special conditions apply to overnight and weekend/holiday closures - closure of intersections must be avoided whenever possible and are not permitted for overnight or weekend/holiday closures).

Except in an emergency, contractors may only close roads after obtaining approval from the County Engineer. The extent, time of closure(s), location of closure(s) and frequency of closure(s) is at the discretion of the County Engineer.

The contractor shall furnish, erect and maintain at their own expense all necessary barricades, suitable and sufficient flashers, signs and any other items necessary to ensure safe road closure procedures. All traffic control devices shall conform to the MUTCD, current

edition. Contractors shall also provide, when necessary and determined by the County Engineer, a sufficient number of certified ~~flaggers~~ ~~flagmen~~ and take necessary precautions for the protection of the work and safety of the public around their construction operations. Details shall be provided in the traffic control plan (See Section 4.6.13).

The contractor may be required to place notification of the pending closure in the news media and/or at appropriate locations along the route to be closed a minimum of five (5) business days prior to the initial road closure. The contractor shall be responsible for notifying any local residents of the closure. Park County shall notify all appropriate school districts, postal offices, utilities and emergency services including, but not limited to County Sheriff's Department, local police departments, fire districts, Search & Rescue and state highway patrol office.

The contractor is responsible for returning the road to a safe and passable condition prior to re-opening the road, including overnight and weekend/holiday openings. Permanent repairs shall be completed according to the requirements of the ROW Permit and these ~~RBS Standards~~. If it is necessary to extend ~~Should~~ the closure ~~need to be extended~~ beyond the approved closure time, advance notification is required and must be approved by the County Engineer. Upon completion of the project and reopening of the road, the contractor is responsible for the prompt removal of all signs, barricades, etc. and notification to the County Engineer.

Road closures that extend beyond the permitted period and ~~were~~ not previously approved by the County Engineer will be in violation of these ~~RBS Standards~~. ~~The matter~~ may also be reported to the ~~BOCC Board of County Commissioners~~ for review and appropriate additional actions.

4.10 PERMIT FEES

The BOCC may authorize fees for various permits administered by the PCPWD. A fee schedule summarizing all the permit fees approved by the BOCC shall be made available with the permit applications and shall be maintained on the County website at www.parkcounty-wy.gov on the Public Works Department page. Permits shall not be issued until all fees have been paid.

~~(5) Minimum fees~~

- ~~(a) Street name sign \$ 150.00~~
- ~~(b) Address marker – new address \$ 150.00~~
- ~~(c) Replacement marker – worn out naturally \$ 0.00~~
- ~~(d) Replacement marker – other causes \$ 50.00~~
- ~~(e) Correction of self-assigned address \$ 250.00~~
- ~~(f) Failure to properly mark driveway \$ 50.00 plus~~
~~mileage~~
- ~~(g) Change by applicant of driveway location or~~
~~address marker \$ 150.00~~
- ~~(h) Exception to charges:~~
- ~~(i) County or Agency required changes~~
- ~~(A) New subdivisions will obligate developers to pay for street name signs.~~
- ~~(6) Enforcement~~

~~(a) No address number(s) shall be released until all required County permits have been issued.~~

~~(b) No address number(s) shall be released~~

5.0 DESIGN & CONSTRUCTION STANDARDS

~~**Purpose and intent:** This~~ These design and construction standards address site development design and construction requirements and ~~section~~ sets forth specific standards for roadway and bridge design in unincorporated Park County. ~~These standards are and is~~ intended for use by County officials and staff, design engineers, land surveyors, and developers as it relates to site development and the CRBS. These *Standards* establish criteria for applicable roadways (See *Section 2.1.2 for applicability*) in the County to be used by the traveling public, to assure ~~their~~ public health, safety and welfare and to assure County resources and funds will not need to be expended to later rectify inadequately designed and constructed facilities.

~~The basis for design used in these Standards is AASHTO, A Policy on the Geometric Design of Highways and Streets. The basis for construction used in these Standards is the WPWSS (Wyoming Public Works Standard Specifications).~~

5.1 DESIGN REFERENCES

Numerous agencies, organizations, technical specifications and guidelines are widely used in site development and roadway bridge design/engineering. Several of these were used as the basis for developing these RBS. All applicable specifications of agencies or organizations listed in **APPENDIX B** ~~Appendix 24~~ are made a ~~part portion~~ of these *RBS Standards* by reference and shall be the latest edition ~~or revised thereof~~. It is recognized that guidelines and standards within various agencies and organizations identified in **APPENDIX B** ~~Appendix 24~~ may identify options or alternatives to these *RBS Standards*. These options or alternatives may be incorporated into designs provided they are approved in advance by and at the sole discretion of the County Engineer.

~~**i. Specification by reference:** All applicable specifications of agencies or organizations listed in Appendix 24 are made a portion of these Standards and Specifications by reference, and shall be the latest edition or revision thereof.~~

5.2 EASEMENTS AND RIGHT OF WAYS

Early in the development and design process for a roadway, the legal status of any existing easements and rights-of-way should be determined. Sufficient easement and/or ROW are needed to construct driveways, roadways, utilities, and drainage infrastructure as described herein. In many cases a developer may need to acquire additional easement and/or rights-of-way from individual landowners to achieve the given design and construction goals.

5.2.1 Easements

An easement is a legal right that grants a person or persons the non-possessory permission to use a specific portion of property (not owned by the grantee) for a defined purpose. It does not transfer land ownership, but it stays attached to the property and transfers automatically when the land is sold. Easements are legally binding and restrict

what can be built on the portion of land covered by the easement. When purchasing property, it is important to check the title to identify any pre-existing agreements.

Easements are typically created and granted based on the type of use. Examples of common easements include:

- *Utility Easements:* The most common type; these grant utility companies the right to enter land to install, maintain, or repair irrigation systems, gas lines, power lines, water pipes, phone lines, or sewer systems.
- *Access Easements:* Grants physical access across land to another person or persons (Grantee). Access easements are typically used when a driveway or roadway must be constructed across a neighboring property to provide access to a landlocked property. This is frequently used when a landlocked property requires a driveway or path that cuts through a neighboring lot.
- *Drainage Easements:* Allows water to flow across a person's property to prevent flooding and erosion or grants an irrigation district access to maintain drainage or irrigation infrastructure.
- *Temporary Construction Easements:* Occasionally temporary construction easements are needed to construct certain improvements on a neighboring property. These types of easements are applicable when the use is temporary and will not require future access or maintenance.

Most easements are created through a written agreement between property owners and usually are recorded with the property deed at the Park County Clerk's Office. In certain situations, an easement can be considered "prescriptive" if an individual continuously, openly and without explicit permission uses a portion of land over a long, legally defined period (typically years).

5.2.2 Right-of-way

A road right-of-way (ROW) is a legally designated strip of land reserved for transportation, public infrastructure, and utility access. It grants the government or public entities the right to build, maintain, and operate infrastructure without transferring ownership of the underlying land from the adjacent private property owner. The ROW encompasses more than just the road surface. It typically includes:

- *The Roadway:* Travel lanes, turn lanes, and shoulders.
- *Pedestrian Paths:* Sidewalks, trails, and bike lanes.
- *Drainage:* Ditches, culverts, and storm drains.
- *Utilities:* Overhead power lines, underground cables, and sewer/water pipes.
- *Safety & Signage:* Clear zones, traffic signs, traffic signals, and guardrails.

The adjacent landowner technically owns the land beneath the ROW, but the municipality, county, or state holds the easement to use and control it. While landowners may maintain the grass, government agencies hold authority over construction, tree removal, and road widening projects within the dedicated ROW. Permanent structures (like walls, fences, or unapproved sheds) cannot be placed in the ROW. Temporary

structures, such as mailboxes, are usually permitted but must comply with these *RBS* (See *Section 5.9.6 for mailbox requirements*).

~~(d) Right-of-Way:~~ The minimum right-of-way widths required for each road classification are specified in *Section 5.3. Table 5-3*. Additional rights-of-way may be required for drainage improvements, *slopes/embankments* (cuts or fills), intersections, curb returns, snow storage and other road appurtenances.

5.2.3 Surveying Requirements for Public Recordation

Surveying required to identify property lines, property corners, retraceable easements, and rights-of-way shall be prepared under the direct supervision of a Wyoming licensed professional land surveyor (PLS). Prepared survey exhibits describing the limits of the roadway easement, record of survey (ROS), or survey plats describing land divisions shall be signed and sealed by the PLS and recorded in the office of the County Clerk.

In situations where a ROS describes the creation, alteration, or vacation of a roadway, the information shall follow the procedures outlined in *Section 2.3*. Upon completion of the statutory process, the County Engineer shall be responsible for updating and maintaining the current County Road information on County Road Plat Maps which are a matter of public record and are available on the PCPWD page on the Park County website (www.parkcounty-wy.gov).

5.2.4 Monumentation of County Roadways

Survey monuments defining the physical limits of a designated County Road rights-of-way shall be set by Wyoming PLS with the appropriate documentation filed with the County Engineer and County Clerk's office consistent with Wyoming State Law. Survey monuments are to be marked and protected during construction and improvements within the designated ROW.

~~g. Survey monument policy:~~ Park County recognizes the need to allow access to survey monuments within County Road rights-of-way. In some cases, access to these monuments may cause damage to the road surface and become a maintenance problem and hazard to the public. It is therefore the desire of PCPWD ~~Park County~~ to coordinate with the surveying community to provide access ~~to~~ and ~~to~~ repair the pavement around the monuments.

5.2.5 Surveying for Road Design

Surveying used as the basis for the design and construction of any roadway covered by these *RBS* (See *Section 2.1*) is to be conducted under the direct supervision of a Wyoming PLS. The surveyor shall establish temporary or permanent benchmarks within the easement or ROW, establishing the survey control system used as the basis for design and construction. Survey control reference monuments should be based on the NAD 1983 (West Central Zone) coordinate system unless an acceptable similar system is approved by the County Engineer.

A minimum of one benchmark location should be provided for every half mile of roadway to be designed and constructed with monuments set at the beginning and ending station

of the project. The benchmark locations and coordinate data should be reflected on the construction plans and should be the basis for grading plans, alignments, profiles, and sections.

In addition to the legal boundary and limits of the road easement and/or ROW, the pre-design survey shall map the following for the benefit of the design engineer:

- *Existing topography (2' contour interval minimum);*
- *Location of existing utilities (including utility easements);*
- *Location of existing drainage conveyances including ditches and berms;*
- *Location, size, material, and invert elevation of existing culverts;*
- *Location and size of existing cattle guards;*
- *Existing access approaches;*
- *Property ownership and property corners;*
- *Other objects or elements existing within the easement/ROW (i.e., fencing);*

5.3 GENERAL DESIGN ELEMENTS

The required design elements for a given roadway project are based on the functional classification of a given roadway (See Section 3.3.2). Roads classified as *Low Volume Rural Roads*, *Rural Roads*, *Local Access Roads*, *Minor Collectors*, *Major Collectors*, *Minor Arterials*, and *Principal Arterials*, must be designed and constructed to the minimum standards described herein. Typical cross sections and relevant required design considerations for these functional classifications are illustrated on the following figures:

- Figure 11- *Low Volume Rural Road*- Typical Section;
- Figure 12- *Rural Road*- Typical Section
- Figure 13- *Local Access Road*- Typical Section;
- Figure 14- *Minor Collector Roads*- Typical Section;
- Figure 15- *Major Collector Roads*- Typical Section;

Roads that are classified as *Minor Arterial Roads* or *Major Arterial Roads* typically fall within the State Highway System and are designed, constructed, and maintained by WyDOT. Therefore, WyDOT will dictate the design and construction requirements and standards for these roadways.

~~(d)~~ Access to subdivisions shall be from roads constructed within dedicated public rights-of-way, private road rights-of-way or recorded perpetual easements. Two (2) points of access into the subdivision shall be provided if the proposed subdivision contains twenty (20) or more lots. ~~(e)~~ Road rights-of-way shall be provided from the proposed subdivision roads or streets to adjacent or adjoining lands if such easements or rights-of-way would improve access to the potentially developable lands or potentially landlocked lands and would facilitate the development of a coordinated road system developing within the area. ~~(f)~~ Roads shall be designed and aligned to join with planned or existing roads.

5.3.1 Design Capacity

~~(a)~~ ~~Design capacities:~~ **Table 1 5-4** presents the range of **Average Daily Traffic (ADT)** anticipated for **the functional** classes of roadways **that make up the CRBS**. If traffic

volumes on a particular road exceed the range specified for its functional classification, the road shall be reclassified to the appropriate category. However, roads may carry lower volumes than stated for their functional classification without being reclassified. In such cases, the function of the road rather than traffic volumes will determine design requirements. All road classifications must be approved by the County Engineer.

5.3.2 Design Speed

The road classification determines the geometric cross section and maximum sustained grades, while the design speed determines minimum or maximum standards for elements of alignment such as stopping and passing sight distances, radii of curvature, tangent lengths and superelevation transition lengths. ~~(b)Design speed:~~ The selection of design speed is influenced principally by the terrain, traffic volumes and appropriate range of design speeds for each road classification. **A roadway's design speed is the selected speed used to determine its geometric features, such as curve radii, sight distances, and lane widths. Design speeds dictate the baseline dimensions required to keep drivers safe at that target speed.**

Engineers shall choose a design speed based on several criteria including:

- **Functional Classification:** Arterials (Freeways and interstates) require higher speeds, while local and collector streets require lower speeds.
- **Context & Land Use:** Urban, suburban, and rural town areas emphasize pedestrian safety, often using lower target/design speeds.
- **Topography:** Flat, open terrain allows for higher speeds, whereas mountainous or steep terrain often necessitates a reduced design speed.
- **Operating Speed:** The anticipated actual speed most drivers will travel.

While design standards vary slightly by state and country, the following ranges are generally accepted:

TABLE 2 - TYPICAL DESIGN SPEEDS BASED ON FUNCTIONAL CLASSIFICATION

Roadway Type	Typical Design Speed Range	Key Design Elements Affected
Interstates & Freeways	50 – 70+ mph	Gentle curve radii, large clear recovery zones
Rural Arterials	50 – 65 mph	Adequate stopping sight distance, superelevation
Urban & Suburban Arterials	35 – 45 mph	Strict considerations, pedestrian intersection sight distances
Collector Roads	30 – 55 mph	Tighter corner radii, traffic calming measures
Local Access Roads	20 – 45 mph	Short sight distances, narrow lanes to reduce vehicle speed
Rural Roads	20 – 45 mph	Lower speeds in residential areas

**Low Volume
Rural Roads**

20 – 30 mph

Narrow road width with no
shoulder dictates lower speeds

Modern engineering for road system design emphasizes aligning the design speed closely with the intended posted speed limit (especially on urban and residential roads). For example, if a road is designed with the dimensions of a 45-mph highway, drivers often feel comfortable driving 45-mph, even if the posted limit is 35 mph.

In setting speed limits for public roadways, the procedures outlined in **APPENDIX T** should be followed to ensure public safety, uniformity, and consistency throughout the CRBS network.

5.3.3 Site Distance

~~Sight distance triangle:~~

~~(a) Determining dimensions and location of sight distance triangles:~~ For safety and visibility purposes, a sight distance triangle shall be maintained at street intersections and where driveways intersect streets. The distances along the legs of the sight distance triangle shall be measured from the corner or intersection point along the rights-of-way lines ~~or at the middle of the travel lane along the edge of the driving surface~~ for driveways ~~and road intersections~~ as shown in **FIGURE 16** ~~Figure 5-14~~. Where a road right-of-way is wider than normal or varies in width because it has been expanded to include cut and fill slopes or drainage improvements, the line along which the legs of the sight distance triangle are measured shall be parallel to the roadway at normal right-of-way width for the type of road under consideration.

No landscape materials, earth berm, signs, structures or other visual obstructions shall be allowed between two and one-half (2½) feet and six (6) feet above the surface within the sight triangle. This regulation is not intended to prohibit the planting of trees or retention of existing trees in the sight distance triangle, if they are pruned so branches are higher than six (6) feet.

5.3.3.1 Site Distance Triangles in Relation to Subdivision Design

~~(b) Incorporating requirements for sight distance triangles into subdivision design:~~ Developers shall incorporate the requirement for maintenance of a sight distance triangle at street intersections and intersections of driveways with streets in the design of subdivisions submitted for County review after the effective date of these **RBS Standards**.

5.3.3.2 Enforcement of Site Distance Triangles with Building/Zoning Permit Issuance

~~(c) Enforcing requirement when Building/Zoning Permits are issued.~~

~~(i)~~ Where a Building/Zoning Permit is filed for property which is unplatted or was platted prior to the effective date of these **RBS Standards**, no Building/Zoning Permit shall be issued for a structure which interferes with maintenance of a sight distance triangle unless application of the requirement would result in peculiar and exceptional practical difficulties to, or exceptional and undue hardship upon, the individual proposing development of the property. The **BOCC with consultation from the** County Engineer shall have authority to waive the requirement for maintenance of a sight distance triangle for such property only when a variance **(See Section 2.2.2)** ~~an administrative relief~~ does not create or compound a safety problem or concern.

~~(ii)~~ Where an Application for Building/Zoning Permit is filed for property which was platted or re-platted after the effective date of these **RBS Standards**, no Building/Zoning Permit shall be issued for any structure which would interfere with the maintenance of a sight distance triangle required by these **RBS Standards**.

~~(d)~~ Property owners shall be responsible for maintaining sight distance triangles free of visual obstructions for the portion of a triangle which falls within the boundaries of their property. When the County Engineer receives a complaint

concerning visual obstructions at a particular intersection or other locations, the County Engineer shall be responsible for inspecting the intersection and for taking the following measures:

(i) *Determine whether the visual obstruction is on public property, then request the appropriate jurisdiction remove the obstruction. If the obstruction is in the County right-of-way, the County may remove the obstruction.*

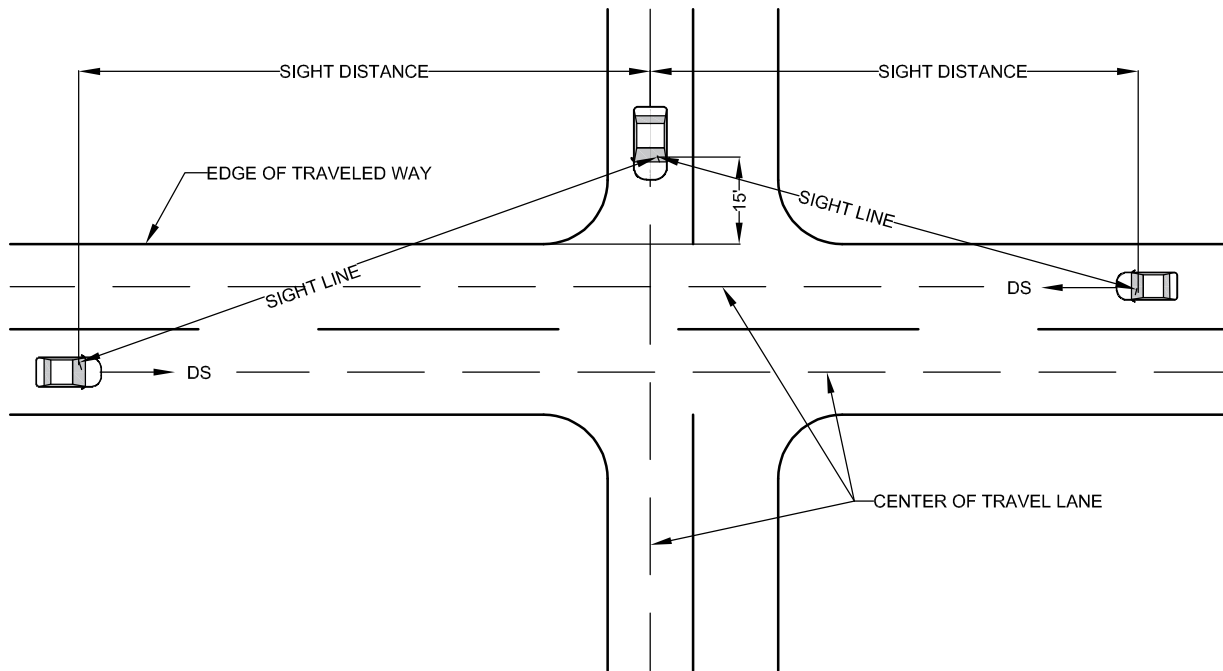
(ii) *If the visual obstruction is on private property, notifying the property owner of the requirement that visual obstructions must be removed within thirty (30) days except as follows:*

(A) *The obstruction is a permanent structure which was built prior to the effective date of these **RBS Standards**, as long as it does not create a hazardous condition; or*

(B) *Where the obstruction is caused by the natural or historic topography of the property and not by earthwork undertaken by the current property owner or his immediate predecessors, the property owner shall not be required to regrade his property to remove the obstruction.*

*If the property owner does not comply within thirty (30) days, further enforcement action shall be taken as provided in **Section 2.2.1 Chapter V, Section-8** of these **RBS Standards**.*

SITE DISTANCE REQUIREMENTS FOR CONTROLLED INTERSECTIONS



GRADE CORRECTION DISTANCE (FT.)				
SPEED	UPGRADE TO		FOR DOWNGRADES	
	3%	6%	3%	6%
25	0	-10	10	20
30	0	-10	10	20
35	-10	-15	10	25
40	-10	-20	10	30
45	-15	-25	15	40

CONTROLLED INTERSECTION	
DESIGN SPEED (DS) (MPH)	SIGHT DISTANCE (FT.)
25	250
30	300
35	350
40	400
45	450
50	500

NOTES:

- FOR CONTROLLED INTERSECTIONS, ROADS SHALL HAVE MINIMUM CORNER SIGHT DISTANCES, AS MEASURED FROM A HEIGHT OF 3.5 FEET ABOVE THE CONTROLLED STREET.
- PUBLIC, PRIVATE DRIVEWAYS ON ROADS SHALL HAVE AN UNOBSTRUCTED SIGHT DISTANCE TRIANGLE MEASURED IN THE SAME FASHION AS CONTROLLED INTERSECTIONS.
- IF THE ROADS ARE NOT LEVEL, USE THE GRADE CORRECTION DISTANCES SHOWN.
- "DESIGN SPEED" SHALL BE THE POSTED SPEED OR THE 85% PERCENTILE OF THE PREVAILING SPEED WHICHEVER IS HIGHER.
- FOR UNCONTROLLED INTERSECTIONS USE THE LATEST AASHTO MANUAL REQUIREMENTS FOR SIGHT DISTANCE.



DRAWN BY: T. Rozmetova
 DESIGNED BY: B. Edwards
 APPVD. BY: B. Edwards
 SCALE: NTS
 DATE: 8/3/2026
 JOB NO.

REV. DATE BY

SITE DISTANCE REQUIREMENTS FOR
 CONTROLLED INTERSECTIONS
PARK COUNTY R&B STANDARDS
 Park County Wyoming

FIGURE
16

5.3.4 Surfacing Requirements

Road surfaces must support varying traffic loads while managing severe weather and drainage. Surface requirements range from well-graded crushed aggregate (for unpaved roads) to asphalt or concrete (for higher-volume roads). All surfaces require adequate structural strength and a properly sloped crown for water runoff.

5.3.4.1 Unpaved and Gravel Roads

For low-volume roads (typically under 200 vehicles per day), unbound aggregate or gravel is standard. The aggregate road surface requires a mix of coarse aggregates (crushed stone/gravel) for strength and fine materials ("fines" like silt or clay) to act as a binder. PCPWD utilizes a WyDOT Grading W material for the crushed gravel surface on unpaved gravel roads.

TABLE 3 – REQUIRED GRADATION FOR GRAVEL ROAD SURFACE MATERIAL

WYDOT GRADING W	
SIEVE SIZE	% PASSING (BY WEIGHT)
1.5"	100%
1.0"	90-100%
1/2"	60-85%
# 4	45-65%
# 8	33-53%
# 30	10-30%
# 200	3-12%

When possible and practical, gravel roads shall maintain a cross-slope (crown) of 4% so water sheds into side ditches rather than pooling and forming potholes.

Fines blowing away lead to wash boarding and rutting. Applying dust palliatives (such as magnesium chloride or calcium chloride) are widely used and encouraged to preserve surface cohesion and control dust.

5.3.4.1 Paved and Treated Roads

As traffic volumes and heavy vehicle loads increase, flexible (asphalt) or rigid (concrete) surfaces are needed. Paved road surfaces must provide adequate macro-texture and micro-texture to ensure tire-pavement friction, specifically so vehicles can channel water away in wet conditions. Paved surfaces require a minimum crown of 2% to minimize the potential for ponding of water and hydroplaning.

Paved surfaces shall be designed to support equivalent single-axle loads (ESALs) without permanent deformation (rutting or fatigue cracking).

Surfacing requirements: All roads serving areas or subdivision roads for which these design standards apply (See *Section 2.1.2*), expected to carry a traffic

volume of two hundred (200) ~~one hundred (100)~~ ADT or greater shall be paved. Other roads may have a gravel or paved surface.

5.3.5 Dimensional Considerations

The dimensions of roadway elements within the designated right-of-way are important considerations and are determined based on the type of use, functional classification, design loading, and future maintenance considerations.

5.3.5.1 Clearances and Separation Requirements

Within the dedicated roadway ROW, there must be sufficient room for the road surface, shoulder, transition slopes, drainage ditches, signage, utilities, and snow storage. The ROW widths shown for the various functional classifications in **FIGURES 11- 15** are considered minimum. Additional easement or ROW may be necessary depending on natural topography, site distance, drainage, and the number and types of utilities that will need to coexist within the ROW.

When possible and practical, public utilities shall be confined to the outside 10 feet on either side of the roadway and in no case shall be any closer than 5 feet to the roadway surface. Similarly, the flowline of drainage ditches shall be at least 7.5 feet from the edge of the roadway surface.

Traffic sign distances from the travel surface and minimum heights shall conform to MUTCD requirements (latest edition).

5.3.5.2 Side Slopes

~~(d) — Side Slopes:~~ Any slope designed steeper than three-to-one (3:1, horizontal:vertical) ~~four-to-one (4:1)~~ shall be certified for stability by a registered engineer qualified in soils analysis. Where heavy snowfall is expected, flatter slopes in cuts on the southern side of the roadway should be used to provide maximum exposure to the sun. Flatter slopes should be used wherever possible to reduce erosion, to decrease maintenance costs, to facilitate plant growth and to provide safer operations.

Transition slopes shall be provided between adjoining cuts and fills and shall be designed for pleasing appearance. Where cut or fill slopes intersect the original ground surface, the cross section shall be rounded to blend the slope into the natural ground surface.

Where the side slopes of the original ground approach one and one-half-to-one (1.5:1), the embankment shall be contained with a suitable retaining wall to avoid long fill slopes. Side slopes in rock will be based on the stability of the formation.

Benching of side slopes should be used sparingly and only where justified by sound engineering reasons, including the following:

- (i) *To stabilize material where benching is more economical than flattening;*
- (ii) *To intercept drainage in long and deep cuts; and/or*

- (iii) *To intercept and store loose material.*

Refer to **APPENDIX U** for general guidelines for earthwork and embankments for roadways (RE: *WyDOT Standard Details*).

5.3.6 Roadway Cross Section

Typical roadway cross sections for the various functional classifications of roadways that comprise the CRBS are illustrated in **FIGURES 11-15**. These sections depict the minimum thickness for the sub-base (pit-run material), road base (crushed gravel), and paved road surface (if applicable). The sections also depict the minimum ROW width, travel lane width, maximum side slopes, and minimum cross slopes.

~~(b) Geometric cross sections:~~

~~(i) Typical sections: A typical cross section is shown on Figure 5-1.~~

~~(ii) Travel lane width: The minimum travel lane widths are provided in Table 5-3.~~

~~(iii) Crown slope:~~ On undivided roads in tangent alignment, the high point of the crown will be centered on the pavement and the pavement sloped toward the edges on a uniform grade. In mountainous terrain, unpaved roads will be sloped toward the cut side of the road on a three (3) percent slope to alleviate surface erosions. On divided multi-lane roads on tangent alignment, each travel way will have a uniform cross slope with the high point at the edge nearest the median. Refer to **APPENDIX U** for general guidelines for roadway earthwork and embankments.

5.3.7 Utilities Protection and/or Relocation

~~Utilities protection:~~ The developer or contractor shall at all times take proper precautions to assure the protection of utilities, service lines or other public or private installations and shall be responsible for the repair of any damage. The developer or contractor shall notify One-Call of Wyoming not less than a minimum of two (2) full business days before excavation begins so the utility company can locate the services (Refer to W.S. §37-12-301 thru 305).

In certain cases, it may be necessary to move or relocate an existing utility to facilitate road construction or improvements. For roads serving subdivisions or industrial/commercial developments, the developer shall be responsible for coordinating the location and relocation (if necessary) all utilities located within the designated easement or ROW. The developer shall also be responsible for all costs to locate and relocate all impacted utilities resulting from the development.

For designated County roadways or public roadways maintained by PCPWD only public utilities are allowed within the designated public road ROW. Public utilities located within the public road ROW are for the greater public good and for the convenience of the utility service provider. If it is determined that an existing utility needs to be moved or relocated to facilitate road construction or maintenance by the County, the cost for relocation shall be the responsibility of the utility service provider. The PCPWD shall provide sufficient notice to the impacted utility to relocate their infrastructure.

5.3.8 Trenching and Backfilling

~~(11) Trenching, backfilling and reconstruction of road surfaces:~~ The method used in trenching for underground utilities and for backfilling trenches shall comply with these *RBS Standards* and the *Wyoming Public Works Standard Specifications (WPWSS)* requirements for Trench Excavation (*latest edition*). Jetting of backfill is not permitted within County rights-of-way.

Upon completion of installation, the roadway shall be repaired or reconstructed as required in these *RBS Standards* for sub-base preparation, base course material thickness and compaction and final surfacing so as to restore the roadway to current construction standards for the given type of road.

5.3.8.1 Trenching and Backfilling Across Gravel Roads

~~(a) Gravel roads:~~ Suitable material excavated from trenches may be used for backfilling, subject to approval of the County Engineer. At no time shall contaminated, wet, soggy, frozen or other unsuitable material be used as backfill. If proper backfill is not available at the site, suitable material shall be imported and unsuitable material removed from the site. Compacted backfill shall extend to the sub-grade of the road or to natural ground. **FIGURE 17 shows typical road trenching and backfilling requirements pertaining to road crossings.**

5.3.8.2 Trenching and Backfilling Across Paved

~~(b) Paved roads:~~ Following approval by the County Engineer, all cuts made in asphalt or concrete surfacing shall be made mechanically, cutting a horizontal and vertical line, and shall be cut twelve (12) inches wider than the edges of the trench or the damaged area. The final pavement cut shall not be made until immediately prior to patching. All excavations made in paved roads shall be completely restored as soon as possible, but in no case longer than thirty (30) days after backfilling is completed.

In the event weather conditions preclude restoration by a hot mix asphalt pavement, temporary repairs may be made by tamping and rolling into place a cold mix asphalt. Such cold mix patches shall be removed and replaced by a permanent hot mix asphalt pavement as weather and availability of materials permit. Permanent hot mix patches shall be no less than four (4) inches in thickness, or not less than the thickness of the existing pavement, plus one (1) inch adjacent to the excavation, whichever is thicker. ~~Permanent patches shall be installed in accordance with Figure 5-5.~~

Damaged pavement shall be repaired by appropriate methods as approved by the County Engineer. In general, cracks shall be filled with an approved crack filler and chip sealed. An overlay with pavement fabric, the full width of the paved surface, shall be required in those instances where, in the opinion of the County Engineer, the ride quality, safety or appearance of the finished roadway has been impaired. Sub-grade failures caused by the *permittee's* operation of heavy equipment shall be rectified by reconstructing the pavement sub-grade layers, replacing the sub-base, crushed base and paving.

The County Engineer shall use the following criteria in evaluating the requirements of an overlay:

- *Four (4) or more cuts are made within a one thousand (1,000) foot section;*
- *There is no more than $\pm$ three-eighths (3/8) inch deviation in the roadway surface in a ten (10) foot span; or*
- *Construction traffic has caused rutting, raveling or shoving of the existing pavement surface.*

5.3.8.3 Trenching for Piping

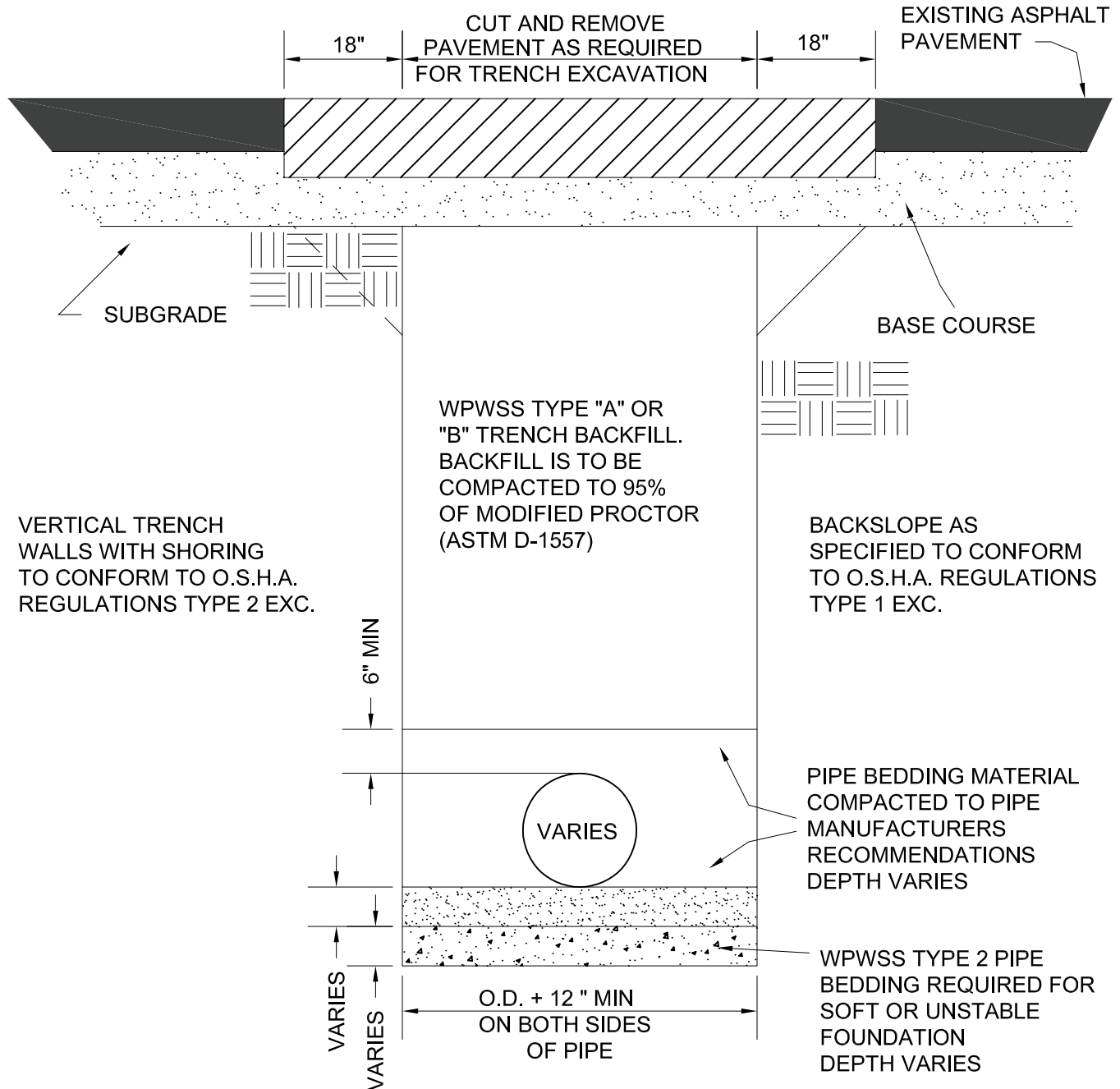
~~Trench excavation:~~ Trenches shall be excavated so pipes can be laid straight at uniform grade, without dips or humps, between the terminal elevations shown on the drawings. ~~(a) — Trench widths:~~ Trenches shall be excavated to a width which will provide adequate working space and side clearances for proper pipe installation, jointing and embedment and compaction. Minimum trench widths at or below an elevation six (6) inches above the top of installed pipe shall not be less than outside diameter (OD) plus twenty-four (24) inches. All excavations ~~activities shall comply with applicable to be in compliance with~~ OSHA regulations including but not limited to 29 CFR 1926 Subpart P.

~~(b) — Excavation below pipe subgrade:~~ Except where otherwise required, pipe trenches shall be excavated below the under-side of the pipe as shown in **FIGURE 17 Figure 5-6** to provide for the installation of granular bedding material. ~~(c) — Bedding material:~~ Approved bedding shall be used for all pipe installations or as specified on the plans. ~~(d) — Placement and compaction:~~ Bedding material shall be spread, compacted and the surface graded to provide a uniform and continuous support beneath the pipe at all points between bell holes or pipe joints.

After each pipe has been graded, aligned and placed in final position on the bedding material and shoved home, sufficient pipe embedment material shall be deposited and compacted under and around each side of the pipe and back of the bell or end to hold the pipe in proper position and alignment during subsequent pipe jointing and embedment operations. Bedding material shall be deposited and compacted uniformly and simultaneously on each side of the pipe to prevent lateral displacement. Approved bedding shall be compacted to a minimum of six (6) inches above the top of the pipe in all areas where compacted back fill is specified.

Whenever crushed rock is used as bedding for thirty-six (36) inch and larger pipe, the portion above the bottom of the pipe shall be vibrated with a mechanical vibratory compactor during placement to ensure all spaces beneath the pipe are filled.

TYPICAL TRENCH DETAIL



5.3.8.4 Backfilling Over Concrete

~~All backfill over concrete shall conform to the following requirements.~~

~~(i) Initial Backfill:~~ To aid curing, No more than eight (8) inches of loose backfill shall be placed over concrete after the concrete has reached its initial set. ~~(ii) Final backfill:~~ Additional backfill shall not be placed over arch encasements or blocking until the concrete has been in place at least fourteen (14) days or until the concrete has reached eighty (80) percent of its ultimate design strength.

5.3.8.5 Compacted Backfill

~~(f) Compacted backfill:~~ Compacted backfill may consist of job excavated material, finely divided and free from debris, organic material, cinders or other corrosive material, and stones larger than three (3) inches in greatest dimension. Masses of moist, stiff clay shall not be used. Job excavated materials shall be placed in uniform layers not exceeding eight (8) inches in uncompacted thickness. The method of compaction and the equipment used shall be appropriate for the material to be compacted and shall not transmit damaging shocks to the pipe. Job excavated material shall be compacted to ninety-five (95) percent of the maximum dry density at $\pm$ ~~two (2)~~ three (3) percent of optimum moisture content as determined by ~~ASTM D-1557 D-698~~. Compacted backfill is required for the full depth and width of the trench above the bedding material within all rights-of-way and easements.

5.3.9 Road Materials & Placement

5.3.9.1 Borrow Sources

~~(7) Borrow:~~ In the event sufficient suitable fill material is not obtainable within the limits of the project to provide all the embankment required, the contractor shall furnish such additional fill material (borrow) to complete the designated embankment. Borrow shall be an acceptable type of embankment material selected by and approved by the County Engineer and developer's engineer before being placed.

5.3.9.2 Structural Fill

~~Structural embankment construction:~~ Embankment construction consists of constructing roadway embankments, including preparation of the areas upon which they are placed, constructing dikes within or outside the rights-of-way, placing and compacting of approved material within roadway areas where unsuitable material has been removed and placing and compacting of embankment material in holes, pits and other depressions within the roadway area. Only approved materials shall be used in the construction of embankment and backfills.

~~(a) Benching:~~ When an embankment is placed and compacted on slopes steeper than ~~three-to-one (3:1)~~ four-to-one (4:1), the roadway shall be continuously benched over those areas. A bench is required at vertical intervals of ten (10) feet

maximum. Benching shall be well keyed and where practical, a minimum of eight (8) feet wide. Each horizontal cut shall begin at the intersection of the original ground and the vertical sides of the previous cuts. Material thus cut out shall be recompacted along with the new embankment material at the contractor's expense. ~~(b) Compaction:~~ Minimum compaction shall be ninety-five (95) percent of the maximum dry density at $\pm$ ~~two (2)~~ **three (3)** percent of the optimum moisture content as determined by ASTM D-1557 ~~D-698~~. Refer to **APPENDIX U** for general guidelines for roadway embankments per WyDOT standard details.

~~(c) Rip-rap:~~ Where embankments encroach on stream channels or lakes, calculations of the flows or wave action shall be made and submitted to the County. Based on these calculations, the developer's engineer shall determine the appropriate size riprap, and riprap shall be placed along the toe of the slope to protect the embankment against erosion from water action. Rip-rap shall be placed where the potential exists for erosion. The developer's engineer shall provide documentation supporting the size of riprap selected.

~~(d) Prohibited materials:~~ Car bodies, waste concrete and asphalt waste products, organic, wet, frozen or other unsuitable materials shall not be used for any structural embankment construction.

5.3.9.3 Subgrade

Site preparation:

~~(b) Clearing and grubbing:~~ All large rocks, brush debris, structures and all other unsuitable material shall be cleared to a depth of at least twelve (12) inches below subgrade or as approved by the County Engineer and replaced with suitable material. Locating suitable disposal sites shall be the responsibility of the contractor or developer, subject to County approval. Trees, except those designated to be saved, and all stumps shall be removed to a depth of at least eighteen (18) inches below the finished subgrade elevation. All trees designated to be saved shall be protected during clearing and subsequent construction operations. Suitable material removed from the excavation may be used in so far as practical, in the formation of embankments, backfilling and for other such purposes.

The bottom of the excavation for the road section, or top of the fill, shall be known as the sub-grade and shall conform to the lines, grades and cross sections shown on the approved plans.

No sub-base or base shall be placed on soft, spongy, frozen or unstable sub-grade which is considered unsuitable by the County Engineer. Where conditions require a road be constructed on unstable ground (i.e. soft, soggy or otherwise unstable or unsuitable ground) a special geotechnical investigation and pavement design shall be required by the County Engineer.

5.3.9.3.1 Subgrade Preparation

Prior to the road being excavated, all service cuts shall be tested to see if the backfill meets density requirements. If deficient, they shall be re-compacted and brought up to the specified density.

After excavation and embankment is completed and the sub-grade brought to final grade, it shall be rolled with a rubber-tire, sheep foot roller or other compaction equipment as required to bring the sub-grade to the required density and stability. All soils shall be compacted to a minimum of ninety (90) ~~five (95)~~ percent of ~~modified standard~~ proctor maximum dry density as determined by ASTM D-1557 ~~D-698~~. The minimum moisture content shall not be less than ~~three (3) two (2)~~ percent below the soil optimum moisture content. ~~“Standard Optimum Moisture.”~~ Additional wetting may be required when the minimum water requirement is not sufficient to produce a stable condition in the sub-grade soil.

5.3.9.3.2 Proof Rolling Procedures for Subgrade Verification

Where the native subgrade soils are variable resulting in difficulty achieving a reliable proctor, the subgrade surface may be proof rolled in lieu of field density testing at the discretion of the design engineer. Proof rolling shall be observed by the design engineer or his/her designee (i.e., geotechnical firm, engineering technician, etc.).

Proof rolling involves driving a loaded, heavy vehicle (typically a 20- to 50-ton tandem-axle dump truck) over a compacted subgrade at 2 to 5 mph. Tire pressures should generally be maintained near maximum manufacturer ratings (often 70 to 90 psi) to properly simulate construction loads. The goal is to visually identify and correct "pumping" (water squeezing out) or excessive rutting (deflection) before placing base layers. Proof-rolling shall not be conducted when the subgrade is saturated or when frozen ground is involved.

A minimum of two complete passes over the subgrade shall be completed under the direct supervision of the design engineer or his/her qualified designee. Each pass shall be offset by no more than one tire width to guarantee complete coverage of the surface.

The subgrade is typically considered failing if it ruts more than 1 inch, exhibits cracking, or visibly pumps soft material upward under the tire load. If unstable areas are found, they shall be marked. The contractor must then excavate the soft or yielding soil, allow it to dry (or blend with drier material), adjust the moisture content, and recompact it. In severe cases, over-excavation and replacement with engineered fill or geogrid may be required. ~~Should the finished sub-grade not be compacted under the oversight of a developer's engineer, heavy construction equipment or loaded trucks (over 50,000 lbs. tandem) shall be driven over the finished sub-grade and deflections noted. Soft and yielding material and portions of the sub-grade which show deflection shall be scarified and re-rolled or shall~~

~~be removed and replaced with approved sub-grade material, then placed and compacted as specified herein.~~ Subgrade shall not be approved for further road construction until it is uniformly stable and unyielding. ~~The County Engineer or designated representative shall be present to inspect the proof rolling operation.~~

5.3.9.4 Subbase Material

~~(9) Sub-base construction, as necessary by design:~~

~~(a) Materials:~~ Subbase material shall be composed of granular material consisting essentially of sand, gravel, rock, slag, disintegrated granite or a combination of such materials. The coarse portions of the material shall be sound fragments of the crushed or uncrushed materials enumerated above. Supplied material shall be a well-graded mixture containing sufficient soil mortar, crushed dust or other proper quality binding material which, when placed and compacted in the roadway structure, shall result in a firm, stable foundation. Material composed of uniform size particles, or which contains pockets of excessively fine or excessively coarse material shall not be acceptable for use.

The material need not be crushed, but shall be graded within the following limits:

TABLE 4 – PHYSICAL PROPERTIES FOR SUB-BASE MATERIAL (PIT-RUN)

STANDARD SIZE OF SIEVE	% BY WEIGHT PASSING SIEVE
2 ½"	100%
2"	95% - 100%
No. 4	30% - 60%
No. 200	5% - 15%
Liquid Limit	35% Maximum
Plasticity Index	6% Maximum

~~(b) General:~~ The specifications presented in this subsection are performance oriented. The objective in setting forth these specifications is to achieve an acceptable quality of roadway structures.

~~(c) Random tests:~~ The County Engineer may order random tests of materials used ~~in the County~~ to verify compliance with material specifications. Any and all materials used to construct public and private improvements that are not from a certified source or are from a certified source and fails one (1) or more random material tests, shall be subject to complete removal. The extent of the material to be removed shall be at the discretion of the County Engineer.

~~(d) Use of materials not listed:~~ Materials listed in this subsection and provided with a set of specifications are those deemed to be the primary structural materials commonly or typically used in public and private improvements. Alternative materials for construction may be proposed for use, except where expressly prohibited. Decisions on acceptability of alternative materials shall be made by the County Engineer.

~~(e) — Construction:~~ The construction of sub-base shall consist of furnishing and placing approved sub-base material to form a stable foundation on which to construct base course, in conformity with the lines, grades and typical cross sections shown on the plans. In addition, sub-base material shall be used to replace unsuitable foundation materials at locations shown on the plans, or as directed by the County Engineer.

Each layer of material shall be placed and spread so after compaction it shall conform to the width and crown of the typical cross sections. The wetting of sub-base layers shall be done with sprinkling equipment of a type which ensures uniform and controlled distribution of the water. All wetting shall be done by uniformly sprinkling each layer of material being placed with only that amount of water needed to obtain maximum density of the material.

Travel may be allowed over sub-base to assist in compaction of the material. Mixing and blading of the sub-base material on the road shall be required if the material is spotty and non-uniform. However, blading shall be held to a minimum to avoid the floating of the heavier rock particles to the surface.

5.3.9.5 Crushed Gravel Road Base

~~(10) Crushed road base course:~~~~(a) Crushed road base course specifications:~~ Placement of crushed road base material shall conform to the lines, grades, cross sections and thickness shown on the approved plans. Crushed road base material shall conform to WyDOT WPWSS Grading “H” or “W” Specifications (See Table 3). ~~Use of Grading “H” is preferred, when available.~~ When placed and compacted, it shall result in a firm, dense, unyielding foundation.

~~(b) Compaction of crushed road base course:~~ Crushed road base material shall be deposited and spread without particle segregation in loose layers not to exceed eight (8) inches in depth or, when compacted, the layer shall have a thickness not to exceed six (6) inches. The material shall be compacted to at least ninety-five (95) percent of the maximum dry density as determined by ASTM-D-698 D-1557 and the moisture content must be no more than $\pm$ ~~two (2)~~ three (3) percent of the optimum for the material. Rolling equipment shall consist of ~~a one (1) or more of the following: rubber tired roller or~~ flat wheel steel roller.

Crushed road base course shall not be placed upon a soft, spongy, frozen sub-grade or sub-base. ~~(c) Wetting operations:~~ The wetting of sub-base layers shall be done with sprinkling equipment of a type which ensures uniform and controlled distribution of the water. All wetting shall be done by uniformly sprinkling each layer of material being placed with only that amount of water needed to obtain maximum density of the material.

5.3.9.6 Pavement

~~Structural sections: Structural sections shall be designated for all new roads, driveways or roads being upgraded due to increased traffic.~~ Pavement designs shall be in accordance with currently acceptable design procedures and must be approved by the County Engineer. Approved procedures include, but are not

limited to, WYPWSS, AASHTO, and the Asphalt Institute. ~~In any case, the minimum pavement thicknesses listed in Table 5-3 must be adhered to.~~

~~Bituminous materials:~~

General: The intent of this section is to specify materials and methods to be used for the construction, overlaying, seal coating and pavement rejuvenating of roads, parking lots, walks and other miscellaneous work. The work covered shall include general requirements applicable to aggregate base course pavements of the plant mix type, bituminous prime coat, bituminous tack coat, rejuvenating applications and asphalt concrete overlay.

All workmanship, materials, and methods of preparation and construction shall be in accordance with specifications in these and WPWSS requirements for *Plant Mix Pavements* and in conformity with the lines, grades, depths, quantity requirements and the typical cross section shown on the plans or as directed by the County Engineer.

~~(2) Hot mix plant asphalt pavement:~~ All pavement shall be hot mix asphalt pavement of the plant mix type unless otherwise approved in writing by the County Engineer. ~~(a) Hot plant mix asphalt pavement:~~ This material shall consist of a mixture of aggregate, filler (if required) and asphalt cement. The job mix formula or mix design shall meet the grading requirements of WYPWSS for Aggregates, Plant Mix Pavements. (i) The job mix formula shall be submitted for review and approval a minimum of seven (7) days prior to placing mix on the project. The mix design shall be performed using either the Marshall or Superior Performing Asphalt Pavement (Superpave) ~~Hveem~~ Procedures as outlined in the Asphalt Institute, *Mix Design Methods for Asphalt Concrete*, current edition.

(ii) An additive may be used to meet the requirement for index of Retained Strength, if necessary. Such additives may be hydrated lime, Type I Portland cement or anti-stripping agent as identified in WYPWSS, Specifications for Plant Mix Pavement (latest edition). (iii) After the formula is established, all mix furnished shall conform to ~~it within~~ the established ~~following~~ range of tolerances as approved by the County Engineer.

~~(b)~~ All commercial testing and laboratory work necessary to establish the job mix formula and all testing necessary to assure conformance of materials and workmanship to the requirements of the specifications throughout the construction period shall be performed at the developer's expense. Two (2) copies of all test reports shall be submitted directly to the County Engineer.

5.3.9.6.1 Tack Coat

~~(3) Tack coat:~~ When tack coat is specified on the approved plans or required by the County Engineer, all materials, application and construction shall be in accordance with the WPWSS requirements for Bituminous Materials and Tack Coats (latest edition). The type of bituminous material, cover aggregate and rates of application shall be as shown on the approved construction plans.

5.3.9.6.2 Tack Coat

~~(4) — Seal coat:~~ When seal coat is required, all materials and construction shall be in accordance with the WPWSS requirements for Bituminous Materials and Seal Coats (latest edition). The type of bituminous material, cover aggregate and rates of application shall be as shown on the approved construction plans.

5.3.9.6.3 Grinding

~~(5) — Grinding:~~ Grinding shall consist of “milling”, “grinding” or “cold planing” the existing pavement surface to establish a new surface profile and cross section in preparation for a bituminous overlay. After grinding, the surface shall have a grooved or ridged finish, uniform and resistant to raveling or traffic displacement. This textured surface shall have grooves of one-fourth ($\frac{1}{4}$) inch plus or minus one-eighth ($\frac{1}{8}$) inch. The existing surface to be ground shall include bituminous pavement, concrete utility patches and a very small amount of concrete pavement.

Grinding around utility castings to the depth of cut shall be included in the area of the pavement surface ground. The contractor may choose to remove the entire existing bituminous pavement around the castings where grinding is not completed and replace it with bituminous surface course, placed and compacted in three (3) inch lifts. The contractor shall vertically cut the limits of the area to be patched, mechanically compact the existing base course and prime the bottom and vertical edges before backfilling.

The contractor shall remove the cuttings immediately behind the grinding machine. The removed material shall be disposed of as approved by the County Engineer. The grinding and cleaning equipment shall utilize a watering system for dust control.

~~e. — Bituminous construction requirements:~~

~~(1) — General conditions: Bituminous plant mix pavement construction shall meet the WPWSS requirements for Plant Mix Pavements.~~

5.3.9.7 Recycled Concrete/Asphalt Material

Recycled concrete may be considered by the County Engineer on a case-by-case basis for use in constructing the subbase layers for parking lot and road construction projects.

Recycled asphalt pavement (commonly known as Reclaimed Asphalt Pavement or RAP) is old asphalt ground up from roads and parking lots. It is used to produce new hot-mix asphalt or compacted directly as a loose gravel substitute. Specialized machines grind up the top layer of existing, deteriorated pavement. Broken asphalt chunks are transported to a facility where they are crushed and screened to achieve a uniform aggregate size. Because the old asphalt binder has been exposed to the sun and air, it can become brittle. Producers often add

"rejuvenating agents" to restore its flexibility and strength before reusing it in new pavement.

Using RAP significantly reduces the need for newly mined rocks, sand, and expensive liquid bitumen, resulting in significant financial savings. When used correctly, pavements featuring recycled asphalt perform just as well—and last just as long—as those made from 100% brand-new materials. The Federal Highway Administration (FHWA) strongly promotes RAP use because it minimizes landfill waste, limits carbon emissions from mining, and reduces energy consumption. RAP will be considered by the County Engineer for use on parking lot and road projects on a case-by-case basis.

5.4 GEOMETRIC DESIGN

~~(a)~~ **Alignment:** The major considerations in alignment design are safety, grade, profile, road width, design speed, sight distance, topography, drainage, ~~and the~~ maneuverability, braking, ~~and~~ performance of heavy-duty vehicles. Alignment should provide safe and continuous operation at a uniform design speed. In mountainous areas, consideration should be given to locating the road so that a southern exposure will be obtained wherever possible to avoid drifting of snow. Road layout should bear a logical relationship to existing ~~or-platted~~ roads ~~in-adjacent-properties~~ and to the principles of good engineering practice.

5.4.1 Horizontal Alignment

~~(i)~~ **Horizontal alignment:**

5.4.1.1 Stopping sight distance

Horizontal alignment must provide at least the minimum stopping distance of the design speed at all points. This includes visibility at intersections as well as around curves and roadside encroachments. ~~The minimum stopping sight distance is the distance required by the driver of a vehicle traveling at the design speed to bring the vehicle to a stop after an object on the road becomes visible.~~

Stopping sight distance (SSD) is the minimum distance required for a driver to see a hazard, react, and bring the vehicle to a complete stop. It is the sum of the "reaction distance" and the "braking distance". Stopping sight distance (SSD) is calculated in accordance with the following formula (Imperial Units- US Customary), or **TABLE 5 Table ?? 5-5**, whichever is greater:

$$SSD = 1.47Vt + \frac{1.075V^2}{a}$$

- SSD = Stopping site distance (ft)
- V = Design speed (mph)
- T = Brake reaction time (assumed standard is 2.5 seconds)
- a= Deceleration rate (assumed standard is 11.2 ft/s²)

$$\cancel{D} = 1.47Vt + \frac{V^2}{30(f + G)}$$

~~V= speed in MPH~~
~~t= reaction time (2.5 seconds)~~

- ~~G = grade, in percent~~
- ~~f = coefficient of sliding friction—with f equaling the following factors based on design speed of roadway, from Table ?? 5-4.~~

TABLE 5 – MINIMUM STOPPING AND PASSING SITE DISTANCES

Table ?? 5-4 FRICTION FACTORS		
DESIGN SPEED (MPH)	f (DESIGN CRITERIA: SNOW PACKED)	
20-40	0.24	
40-50	0.22	
50-60	0.21	
60-70	0.20	

Table ?? 5-5 MINIMUM STOPPING AND PASSING SIGHT DISTANCE		
DESIGN SPEED(MPH)	STOPPING SIGHT DISTANCE (Feet)	PASSING SIGHT DISTANCE (Feet)
15	125	700
20	115 125	710 800
25	155 150	900
30	200	1,090 1,100
35	250	1,280 1,300
40	305 325	1,470 1,500
45	360 400	1,625 1,650
50	425 475	1,835 1,800
55	495 550	1,985 1,950
60	570	2,135
65	645	2,285
70	730	2,480

RE: FHWA AASHTO (A Policy of Geometric Design of Highways and Streets)

*Assumes a presumed reaction time of 2.5 seconds, a Driver eye height of 3.5 feet, and an object height of 2 feet;

**Assumes wet roadway (accounts for reduced friction during inclement weather).

*** Assumes level roadway (SSD should be adjusted on downward inclines greater than 3% grade)

RE: For Passing Site Distance; AASHTO Greenbook Guidelines

Offset clearance to achieve stopping sight distance on horizontal curves shall be in accordance with current AASHTO Policy. The centerline of the inside lane is to be used, with the offset distance measured from the centerline of the inside lane to the obstruction.

5.4.1.2 Passing Site Distance

~~Passing sight distance:~~ Passing sight distance is the minimum sight distance that must be available to enable the driver of a vehicle to pass another vehicle safely and comfortably without interfering with oncoming traffic traveling at the design speed. Two-lane roads should provide adequate passing zones. Required passing sight distance for given design speeds is stated in **TABLE 5. 5-5**.

5.4.1.3 Curvature

The minimum radius of curvature (R_{min}) prevents vehicles from sliding off the road when traversing curves. It is calculated using the vehicle's design speed, the maximum allowed pavement banking (superelevation), and the coefficient of side friction.

$$R_{min} = \frac{V^2}{15 \times (e_{max} + f_{max})}$$

- V = Design speed (mph)
- e_{max} = Maximum roadway superelevation (expressed as a decimal)
- f_{max} = Maximum side friction factor (varies depending on speed)

~~(C) — Curvature:~~ **TABLE 6** ~~Table 5-6~~ specifies the minimum centerline radius of curvature for specific design speeds. This table is based on speed alone and does not take into consideration sight distance factors. Every effort should be made to exceed the minimum values.

Consistency in design speed and curve radius should be used to avoid surprising the driver. Where changes in the design speed are necessary, the design speed between approach tangents and curves will not change by more than ten (10) MPH. Under no condition will a low-speed curve be introduced at the end of a long tangent where high approach speeds are anticipated. Compound curves should be avoided. Reversing curves without an intervening tangent will not be permitted where design speeds exceed twenty-five (25) MPH. The minimum lengths of such tangents are specified in **TABLE 6. Table 5-6**.

Where an object off the pavement restricts sight distance, the minimum radius of curvature is determined by the stopping sight distance, but in no case will it be less than as specified in **TABLE 6. Table 5-5**.

TABLE 6 – MINIMUM RADIUS OF CURVATURE BASED ON DESIGN SPEED

		Table ?? 5-6 MINIMUM RADIUS OF CURVATURE	
DESIGN SPEED (MPH)	MINIMUM CURVATURE RADIUS (Feet)	MAXIMUM SIDE FRICTION, f_{max}	MINIMUM TANGENT LENGTHS (Feet)
15	400		50
20	450		75
25	225		100
30	230 300	0.17	150
35	450		200
40	460 600	0.15	250
45	775		250
50	800 950	0.14	250
55	1,200		250
60	1,310	0.12	
70	2,050	0.10	

RE: AASHTO based on e_{max} of 6%

5.4.2 Vertical Alignment

5.4.2.1 Minimum and Maximum Grades

~~(A) — Minimum and maximum grades:~~ Minimum and maximum sustained grades shall be as provided on **FIGURES 11 through 15** depending on the functional classification of the roadway. ~~listed in Table 5-3 except as provided in Item (B) below.~~ The minimum and maximum design grade should be used infrequently rather than as a value to be used in most cases. ~~Exceptions to maximum grades:~~

A local access, rural road or low volume rural road may have sections with a maximum grade of ten (10) percent, provided all of the following conditions are met:

- The section shall be no longer than five hundred (500) feet.
- The section shall have a horizontal radius of fifteen hundred (1,500) feet or greater;
- Grades shall not exceed six (6) percent for five hundred (500) feet on either end of the section.
- Curves with a horizontal radius of less than six hundred (600) feet shall not be within five hundred (500) feet on either end of the section; and

~~(V) — Land on each side of the section must be designated permanent open space.~~

5.4.2.2 Vertical Curves

~~(C) — Vertical curves:~~ Vertical curves must be designed to provide adequate stopping and passing sight distance, headlight distance, driver comfort and good

drainage. Minimum lengths of crest vertical curves are controlled by stopping sight distance requirements. The minimum length for sag and crest vertical curves shall be determined by current AASHTO criteria.

Vertical curves that are long and flat may develop poor drainage and should therefore be avoided. Vertical curves are not required where the algebraic difference in grade is less than two-tenths (0.2) percent ($A < 0.2$). ~~(D)~~ ~~———— Sight~~
~~Distance:~~ The grade line must meet sight distance requirements for the design speed.

Table 5-3 SUMMARY MINIMUM OF ROAD DESIGN ELEMENTS

DESIGN ELEMENT	PRIMARY	SECONDARY	RESIDENTIAL	LOCAL ACCESS	RECREATIONAL
Right-of-Way (min) (ft)	80	60	60	60	40
Minimum Recommended Design Speed (mph)	65	60	40	40	30
Number of Lanes	2-4	2	2	2	2
Lane Width (ft)	12	12	12	12	10

Shoulders (ft)	6	4	2	NA	NA
Maximum Sustained Grade	6%	6%	6%	8%	10%
Bridge Width (min) (ft)	36	32	24	24	24
Design Loading	HS20-44	HS20-44	HS20-44	HS20-44	HS20-44
Return Radius (ft)	35	30	25	20	20
Typical Speed Limit	65	55	30	30	30
Cross Slope w/o Super Elevation (%)	4 (max)	4 (max)	4 (max)	2 (min)	2 (min)
Max. Super Elevation (%)	4	4	4	4	4

Minimum Road Grade (%)	0.5	0.5	0.5	0.5	0.5
Maximum Road Grade (%)	6	6	6	6	8
Maximum Grade at Intersection	2% for 40'	2% for 300'	4% for 150'	4% for 100'	4% for 100'
Minimum Pavement Sections: Full Depth HMA Composite (HMA/Crushed Gravel)	8" N/A	6" 3"/8"	6" 3"/6"	6" N/A 1/6"	N/A N/A

NOTE: Variations may be approved by County Engineer based on generally accepted engineering practices, references and standards.

5.4.3 Switchbacks

Road switchbacks in rural areas require a delicate balance between vehicle navigation, environmental conservation, and topography. The primary goal is to safely maneuver the curve while controlling grades, ensuring adequate sight distances, and safely routing water away from the road structure to prevent erosion. Specific geometric guidelines should be followed when designing rural switchbacks.

~~(iii) — Switchbacks:~~ A switchback is defined as a curve with a delta greater than one hundred twenty (120) degrees and a radius of less than one hundred (100) feet. ~~(A) — Use of switchbacks:~~ Switchbacks will not be allowed on ~~arterial or collector primary or secondary~~ roadways. On residential, local access roads, rural roads, low volume rural roads or primitive roadways when other alternatives may cause significant adverse impacts, the use of switchbacks may be allowed on a case-by-case basis with approval from the County Engineer.

~~(B) — Minimum standards:~~ Switchbacks shall be designed for speeds of not more than ten (10) to fifteen (15) MPH. Maximum centerline grades within twenty-five (25) feet of a switchback curve and throughout the curve shall not exceed four (4) percent. ~~Curve widening shall be in accordance with Chapter V, Section 2-c(3)(b)(vii) of these Standards.~~ Minimum centerline radius of the curve will be fifty (50) feet. Adequate area for snow storage shall be provided. The radius on the curve's centerline must accommodate the design vehicle (such as a delivery truck, fire engine, or passenger car). While a standard passenger car may negotiate a 50-foot radius, larger rural service vehicles typically need a minimum of 45 to 50 feet for the *inside* turning radius.

Inside lanes should be widened through the curve by 2 to 4 feet (making the lane 12 to 14 feet wide) to prevent off-tracking. The shoulder can be utilized or modified to assist with the turning paths of long-wheelbase trucks.

The longitudinal grade must be flattened as it approaches and enters the switchback curve. A flat landing area of 0% to 2% is highly recommended through the middle (apex) of the 180-degree turn to prevent vehicles from losing control. Grades entering the switchback should ideally remain under 8% to 10%. Any grade exceeding 15% is extremely difficult for maintenance vehicles to plow and sand during winter conditions.

Instead of heavily superelevating the curve for switchbacks (which can increase rollover risks for slow-moving heavy trucks), designers may use a flat or slightly "insloped" cross-slope to help manage water drainage toward the cutbank.

The curve must provide sufficient Stopping Sight Distance (SSD) so drivers can see upcoming vehicles or hazards. Cut slopes on the inside of the turn should be graded back to ensure an adequate line of sight around the apex.

Advanced warning signage, such as a Turn or Curve warning sign paired with an advisory speed plaque, ideally meeting Manual on Uniform Traffic Control Devices (MUTCD) specifications shall be incorporated into the design and construction.

An adequate clear zone shall be maintained through the switchback(s) and W-beam guardrails shall be installed along steep, high-fill embankments on the outer edges of the turn. Drainage shall be controlled through the switchback(s) by preventing water from flowing straight down the road surface by incorporating rolling dips and water bars leading into and out of the switchback. In some cases, armored ditches or rip-rap may be required to slow the water and prevent erosion.

5.4.4 Alignment Coordination

~~(iv) — Alignment coordination:~~ When vertical and horizontal curves are superimposed, the superelevation may cause distortion in the outer pavement edges. Where this may be the case, edge of pavement profiles shall be plotted and smooth curves introduced to remove any irregularities. Sharp horizontal curves should not be introduced at or near a pronounced summit or sag.

5.4.5 Super Elevation

Superelevation is the banking of a roadway curve by raising the outer edge higher than the inner edge. This design practice counteracts centrifugal force, helps vehicles safely and comfortably navigate turns at speed, and assists with surface drainage., ~~t~~The maximum superelevation will be limited to four (4) percent to account for snow and ice conditions which occur frequently in Park County—~~(see Table 5-3)~~. The axis of rotation of undivided roadways is usually the centerline. For curves following long, level tangents, the axis of rotation may be taken at the inside edge of the pavement, with approval from the County Engineer.

5.4.6 Super Elevation Transitions

~~Superelevation transition:~~ Superelevation transition is the progression of the roadway from the normal crown section to a fully superelevated section. ~~T~~he length required to affect the transition should be adequate for the likely travel speeds ~~to meet the requirements of safety and comfort. Suggested minimum tangent lengths are given in Table 5-6.~~ It is recommended that sixty (60) percent to eighty (80) percent of the superelevation runoff be on the tangent.

5.4.7 Spiral Curves

~~(vi) — Spiral curves:~~ A spiral curve is a transition geometry with a continuously changing radius used between a straight road tangent and a circular curve. It allows drivers to steer smoothly, safely matches the banking (superelevation) of the roadway to centrifugal forces, and prevents sudden jerks or skidding at high speeds. Where the alignment includes spiral curves, superelevation is applied entirely on the easement curve.

When spiral curves are incorporated into the design, the following components shall be computed and illustrated on the design plans.

- **TS (Tangent to Spiral):** The point where the straight road starts turning into the spiral.

- **SC (Spiral to Circular):** The point where the spiral ends and meets the main circular curve.
- **CS (Circular to Spiral):** The point where the circular curve ends and transitions back into a tightening or exiting spiral.
- **ST (Spiral to Tangent):** The final point where the spiral straightens back out into a regular tangent.
- **Radius (R):** Starts at infinity (straight line) at the TS and gradually shrinks down to match the exact radius of the circular curve at the SC

5.4.8 Curve Widening

~~(vii) Curve widening:~~ Curve widening is the physical or digital enlargement of a road lane on a horizontal curve. It compensates for mechanical vehicle off-tracking and driver psychological behavior, ensuring large vehicles safely navigate turns without crossing centerlines.

Curve widening shall be incorporated in situations where there is a potential for rear wheels to track inside the front wheels on a curve requiring a wider swept path for large trucks. Drivers naturally steer further away from lane edges and oncoming traffic when navigating turns. Curve widening should be incorporated when deemed necessary by the design engineer to reduce roadway departure crashes and to minimize the potential for accidental lane crossovers on tight radiuses. Smaller curve radii requires greater road surface enlargement.

Curve widening (when used) shall consider the road classification and type of traffic that will use the road. For example semi-trailers dictate heavier widening adjustments. The design speed shall also be considered. Higher speeds influence transition lengths and comfort allowances.

Curves will be widened on the inside radius in accordance with ~~the~~ current AASHTO criteria.

5.4.9 Intersection Design

~~(D) Curb returns:~~ Minimum curb returns or pavement rounding radii at intersection corners are based on the functional classification of the roadways involved. Refer to Figures 11-15 for the minimum curb returns and surface curve radii for the given road functional classifications. ~~as follows:~~

ROAD CLASS	CURB RETURN RADIUS
Primary road	35 Feet
Secondary Road	30 Feet
Residential Street	25 Feet
Local Access	20 Feet

Additional right-of-way may be required to provide a minimum clear distance for fifteen (15) feet between the curb or edge of pavement and the right-of-way limit.

~~(E) Intersections:~~ The minimum distance between intersections for various road classifications is as follows:

TABLE 7 – MINIMUM SPACING BETWEEN ROAD INTERSECTIONS

ROAD FUNCTIONAL CLASS	DISTANCE
Principal Arterials	2-6 Miles
Minor Arterials	½ to 1 Mile
Collectors (Major)	1,300 feet
Collectors (Minor)	750 feet
Local Access Roads	500
Rural Roads	300
Low Volume Rural Roads	300
Primitive/Unimproved Roads	200 feet
Primary Road	1,300 Feet
Secondary Road	600 Feet
Residential Street	300 Feet
Local Access	150 Feet

Distance is measured from the inside edge of each right-of-way. When roads of two different functional classifications exist. The minimum spacing for the higher road classification shall be used.

5.5 GEOTECHNICAL CONSIDERATIONS

In constructing a new road on a new road alignment or when a new bridge is planned, the condition and physical properties of the underlying soils shall be considered. For new County Roads, roads to be dedicated for general use of the public, and bridges which will be used by the public, a geotechnical investigation shall be conducted. The purpose of the geotechnical investigation is to evaluate subsurface soil/rock physical properties and groundwater conditions (if applicable) along a proposed alignment. This information is necessary for determining subgrade strength, basis for pavement and foundation design, and geological hazards. Investment in a geotechnical investigation can prevent future problems with long-term road or bridge failures including settling, foundation rotation, cracking, and rutting.

When required, the geotechnical investigation shall include the following:

- **Desk Study:** Review geological maps, historical land use, and aerial photos.
- **Site Reconnaissance:** Walk the route to note surface drainage, rock outcrops, and soft spots.
- **Subsurface Exploration:** Drill test borings, dig test pits, or use cone penetrometers. The number, spacing, and depth of test pits and boring locations shall be determined by the design engineer and shall be adjusted as necessary to consider the type of road and local conditions.
- **Laboratory Testing:** Analyze soil samples for soils classification, moisture content, plasticity, density, shrink/swell potential, and bearing capacity. The number and frequency of laboratory test samples shall be at the discretion of the design engineer based on the type of road and local conditions.

When shallow groundwater or perched subsurface water is anticipated or encountered, the horizontal and vertical extent of the seasonal high-water table shall be identified so that adequate subsurface and surface drainage measures can be incorporated into the design.

5.6 DRAINAGE SYSTEMS

~~(e) ——— Drainage:~~

~~(i) General:~~ The primary objective of drainage design as it relates to these RBS is protection of the County roads infrastructure (roads, bridges, etc.) and adjacent properties property while minimizing the possible flood damage to surrounding properties and structures. Water flowing in a roadside ditch shall be diverted away from the road as quickly as possible. Similarly, property should be developed with sufficient grading, ditches, berms, culverts, and other drainage conveyances to divert drainage away from buildings, structures, and other developed infrastructure without adversely impacting adjacent property.

5.6.1 Drainage and Erosion Control Plans

A Drainage and Erosion Control Plan (DECP) shall be prepared by a Wyoming licensed professional engineer and approved by the County Engineer in the following circumstances:

- For Residential Developments:
 - When a Grading Plan is required (See Figure 1);
 - Development will result in a net initial disturbance of more than 1 acre;
 - Major subdivisions (See DSR);
- For Commercial/Industrial Developments:
 - When a Grading Plan is required (See Figure 2);
 - Development will result in a net initial disturbance of more than 1 acre;
 - Development will involve a new structure that is accessible to the public with a footprint of 10,000 square feet or more;
- For Roads (ADT >59 vehicles per day):
 - New roads to be dedicated for public use or major improvements to public roads;
 - New County Roads or major improvements to County Roads;

When required for residential, commercial, or industrial development, the DECP shall compare the calculated site specific storm-water runoff rates for both pre-development and post-development conditions. The runoff rates shall be compared and shall demonstrate no net increase in storm water runoff to adjacent properties or public roadways. Onsite stormwater detention (ie, detention ponds) may be incorporated into the design to achieve the no net increase in storm water runoff requirement. DECPs (when required) shall be sealed and signed by a Wyoming licensed professional engineer.

~~(ii) — Subdivision Drainage Facilities:~~

~~(A) — Runoff: Runoff from a project site after construction shall not exceed the level of runoff which occurred prior to construction. Any runoff in excess of pre-construction levels shall be detained on-site and infiltrated or evaporated. The entire drainage area~~

~~upstream from the project site up to a minimum of two hundred (200) acres shall be considered when determining runoff quantities, whether or not the two hundred (200) acre area is part of the project site.~~

DECP prepared in conjunction with site development shall include (B) ~~The applicant shall prepare a drainage plan and report addressing the historic and developed flows from the proposed development. The~~ a report and drainage plans that shall, at a minimum, provide a written graphical representation of the project pre and post development. At a minimum, the drainage plan and report shall address the following:

- (I) Historic vs. developed flows ~~as outlined in Section 5.6.1.2. for the one hundred (100) year and twenty five (25) year events.~~
- (II) Developed flows must be shown going to a defined drainage capable of handling the developed flows on site.
- (III) Irrigation flows must be considered separately from stormwater flows.
- (IV) All calculations used in the report, including detention/retention requirements, control structure requirements, all hydrology and hydraulics, pipe, ditch and conveyance calculations.
- (V) Topography (historic and developed) of the proposed development and the drainage basin.
- (VI) Construction plans showing all necessary information for the construction and maintenance of the facilities required to implement the drainage plan improvements.

5.6.1.1 Design Storm Intensity

Stormwater runoff analyses for site design shall consider the high intensity short duration storm events which are most common in the Rocky Mountain West. Rainfall intensities for 25-year occurrence for 5-minute, 10-minute, 15-minute, 30-minute, and 60-minute durations shall be considered in the design using the rainfall intensity data derived from NOAA Atlas 14 Volume 12 (latest edition). These high-intensity short duration storm events shall be considered sufficient (for practical purposes) for design of drainage ditches and culverts. When designing stormwater detention basins, the 25-year, 24-hour design storm intensity shall be used as the basis for design with inflow and outflow hydrographs considered.

For road and bridge systems the design storm intensity shall be determined based on the functional classification of the roadway. Roads and bridges along a road alignment that is associated with the following functional classifications shall be designed to consider runoff rates from a 25-year return frequency.

- Low Volume Rural Roads
- Rural Roads
- Local Access Roads
- Minor Collectors

Roads and bridges that are critical in providing emergency access connectivity including Major Collectors, Minor Arterials, and Principal Arterials should be designed to consider a 100-year return frequency when possible and practical.

5.6.1.2 Calculation of Peak Run-off Rates

~~Design Events shall be one (1) hour duration, twenty-five (25) year frequency.~~

Depending on the size, scope, and complexity of the development or project (i.e., County roads and bridges), different methods may be used at the discretion of the design engineer for computing peak discharge rates (reported in cubic feet per second). The Rational Method is generally considered acceptable for use for sites that are 20 acres or less with minimal storage. The Modified Rational Method can be adapted for use for sites to estimate simple runoff hydrographs and storage volumes for small detention facilities. NRCS TR-55 and TR-20 are widely accepted and used computer models that are considered acceptable in many cases to determine runoff rates based on assignment of runoff curve numbers (CN) and time of concentration for various water sheds. Other numerical and computer models may also be considered by the design engineer with prior approval by the County Engineer.

When runoff curve numbers are used (i.e., Rational Method), the runoff curve numbers listed in the HEC-HMS Technical Reference Manual (latest edition) shall be used (See **APPENDIX V**). Alternative curve numbers based on special land use and circumstances will be considered by the County Engineer on a case-by-case basis.

For large complex watersheds involving stream channels, bridges, large culverts (i.e., multi-plate and box culverts) more advanced numerical and computer hydrology models may be necessary. EPA's Storm Water Management Model (SWIMM), the Corps of Engineers Hydrologic Engineering Center (HEC-HMS), and other models may be used at the discretion of the design engineer.

Hydrology analyses shall also consider the potential for flooding and elevated runoff rates from melting snow during rapid spring warm-up periods when possible and practical. Flooding pertaining to events of this type are not uncommon in Park County and should be considered by the design engineer.

~~(A) — Storm runoff estimates: The following methods may be used for estimating peak flows:~~

~~(I) — Runoff from stream flow records;~~

~~(II) — HEC-1 Computer Program from the Corp of Engineers;~~

~~(III) — "Streamflows in Wyoming", H.W. Lowman, U.S. Geological Survey, Water Resources Investigation Report 88-4045;~~

~~(IV) — The Rational Method, as follows:~~

~~$$Q = CIA$$~~

~~Where~~

~~Q=runoff in cubic feet/second~~

~~C=coefficient of runoff (see Table 5-7)~~

~~I=average intensity of rainfall in inch/hour for a duration of the time of concentration~~

~~A=drainage area in acres~~

~~The rational method should be used only on areas of less than ten (10) acres;~~

~~(V) Tabular method as per Technical Release MO55 from the Engineering Division of the U.S. Department of Agriculture; or~~
~~(VI) SCS Method; or~~
~~(VII) Other methods, if approved by County Engineer~~

Table 5-7 CO-EFFICIENT OF RUNOFF	
TYPE OF SERVICE	VALUE OF C-RAINFALL
Roofs	0.97
<u>Pavements</u>	
—Concrete or Asphalt	0.97
—Gravel, from clean loose to clayey and compact	0.60
<u>Earth Surfaces</u>	
—Sand, from uniform grain size, no fines, to well graded,	
—Some clay or silt:	0.60
—Bare	0.45
—Light Vegetation	0.35
—Dense Vegetation	
—Clay, from coarse sandy or silty, to pure colloidal clays	0.70
—Bare	0.50
—Light Vegetation	0.40
—Dense Vegetation	

5.6.2 Design of Drainage Improvements

~~(C) Design of Drainage Improvements:~~ Drainage improvements shall be designed and constructed in accordance with current **industry accepted** professional **engineering** standards. All drainage installations shall be designed to permit free, unobstructed passage of debris and silt or provide for their deflection and/or collection at a point upstream that will not create an expensive maintenance problem. Settlement basins shall be provided when a silting problem exists downstream. Modification of natural channels or transferring runoff from one basin to another is not permitted except where no reasonable alternative exists and where the proposal has been reviewed and approved by the County Engineer. ~~In addition, the following design requirements may be required to be met and documented.~~

~~(ii) Use of culverts at access point to roads:~~ Driveways or road connections to a County road **or road dedicated for public use** shall not be constructed in such a way as to impede the normal flow of drainage in roadside ditches, culverts, underdrains, bridges or other drainage works, or to cause such drainage to flow onto or across the driving surface of **the a-County** road. In the event such an impediment results in damage to **the a-County road**, the Road and Bridge Department will remove the impediment and bill the property owner for the costs of repairs to the road, including labor, equipment and materials.

Materials including, but not limited to, culverts, signs and conduits, shall be new unless approved by the County Engineer.

5.6.2.1 Site Drainage from Adjacent Properties

Drainage pertaining to new development is intended to be maintained in a manner that results in no-net increase in runoff as described in Section 5.6.1. However, in some cases, historic, or natural topography and drainage is such that storm water runoff from an upgradient adjacent property (or properties) is naturally routed through a site being considered for development. Site drainage calculations and plans shall consider storm-water run-on from adjacent properties when applicable. When existing run-on is known to occur, provisions shall be incorporated into the site design such as ditches, culverts, and berms to divert drainage around or away from the developed portion of the site.

5.6.2.2 Ditches/Berms

~~(C) — Open channels and ditches:~~ Channels and ditches shall be designed to avoid roadside safety hazards and divert water away from site facilities and structures. The minimum flowline slope shall be one (1.0) percent when possible and practical. Maximum slopes shall be controlled by the maximum permissible velocities based on the calculated flow rate, peak shear velocity, and channel surface. ~~given in Table 5-9.~~

Manning's equation shall be used to estimate velocities in channels and ditches.

$$V = 1.486 \frac{R - S^{1/2}}{n}$$

where

V=velocity of flow in channel in feet per second
 n=roughness coefficient (RE: See APPENDIX V)
 R=hydraulic radius in feet per foot

Where the channel is comprised of a combination of the materials given in Table 5-9, the maximum permissible velocity selected should prevent undue scouring of the finer materials.

~~(III) — Ditch and Open Channel Sizing:~~ Ditches shall be sized to handle the anticipated quantity and debris flows anticipated for the drainage. Consideration should be given to erosion, scour the flow velocity and the ditch capacity and maintenance.

Table 5-9 MAXIMUM PERMISSIBLE VELOCITIES		
CHANNEL MATERIAL	"n"	VELOCITY (ft/sec)
Lines or well-established grass	.05	5
Bunched grasses with exposed soil	.04	3
Fine sand or silt	.02	4
All other bare soils	.03	2

5.6.2.3 Culverts

~~The Design Engineer shall utilize~~ Culverts under all roads for which these RBS apply shall be designed to accommodate runoff from a twenty-five (25) year

frequency storm ~~runoff~~ utilizing the maximum available head. The maximum available head shall be determined by the uppermost ponding elevation chosen to prevent flood damage to upstream properties. Inlets and other facilities draining the road surface shall accommodate the twenty-five (25) year frequency storm runoff. All roads shall remain free of ponding. ~~At least one (1) travel lane shall remain open during a one hundred (100) year design storm.~~ Velocities of flow in culverts shall be calculated using acceptable design charts or formulas **including the Mannings Equation**. Where the Manning Equation is used, the **following “n” values provided in APPENDIX V will be used.** ~~Dwill apply (see Table 5-8)~~

~~(H) Culvert Sizing:~~ Culverts shall be sized to handle the anticipated quantity and debris flows anticipated for the drainage. Consideration should be given to erosion, scour, flow velocity and the culvert capacity and maintenance. Head-wall(s), wing-wall(s) and/or flared end sections may be required.

Table 5-8 MANNING EQUATION OF “n” VALUES	
MATERIAL	MANNING EQUATION “n” VALUES
Corrugated Steel Pipe	0.027
Reinforced Concrete Pipe	0.013
Concrete	0.013 to 0.020
Asphalt	0.016

5.6.2.3.1 Culvert Location and Placement

~~(B) Culverts:~~ Culverts shall be located at each natural draw or water course as conditions warrant to prevent excessive accumulation of flow in roadside ditches or along the toe of slopes. Draws and water courses shall be cleared of debris for a distance of one hundred (100) feet upstream from all culvert inlets.

Inverts at the inlet shall be slightly elevated above the normal flow line in steep or natural draws to avoid plugging by debris. Inlets shall not be elevated in those instances where ponding or accumulation of backwater would be objectionable (stagnation, irrigation ditches, etc.).

The culvert shall slope downward in the direction of natural flow and be designed to be self-cleaning. The outlet shall be designed not to discharge on unprotected fills or unstable material or at adverse angles to streams or open channels. Head-wall(s), rip-rap or other approved means of protection are required at inlets and/or outlets where erosion might occur.

New driveways, approaches, or connecting roads off a dedicated County Road or road designated for use by the public shall incorporate a drainage culvert unless a waiver is provided by the County Engineer through the ROW Permit process. In certain instances, a culvert may not be required by virtue of the topography. This shall be determined during the ROW Permit process. Culverts for driveways and approaches shall have a minimum diameter of 18”.

Culvert diameters shall be sized to handle the computed peak runoff rates as determined in Section 5.6.1.2. Minimum diameter for round pipe shall be eighteen (18) inches. Arch pipe openings, at a minimum, shall be equivalent to an eighteen (18) inch round pipe.

When a battery of pipes is used, a clear spacing of one-half (½) the pipe diameter (one (1) foot minimum, four (4) foot maximum) must be provided between pipes. Maximum and minimum cover, pipe metal gauge and strength classification shall be as specified on culverts.

Manholes shall be used for cleanouts. Cleanout access shall be provided at least every two hundred (200) feet for pipes twenty-four (24) inches in diameter or less, and at least every four hundred (400) feet for larger pipes. Cleanout access shall also be provided at each angle point and at each change in grade.

Materials ~~including, but not limited to, culverts, signs and conduits,~~ shall be new unless approved by the County Engineer.

5.6.2.3.2 Reinforced Concrete Pipe (RCP)

Reinforced Concrete Pipe (RCP) shall be used for all installations which cross a paved County road ~~or are~~ within a County right-of-way. Corrugated metal pipe (CMP) may be used for unpaved roads and driveway crossings. Alternative materials, including PVC, HDPE, etc., may be approved by the County Engineer provided supporting documentation and justification addresses load analysis, corrosion protection, erosion/abrasive resistance and soil characteristics such as ph. ~~PVC or HDPE Plastic type~~ pipe will not be allowed where there exists a reasonable expectation of vegetation burning.

5.6.2.3.3 Corrugated Metal Pipe (CMP)

~~(6) — Culverts for driveway crossings: This section covers corrugated metal pipe culverts used for driveway crossing. Culvert sizing shall be in accordance with Chapter V, Section 2.c.(3)(e) of these standards and regulations.~~ Corrugated metal pipe (CMP) shall be furnished and installed with all jointing materials, accessories and appurtenances as indicated on the drawings and as specified. ~~(a)Material: Materials used for driveway crossings shall conform to the applicable WPWSS for Storm Drains and Culverts (i)Circular pipe:~~ Corrugated metal pipe shall be galvanized, and corrugations may be annular or spiral with annular ends. ~~, as shown in Figure 5-7. (ii) Coupling bands:~~

All field joints in corrugated metal pipe shall be made with coupling bands, fabricated from the same material as the pipe. Coupling bands for field joints in corrugated metal pipe for all culverts shall be the pipe manufacturer's standard coupling band type.

5.6.2.3.4 Culvert End Sections

~~(iii) — End sections: Flared metal~~ End sections shall be provided on all culverts unless otherwise specified by the County Engineer. For CMP, the end sections shall be fabricated from sixteen (16) gauge galvanized sheet metal for thirty (30) inch diameter and smaller pipe, fourteen (14) gauge for thirty-six (36) inch through forty-eight (48) inch, and twelve (12) gauge galvanized sheet metal for fifty-four (54) inch diameter and larger pipe. The CMP end sections shall be provided with a rolled reinforced edge and a galvanized top finish plate.

5.6.2.3.5 Culvert and Accessory Handling

~~(b) — Handling:~~ Pipe, couplings and accessories will be handled in a manner that will ensure installation in sound, undamaged condition. Equipment, tools and methods used in unloading, reloading, hauling and laying pipe will be such that the pipe is not damaged. ~~(c) — Cleaning:~~ The interior of the pipe and any couplings shall be thoroughly cleaned of all foreign matter before being installed. Before jointing, all joint contact surfaces shall be wire-brushed, if necessary, wiped clean, and kept clean until jointing is completed.

5.6.2.3.6 Culvert Installation Requirements

~~(d) — Installation:~~

~~(i) — Installation requirements:~~ Corrugated metal pipe (CMP) shall be laid true to the grade required by the drawings and shall be installed in accordance with the following requirements.

~~(A) Pipe: The pipe shall be installed in accordance with the details indicated on Figures 5-6 and 5-7 and the applicable portions of Chapter V, Section 3.~~ The pipe shall be protected from lateral displacement by means of an approved bedding material as specified for trench backfill. The minimum cover for corrugated metal pipe is twelve (12) inches.

(B) Sections of the corrugated metal pipe shall be joined together using metal coupling bands, centered on the joint, with the pipe sections as close together as possible. Each coupling band shall be bolted in place and tightened sufficiently to ensure a tight joint and to form a continuous conduit capable of resisting all stresses.

(C) Flared end sections: ~~The end sections~~ shall be attached to the culvert by threaded rod and connecting lug (for CMP).

~~(D) Rip-rap:~~ If required, culverts shall have a rip-rap bed of ten (10) feet by ten (10) feet at the inlet and outlet for erosion control. The rip-rap shall consist of hard, dense, sound, rough fractured stone as nearly cubic as practical and a minimum $d_{50} = 18"$. Slab type stones and flaking rock shall not be used. Alternate rip-rap size may be approved by County Engineer

provided developer's engineer provides supporting documentation or reference for a different size rip-rap.

5.6.3 Ice Build-up

A recurring problem on Park County roads is ice build-up in winter. Drainage design shall anticipate areas of potential ice build-up. Additional design considerations may be required in these areas.

5.6.4 Stormwater Detention Facilities

~~(I) Detention and retention:~~ Detention systems shall be designed to store the difference between the developed volume and the historic volume of runoff during the twenty-five (25) year event. Discharge from the detention system will be at twenty-five (25) year historic rates. Retention systems shall be designed to store the entire twenty-five (25) year twenty-four (24) hour event from the developed site.

5.6.5 Infiltration Trenches

~~(IV) Gravel Trenches:~~ Gravel trenches shall be designed to store the entire volume in excess of historic rates of runoff, and to allow this runoff to percolate into the soil. In designing gravel trenches, the assumption shall be made that gravel has twenty percent (20%) priority. The percolation rate of the soil shall not be slower than sixty (60) minutes per inch.

5.6.6 Subsurface Drainage

~~Subsurface drainage:~~ Subgrades subject to poor drainage, underground seepage or a high water table shall be adequately drained for roadbed stabilization. Drains shall be installed to prevent the high ground water level from coming within four (4) feet of the roadway ~~surface pavement~~. Perforated pipe shall be used to carry away collected water. Other drainage systems may be approved by the County Engineer. French drains which contain no pipe are unacceptable.

5.6.7 Drainage Infrastructure Construction and Maintenance Responsibility

~~(V) Facility capacity and maintenance.~~

~~(VI) Storm Flow Routing: As required to a defined and acceptable drainage way.~~

~~(VII) Other Methods: Other methods for controlling runoff may be approved or required by the County Engineer.~~

~~(D) Required Improvements:~~ Approval of any final plat shall include the requirement drainage improvements be constructed in accordance with these **RBS Standards**. The improvements shall be included as part of the subdivision improvement agreement and any financial guarantee **required by the DSR**.

~~(E) Responsibility:~~ All drainage detention, retention, filtration and facilities **constructed as part of site development** including, but not limited to, culverts or evaporation areas are the responsibility of, and shall be maintained by, the subdivision and/or the Homeowner's Association (**HOA**).

The developer/contractor is responsible for obtaining and complying with all applicable local, state and federal permits.

5.7 TRAFFIC SAFETY CONSIDERATIONS

According to the United States Department of Transportation (DOT), rural roads account for over 40% of U.S. traffic fatalities despite carrying lower traffic volumes. Safety challenges include roadway departures, high-speed limits, narrow lanes without shoulders, variable weather conditions, lack of lighting, unpaved drop-offs, and longer emergency response times.

Physical and infrastructures impact the number, type, and severity of vehicle related accidents pertaining to rural roadway systems. The absence of clear recovery zones, guardrails, or sufficient shoulder widths can contribute to roadway departures. Similarly, unsignalized rural intersections combined with high speeds create severe collision risks. Narrow lanes, unpaved edges, and steep drop-offs can adversely impact vehicle control. Sharp vertical or horizontal curves (See Section 5.4.1 and 5.4.2), overgrown brush, and lack of overhead lighting obscure hazards in many cases. The design, construction, and maintenance of the CRBS should consider these factors in the interest of improving traffic safety.

Driver behavior also impacts traffic safety on rural roads. Higher speeds increase crash severity. Agricultural vehicles, large machinery, livestock, and wildlife can create sudden blind spots or passing hazards.

5.7.1 Traffic Studies

~~Future planning:~~ Prior to the design of a new road, projections of future development and densities, estimates of **current and** future traffic volumes and appropriate classifications and design speeds shall be determined if required by the County Engineer. **For roads associated with development, the following factors will be considered by the County Engineer in determining when a traffic study is warranted.**

5.7.1.1 New Road Serving Residential Development

Applications submitted to the Planning and Zoning Department involving major subdivisions with 10 or more residential lots shall include a traffic study prepared by a Wyoming licensed professional engineer. The traffic study shall consider the volume, type, and characteristics of existing traffic in the area and the projected increase in traffic resulting from development. The traffic study shall include an assessment of existing infrastructure and the impacts the proposed development will have on infrastructure. The study shall also include an analysis of site distance, turning movements, and vehicle speeds with recommendations for physical improvements (signage, shoulders, turn lanes, deceleration lanes, etc.) to maximize traffic safety.

5.7.1.2 New Road Serving Industrial or Commercial Development

Applications to the Planning and Zoning Department for commercial/industrial development where a site plan is required (refer to DSR) or applications submitted associated with Special Use Permits (refer to DSR) shall include information regarding the estimated type and volume of daily traffic that will use the proposed development. This information will be used by the County Engineer to determine whether a traffic study is required. In the absence of actual traffic information, **TABLE 8** can be used to estimate the volume of traffic based on the type and scope of development contemplated.

Commercial/industrial development that is projected to generate 100 vehicles per day or more will require a traffic study. In addition, RV Parks or commercial/industrial development that is expected to result in more than ten (10) RV or ten (10) heavy truck/trailer combinations per day shall prepare a traffic study. Traffic studies shall be prepared by a Wyoming licensed professional engineer. The traffic study shall consider the volume, type, and characteristics of existing traffic in the area and the projected increase in traffic resulting from development. The traffic study shall include an assessment of existing infrastructure and the impacts the proposed development will have on infrastructure. The study shall also include an analysis of site distance, turning movements, and vehicle speeds with recommendations for physical improvements (signage, shoulders, turn lanes, deceleration lanes, etc.) to maximize traffic safety.

5.7.1.3 Design Period for Traffic Studies

~~(a) — Design period:~~ Roadway design shall be based on the projected needs twenty (20) years after construction. ~~(b) — Projected development:~~ Projections of development over the design period shall be based on zoning, existing land use, proximity to developed areas, historic growth and other factors, such as County or Municipal development plans which can be expected to influence development.

5.7.1.4 Projected Traffic Volumes

~~(c) — Projected traffic volumes:~~ ~~Table 5-2~~ **TABLE 8** illustrates traffic generated for various types of development. For example, residential property can be expected to generate an ADT count of ten (10) trips per unit. These per unit ADT counts are applied to the projected development to generate estimates of the design ~~year~~ traffic volumes. When per unit ADT counts are not listed for a type of development, or an ADT has not been established for a particular category or location by the County Engineer, the design engineer shall use an acceptable reference approved by the County Engineer such as the ITE “Trip Generation Handbook,” current edition, to obtain the appropriate ADT count.

TABLE 8 - PER UNIT AVERAGE DAILY TRAFFIC ESTIMATES

TYPE	UNIT	PER UNIT ADT
Residential	Per dwelling unit	10.0
Condominium/Townhouse	Per dwelling unit	7.0
Mobile Home Park	Per mobile home	5.0
Hotel	Per room	9.0
Restaurant	Per 1,000 S.F. gross	90.0
Commercial	Per 1,000 S.F. gross	115.0
Office	Per 1,000 S.F. gross	11.0
Campground	Per space	7.0
RV Park	Per space	7.0
Super Market	Per 1,000 S.F. gross	102.0
Other uses as referenced in the Park County Development Standards and Regulations, Table 2.1, Schedule of Uses		Per Unit ADTs shall be approved by the County Engineer

5.7.2 Site Distance

When conducting traffic studies or designing traffic systems to maximize safety, site distance is a critical consideration. Refer to Section 5.3.3 for specific site distance design considerations.

5.7.3 Establishing Speed Limits

Speed limits for unposted public roads are set by Wyoming State Statutes. Under W.S. 31-5-301, the default maximum speed limit for rural unpaved roads is 55 miles per hour, while paved two-lane highways and state/county roads default to 65 mph unless otherwise signed. The speed limit for divided multi-lane highways is 70 mph. Under W.S. 31-5-301(a), drivers must operate at a safe speed regardless of the posted limit, factoring in weather, reduced visibility, ice, loose gravel, or winding alignment.

Prior to posting a new speed limit or altering a speed limit for public roads including dedicated County Rods, a speed study must be conducted. Speed studies shall follow the guidelines provided in **APPENDIX T** and are intended to comply with WYDOT Chapter 30 procedures for establishing speed limits on local roads. All speed studies shall be approved by the County Engineer before new speed limits are posted.

5.7.4 Traffic Control Devices

~~e. —Traffic safety:~~

~~Traffic control devices:~~ All signs, striping, markers, delineators, signals and other traffic control devices shall conform to the requirements of the Manual on Uniform Traffic Control Devices, latest edition, hereinafter referred to as the MUTCD, published by the U.S. Department of Transportation, Federal Highway Administration. In new developments, all required street sign names, speed limit signs, stop signs and other traffic control devices shall be paid for by the developer, and installed by the County unless otherwise approved by the County Engineer. Non-standard signs or other traffic control devices are subject to rigid control and approval by the County Engineer shall be obtained for their use. All signing and striping plans shall be submitted to the County Engineer for approval.

5.7.4.1 Signage

Signs within subdivisions, except for road name signs, are to be maintained by the developer or homeowners. Typical placement for two-lane rural roads requires signs to be set 12 feet laterally from the edge of the traveled way, with a bottom height of 5 feet above the road surface. Sign spacing, location, sizes, colors (lettering and background), and materials shall conform to MUTCD requirements.

Regulatory signs (ie, Stop, Yield, or Speed Limits) are typically black-and-white or red-and-white and dictate legal obligations. Establishment of regulatory signs on public roads shall be designed by a Wyoming licensed professional engineer and shall require approval by the County Engineer.

Warning signs are typically diamond-shaped yellow-and-black signs alerting drivers to sharp curves, narrow bridges, merging lanes, or changes in road surface such as loose gravel or pavement edges. Guide and street name signs are used to assist emergency responders and for local traffic navigation. These signs are to have white lettering on green backgrounds. See **FIGURE 4** for typical County Road name signs and **FIGURE 5** for typical non-county road signs.

Private signs, commercial advertisements, and political campaign signs are strictly prohibited within public road rights-of-way including designated County roads. Unauthorized signs placed within a public road right-of-way will be confiscated by the Road & Bridge Division of the PCPWD.

5.7.4.2 Striping

Centerline striping is required for roads with an ADT of 1000 vehicles per day or greater and for roads with functional classifications that include Principal Arterials, Minor Arterials, and Major Collectors. At the County Engineer's discretion, striping on paved Minor Collectors or Local Access Roads (200 to 1000 vehicles per day ADT) may be considered in certain locations and in special circumstances where site distance and lane departure is a documented concern. Edge or road or shoulder striping will only be employed on roads that have an ADT in excess of 1000 vehicles per day (Major Collectors, Minor Arterials, and Principal Arterials).

Striping width, colors, materials, locations, and transitions shall conform to MUTCD standards.

5.7.4.3 Traffic Calming Devices (ie speed bumps)

Traffic calming standards may be employed at the discretion of the County Engineer to consider transition zones where high-speed roadways enter small towns or rural communities. Standard devices include driver feedback signs and transverse rumble strips. Transition zones may incorporate specialized signage or median islands to alert drivers that they are entering a community where slower traffic is necessary to minimize pedestrian conflicts.

Speed humps or speed tables are prohibited on public roads including dedicated County Roads without prior approval by the County Engineer. Rare situations where speed humps may be considered include school zones or bus loading/unloading zones that are associated with poor site distance. In no cases will speed humps be considered when the speed limit for the road section is greater than 30 mph based on an analyses of 85th percentile speed data.

When traffic calming devices are incorporated, the type of traffic (ie, agriculture, etc) and maintenance (ie, snow plowing) shall be considered.

5.7.5 Temporary Traffic Control Plan

Temporary Traffic Control Plans (TTCPs) for work zones must comply with the minimum standards outlined in the Manual on Uniform Traffic Control Devices (MUTCD), latest edition. At a minimum, TTCPs require clear design layouts specifying advance warning signs, transition tapers, protective buffer spaces, active work areas, and provisions for pedestrians, cyclists, and motorists. Minimum guidelines for traffic control in work zones for 2 lane roadways shall conform with WYDOT guidelines (See **APPENDIX R**).

A compliant temporary traffic control zone is divided into distinct sequential areas:

- **Advance Warning Area:** Tells road users what to expect ahead using regulatory, warning, or guide signs.
- **Transition Area:** Moves traffic safely out of its normal path using tapers and channelizing devices.
- **Activity Area:** Contains the actual workspace, traffic space, and buffer spaces separating workers from moving vehicles.
- **Termination Area:** Allows traffic to resume normal driving conditions past the site.

Only approved channelizing devices (cones, drums, barricades) shall be used. These include flashing arrow panels, and changeable message signs aligned properly and spaced closer within tapers. All traffic devices in the work zone shall be in new or good condition and be highly visible to motorists.

All workers in the work zone shall wear high-visibility safety apparel meeting ANSI/ISEA performance classes for all on-foot personnel, alongside positive protection barriers or shadow vehicles where high-speed risks exist. Project Foreman responsible for safety

in the work zone shall conduct routine day and night checks of all traffic control elements to maintain alignment, cleanliness, and device visibility. Devices shall be promptly removed as soon as work operations conclude.

5.8 DEAD END ROADS AND CUL-DE-SACS

~~(viii) Cul-de-sacs and turnarounds:~~ Using cul-de-sacs should be avoided. Where cul-de-sacs are the only alternative, turnarounds shall be provided. An alternative to the bulb type turnaround is the use of a hammerhead turnaround. Hammerhead turnarounds will only be allowed when, in the opinion of the County Engineer, a standard cul-de-sac is not practical. ~~Figures 5-2 through 5-4~~ **FIGURES 18 and 19** illustrate acceptable cul-de-sac ~~and hammerhead configurations.~~ Other configurations may be approved by the County Engineer. Whenever possible, roadway systems shall provide at least two (2) access points to lots platted for development.

The maximum length of roads ending in turnarounds shall be one thousand (1,000) feet. Adequate snow storage shall be provided to keep turnarounds clear. Dead end roads which do not have turnarounds are not allowed.

5.9 SITE AND ROADSIDE IMPROVEMENTS

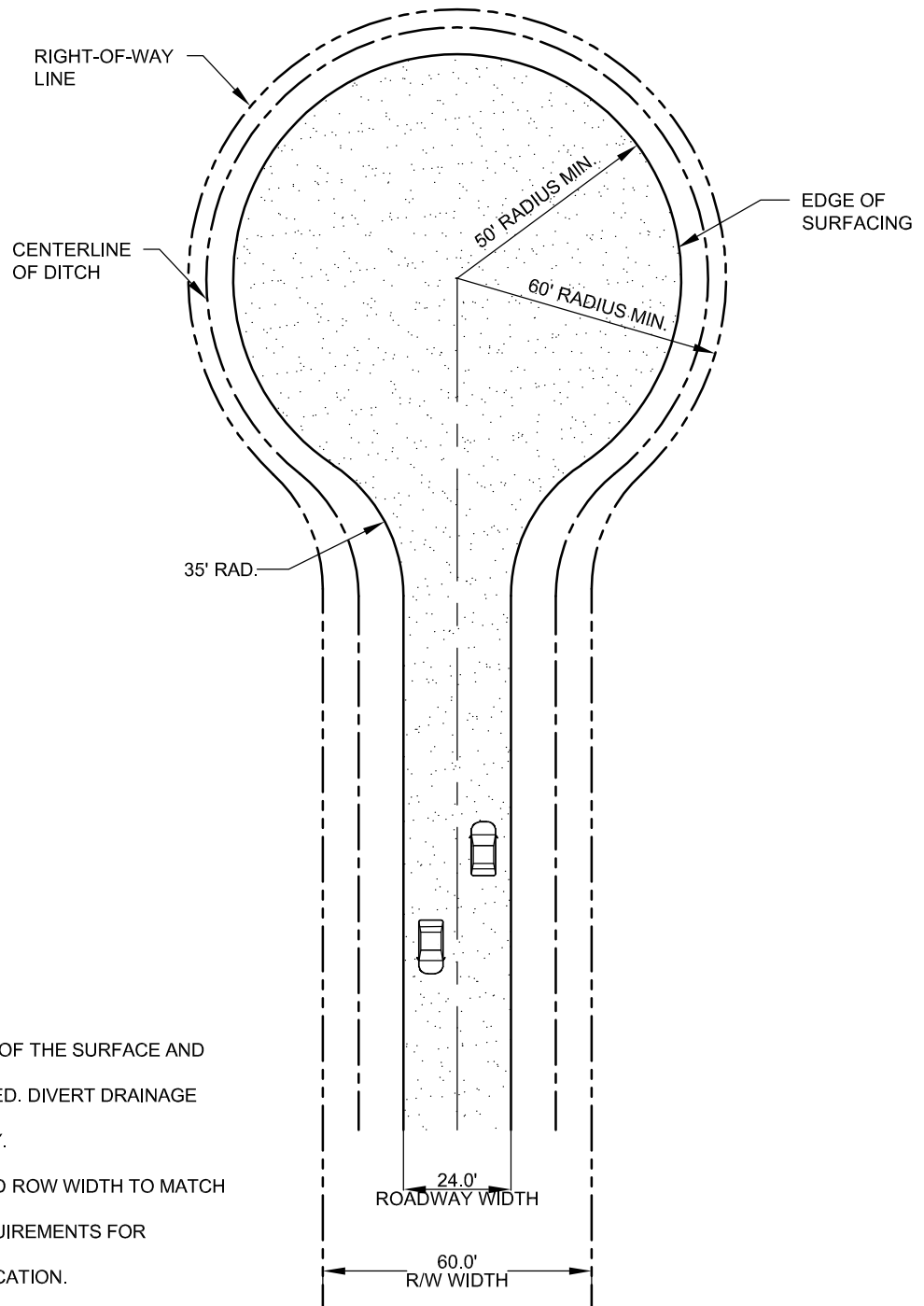
Site improvements for development and roadside improvements (when applicable and required) shall comply with the minimum design standards outlined herein. Refer to **FIGURES 1** and **2** for applicability of these design standards as it relates to residential and commercial/industrial development.

5.9.1 Site Grading Requirements

Site grading for commercial development shall ensure positive drainage away from buildings and parking areas. Similarly grading for roadside improvements shall ensure positive drainage away from the roadway and towards the stormwater collection network (series of ditches, berms, and culverts). Building pads and parking areas shall be slightly elevated compared to the natural terrain to accomplish positive site drainage. Site grading for commercial/industrial development that will be accessible by the public shall also consider ADA requirements as it relates to building pad elevations as well as walkway, and parking lot grades. Generally speaking ADA ramps and pathways shall have a maximum slope of 2%.

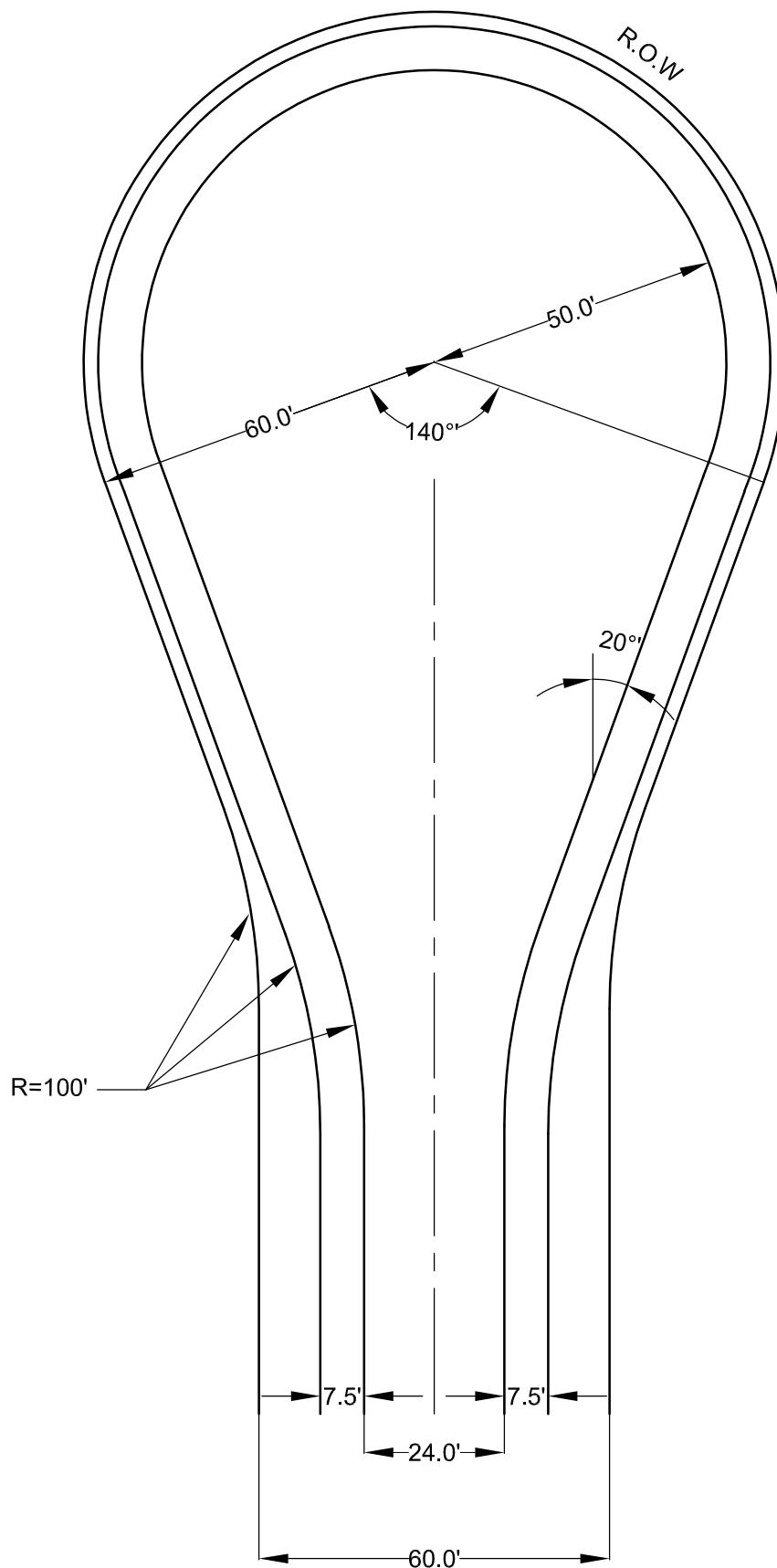
Unpaved areas shall be graded to a minimum grade of 2%. Paved areas shall have a minimum slope of 1%. Maximum slopes for earthwork shall be 3:1 (horizontal:vertical). In some cases, retaining walls may be incorporated into the site design in situations where it is not possible or practical to achieve the maximum slope requirements. Retaining walls higher than 4 feet shall be designed by a Wyoming licensed professional engineer.

STANDARD CUL DE SAC LAYOUT



NOTES:

1. ADEQUATE DRAINAGE OF THE SURFACE AND DITCH LINE IS REQUIRED. DIVERT DRAINAGE AWAY FROM ROADWAY.
2. ADJUST ROADWAY AND ROW WIDTH TO MATCH MINIMUM DESIGN REQUIREMENTS FOR FUNCTIONAL CLASSIFICATION.



A site grading plan is required for new roads or major improvements to roads that are expected to have an ADT of 60 vehicles per day or more. Similarly, site development will require a grading plan under one of the following circumstances:

- Development is expected to generate an ADT of 60 vehicles per day or more;
- Development will include a new structure or addition to an existing structure that has a building footprint of 10,000 square feet or more;
- Development will result in a site disturbance of 1 acre or more;

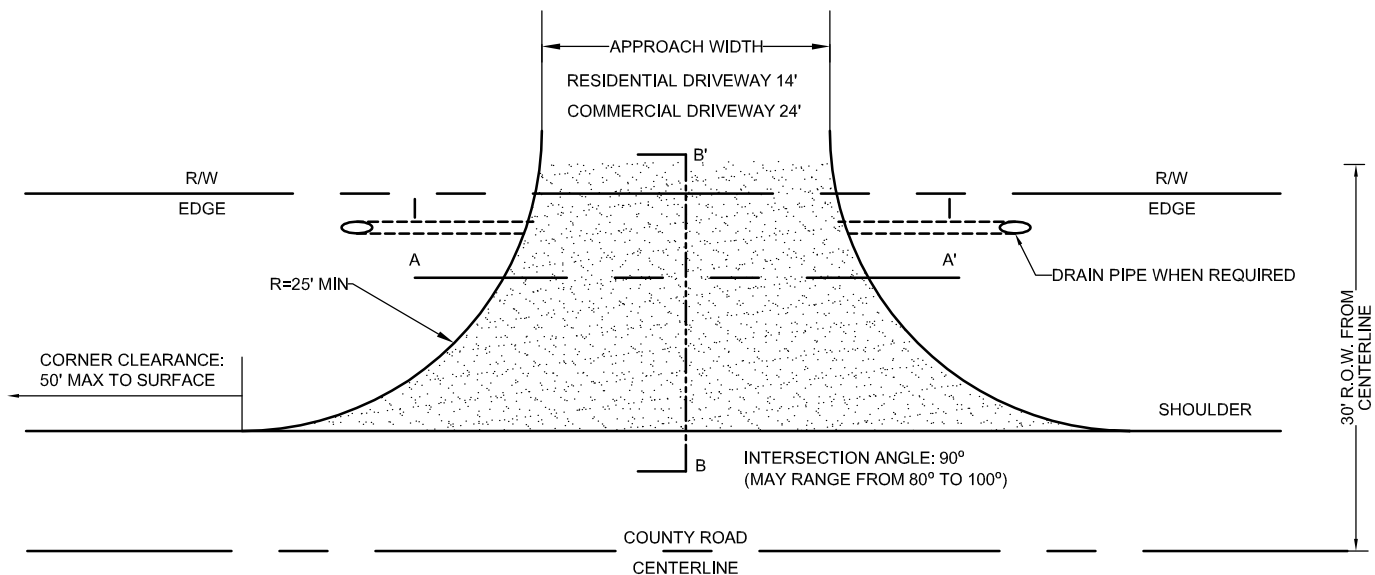
Site grading plans shall be prepared by a Wyoming licensed professional engineer and shall show existing contours and proposed contours. A minimum 2-foot contour interval shall be shown on the plan with all contours properly annotated. Grading plans show the horizontal and vertical extent (finished floor elevation) of any new structures or buildings. Grading plans shall also show all earthen embankments, drainage ditches, berms, culverts, etc. and shall show existing and proposed invert elevations. Grading plans shall also show property lines and existing utilities that could be impacted by site grading plans. Grading plans shall be signed and sealed by the engineer of record and shall be submitted to the County Engineer for review and approval.

5.9.2 Driveways, Accesses and Approaches

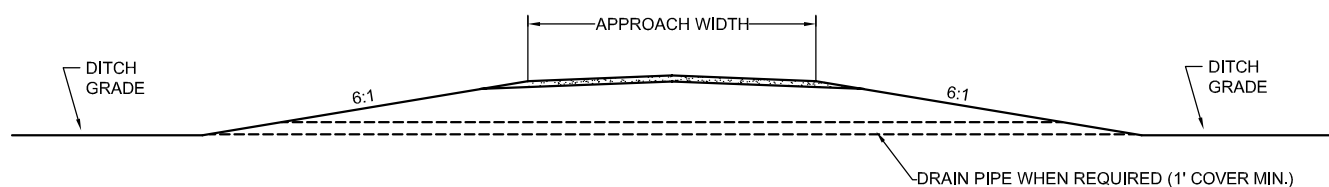
~~Driveways and parking areas:(1)—Requirement for ROW permit:~~ Whenever a property owner, developer, contractor or other individual proposes to connect a driveway or parking area to a public roadway, they must obtain a ROW Permit from the County Engineer prior to commencing construction. The submittal requirements and procedures for obtaining ROW Permits ~~outlined in Section 4.6 shall be followed. are outlined in stated in Chapter V, Section 4 of these Standards.~~

~~(2)—Standards for driveway design:~~ A driveway is an access for vehicles providing a connection from a public or private roadway to either individual parcels, residences, multiple residences, commercial businesses, recreational, institutional or industrial land uses or a combination of land uses. An access way serving a ranch or farm and any associated residence, regardless of length, shall be considered a driveway, and shall meet only such standards as are necessary for public health and safety. Maintenance of driveways, including but not limited to culverts, drainages and surfacing shall be the responsibility of the property owner.

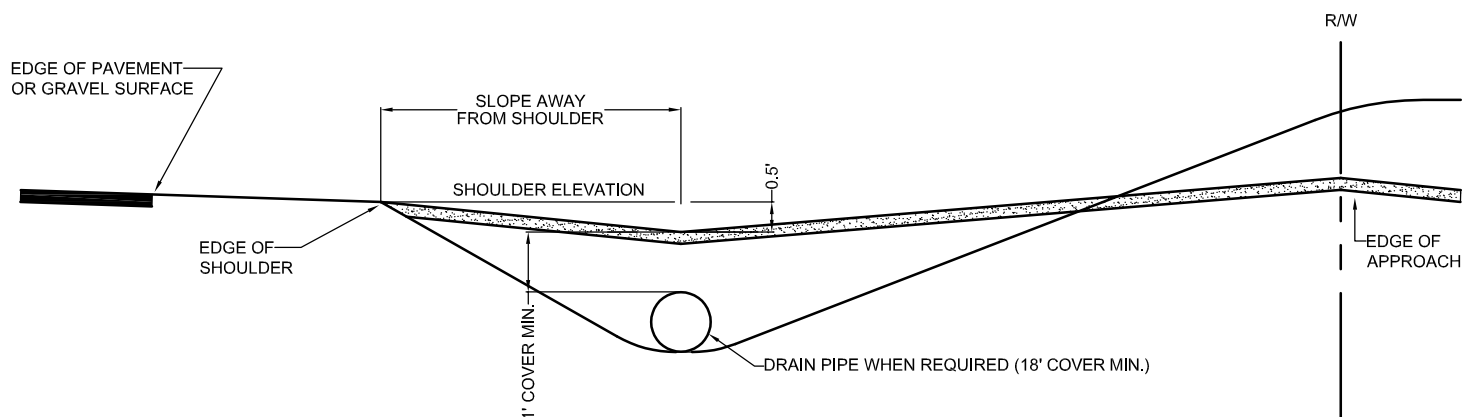
These ~~is~~ standards shall only apply to that portion of the residential driveway that is within two hundred (200) feet of the centerline of the public roadway. ~~Accesses to~~ ~~serving~~ uses, including multiple residences, commercial business, recreational, institution, industrial or a combination of land uses, shall conform to the minimum requirements outlined in **TABLE 9 and FIGURE 20 of these RBS.** ~~and these Standards.~~ The actual location, width and construction requirements are dependent on the use of the driveway(s), including circular and/or multiple accesses and will be reviewed and approved by the County Engineer during the ROW Permit process. ~~(d) Driveway widths: The dimensions of driveway widths, openings, and radii shall be as shown in Table 5-10.~~



TYPICAL DRIVEWAY APPROACH PLAN VIEW AND DIMENSIONS



SECTION A- SECTION VIEW ACROSS APPROACH



SECTION B- SECTION THROUGH CENTER LINE OF APPROACH

TABLE 9 – DRIVEWAY/CLEAR ACCESS SURFACE WIDTHS

TYPE OF SERVICE	MINIMUM DRIVEWAY WIDTH	RETURN RADIUS MAXIMUM
Commercial/Other	20 feet	*
Field Entrance	10 feet	10 feet
Individual Residence	10 feet	10 feet
Duplex or Multiple Residences	18 feet	10 feet
Multi-family	18 feet	10 feet

* To be determined at time of site plan review

Note: Actual driveway widths shall be provided on ROW Permit and shall be approved by County Engineer or designee prior to construction.

5.9.2.1 Driveway spacing from Road Intersections

~~(a) Location of driveways relative to intersections:~~ Driveways shall be placed so the following minimum distances are maintained to any street intersection, including a T-intersection on the opposite side of the street from a property where a driveway is proposed. **Also see Section 5.4.9 as it relates to road intersections.**

(i) Where the driveway connects to a **Low Volume Rural Road or Rural Road** ~~local access or low volume road~~, a minimum distance of fifty (50) feet from curve return to curve return shall be maintained.

(ii) Where a driveway connects to a Local Access Road, Collector (Minor or Major), or Arterial (Minor or Principal), ~~collector or larger road~~, a minimum distance consisting of the left turn stacking distance plus twenty (20) feet as measured from curve return to curve return shall be maintained. The left turn stacking distance shall be determined by the County Engineer based on available data from an acceptable traffic study.

5.9.2.2 Spacing of Driveways

~~(b) Spacing of driveways:~~ Driveway openings shall be separated by at least thirty (30) feet as measured from curve return to curve return, or else shall be combined. **More-Additional** spacing may be required for traffic safety and proper traffic operation.

5.9.2.3 Shared Driveways

~~(c) Shared driveways:~~ Developers or property owners proposing the use of shared driveways shall record an easement defining the location of the driveway and either a covenant or deed restriction requiring construction of the driveway at the location. **Shared driveways are encouraged when possible and practical.**

5.9.2.4 Driveway Grades

~~(e) — Driveway grades:~~ Driveways shall have a maximum grade of eight (8) percent, and shall provide a reasonable transition in terms of grade between the driveway and the roadway it joins over a distance of not less than twenty-five (25) feet. For single family residences and duplexes located on lots having difficult terrain, driveway grades may exceed eight (8) percent **if approved by** as long as approved by the County Engineer **and Fire Marshal**.

5.9.2.5 Vehicle Turnarounds

~~(f) — Vehicle turnarounds:~~ All driveways exiting onto roadways with average daily counts greater than **one hundred (100)** ~~five hundred (500)~~ VPD shall be designed to avoid vehicles having to back onto the roadway when exiting.

5.9.2.6 Driveway Surfacing

~~(g) — Surfacing of driveways:~~ Driveways serving single family residences or duplexes may be either graveled or paved. Where a driveway is to be graveled, the surface shall be constructed of six (6) inches of crushed road base. ~~compacted to ninety-five (95) percent standard proctor~~. Where a driveway is to be paved, the surface shall be constructed with a minimum of six (6) inches of crushed road base compacted to ninety-five (95) percent **of the modified proctor maximum dry density** and two (2) inches of asphalt. **Driveways serving multi-family residences or commercial uses shall be designed and constructed while adhering to the road surfacing requirements outlined in Section 5.3.4. in accordance with Chapter V, Section 2 of these Standards.**

5.9.2.7 Drainage Considerations for Driveways

~~(h) — Provisions for drainage:~~ Driveway design shall make adequate provision for drainage and prevention of erosion. All driveways shall have minimum eighteen (18) inch diameter culverts to handle roadside drainage unless otherwise approved by the County Engineer.

5.9.2.8 Site Distance for Driveways

~~Minimum sight distance:~~ Driveway shall be designed and located to provide a minimum sight distance clear of all obstructions, natural or man-made, for at least two hundred (200) feet in either direction on **Low Volume Rural Roads or Rural Roads**. ~~local access roads and~~ For Local Access Roads, Collectors, or Arterials the location and elevation of the driveway shall have a minimum site distance of four hundred (400) feet. **See FIGURE 16 for minimum site distance triangles for controlled intersections. on collector roads. See Figure 5-11.**

5.9.2.9 Driveway Access Approach to Public Road

~~(j) — Driveway Access Approach to Public Road:~~ Driveways shall be constructed to access perpendicularly to the public road. If perpendicular access to the public road is not feasible, then a request for waiver of the perpendicular

requirement shall be submitted in writing to the County Engineer for review and approval.

5.9.3 Parking Areas and Pullouts

~~(3) Standards for parking areas~~

~~(a) Parking area grades:~~ Parking areas shall have a maximum grade of four (4) percent, and a minimum grade of (1) percent to facilitate drainage. ~~(b) Surfacing of parking areas:~~ All parking areas that will serve the general public shall be constructed with a minimum of six (6) inches of crushed road base compacted to ninety-five (95) percent modified standard proctor. Paving of two (2) inches of asphalt will generally be required when fifteen (15) or more parking spots are established for commercial or industrial facilities.

Due to the frequency of use and specific function, the design and construction of parking lots comprised of fifteen (15) or more parking spaces shall be approved by the County Engineer. See **FIGURE 21** for typical parking lot guidelines.

~~(c) Provision for drainage in parking areas:~~ Parking area design shall make adequate provision for drainage and prevention of erosion (See Section 5.6).

~~(d) Placement of parking areas on fill:~~ If a parking area is to be placed on fill, the fill used shall be suitable material as specified by a registered professional geotechnical engineer. The fill shall be compacted to ninety-five (95) percent modified standard proctor with slopes at no more than three-to-one (3:1) and protected to prevent erosion. Parking areas on fill may be designed using retaining walls as an alternative in accordance with the County's Zoning Regulations, and approved with approval by the County Engineer.

5.9.4 Fences

~~c. Fence policy:~~ Park County does not install or maintain fences along its rights-of-way. Should a property owner wish to install a fence along the County rights-of-way, the fence must be installed in accordance with these RBS Standards. Maintenance of the fence shall be the responsibility of the property owner. Generally, fences are to be installed beyond the County rights-of-way. If a property owner requests fence to be installed within the County rights-of-way system, per W.S. §11-28-105, the BOCC Board of County Commissioners shall consider the request and recommendations of the County Engineer and/or Road and Bridge Foremen and may authorize the construction. The BOCC Board of County Commissioners shall be responsible for determination if owner or County shall be responsible for payment.

~~(1) Standards:~~

~~(a)~~ All fences within County rights-of-way shall be installed in accordance with WYDOT Standard Specifications, Section 607. Any deviations from these WYDOT Standard Specifications must be reviewed and approved by the County Engineer and the Road & Bridge Foremen. ~~(b)~~ All fences within County rights-of-way shall be designed, constructed and installed in conformance with all applicable sections of the RBS these Standards, including but not limited to, requirements to obtain a ROW Permit, sight triangle and distance restrictions and traffic safety requirements.

Fences installed or required in conjunction with development shall not restrict visibility for access roads or driveways in a manner that would impact traffic safety.

5.9.5 Cattleguards

~~d. — Cattle guard policy:~~ Park County maintains existing cattle guards on designated County maintained public roadways. ~~the County Road System.~~ If a property owner requests a new cattle guard within a designated ~~the County Road rights-of-way system,~~ the BOCC ~~Board of County Commissioners,~~ per W.S. §11-28-105, shall consider the request and recommendations of the County Engineer and/or Road and Bridge Foremen and may authorize the installation. The BOCC ~~Board of County Commissioners~~ shall be responsible for determination if owner or County shall be responsible for payment. The property owner is responsible for all fencing to connect to the new cattle guard.

~~(1) — Standards:~~~~(a)~~ All cattle guard installations shall conform to the following minimum standards:

Low Volume Rural Roads:	BLM or USFS Standard Specifications
Rural Roads:	BLM or USFS Standard Specifications
Local Access Roads:	WYDOT Standard Specifications
Collectors (Minor and Major):	WYDOT Standard Specifications
Arterials (Minor and Major):	WYDOT Standard Specifications

~~requirements shown on FIGURES _____, _____, _____. Figures 5-12, 5-13, 5-14 and 5-15.~~ Any deviations from these RBS ~~Standards~~ must be reviewed and approved by the County Engineer and Road & Bridge Foremen. All cattle guard installations must be designed, constructed and installed in conformance with all applicable sections of these RBS ~~Standards~~, including but not limited to, requirements to obtain a ROW Permit (See Section 4.6) and traffic safety requirements (See Section 4.6.14).

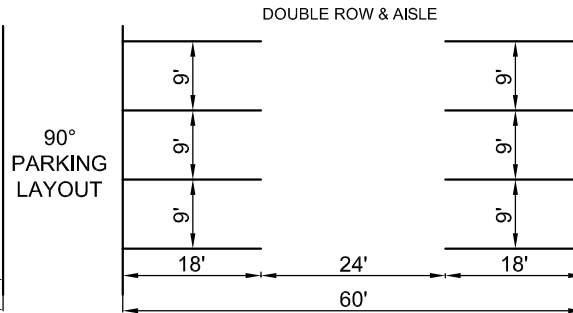
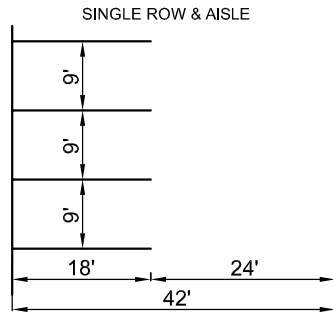
5.9.6 Mailboxes

~~b. — Mailbox policy:~~~~(1)~~ All mailbox installations shall require the approval of the U.S. Postal Service, and the County Engineer. All mailbox installations shall conform to the requirements shown on **FIGURE 22.** ~~Figures 5-9 and 5-10.~~ Any deviations from these requirements shall be reviewed and approved by the County Engineer and the Postmaster.

~~(2)~~ Mailboxes shall have pullouts constructed as a joint project of the box holder, Park County and the U.S. Postal Service. All mailboxes and turnouts must be designed, constructed and installed in conformance with all applicable sections of these Standards, including but not limited to requirements to obtain a ROW Permit (See Section 4.6), sight triangle and distance restrictions (See Section 5.7.2) and traffic safety requirements (See Section 5.7).

(a) ~~Mailbox~~**Box** holders may remove and replace their existing mailboxes and stands at their own expense, provided the design of their structure is not deemed a safety hazard by the U.S. Postal Service or the County Engineer and the required ROW Permit is acquired.

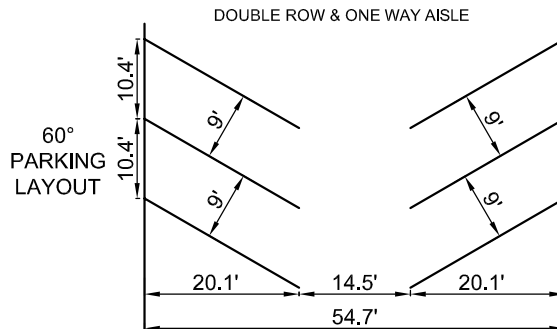
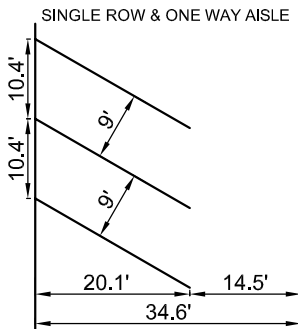
PARKING LOT LAYOUT DIMENSIONS



(a) *Space size.* The following minimum standards shall apply to the width and length of parking spaces.

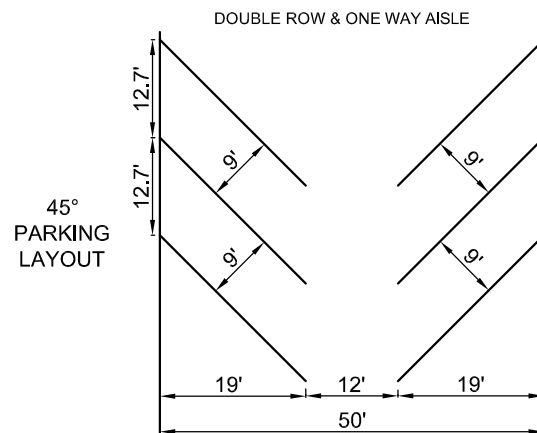
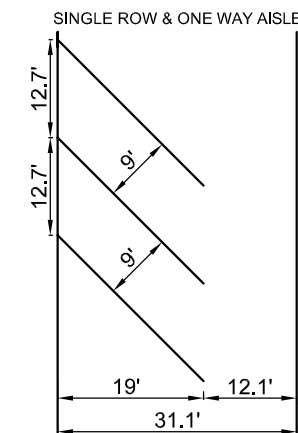
Type	Width	Length
Standard parking space	9 feet	18 feet*
Parallel parking space	8 feet	22 feet

* Parking spaces adjacent to landscape areas may project into the landscape area and be reduced to 16 feet in length when separated from the landscape area by curbing or approved wheel stops.

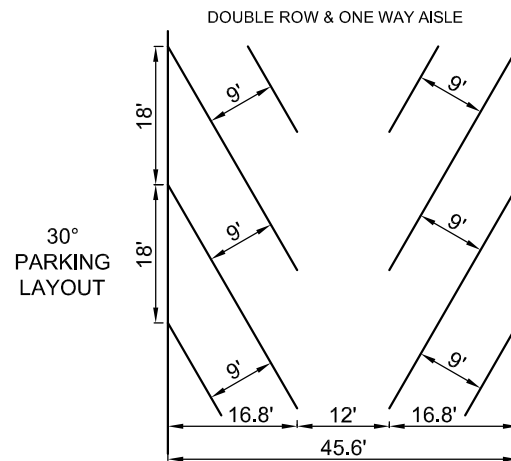
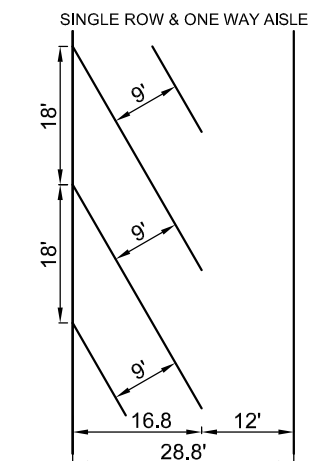


(b) *Angle parking size.* The standards for the minimum width of parking spaces plus the aisle are shown in the following table. These standards apply to a single row of head-in parking or two rows of head-in parking sharing an aisle. See the illustrations below.

Angle	Width: 1 Row Sharing Aisle	Width: 2 Rows Sharing Aisle
90 degree angle parking	42.0 feet	60.0 feet
60 degree angle parking	34.6 feet	54.7 feet
45 degree angle parking	31.1 feet	50.0 feet
30 degree angle parking	28.8 feet	45.6 feet



- (1) When parking lot driveways are less than 20 feet in width, marked separate entrances and exits shall be provided so that traffic shall flow in one direction only.
- (2) The location of ingress and egress driveways shall be subject to approval of the County Engineer.
- (3) Driveways for parking areas shall be maintained to allow access by emergency services vehicles.
- (4) Adequate room for maneuvering for longer vehicles, truck/trailer combinations, etc. shall be located completely off the right-of-way of any public road;
- (5) Parking areas shall have a crushed gravel base surface if the business, commercial, or industrial development is expected to service 60 vehicles per day but less than 200 vehicles per day. Parking areas that will service more than 200 vehicles per day for business, commercial, or industrial development shall be paved;
- (6) Any parking lots for commercial businesses that will be accessible to the general public shall comply with ADA requirements.
- (7) Parking space delineation (i.e., striping) is only required when the parking areas are paved;



DRAWN BY: T. Rozmelova
DESIGNED BY: B. Edwards
APPVD. BY: B. Edwards
SCALE: 1"=30'-0"
DATE: 7/29/2026
JOB NO.

REV. DATE BY

PARKING LOT LAYOUT DIMENSIONS
PARK COUNTY R&B STANDARDS

Park County

Wyoming

FIGURE

21

FIGURE 22- MAILBOX STANDARD DETAILS

(Insert in place of this page)

~~(b)~~—When Park County has a road reconstruction, repair or upgrade project, the ~~PCPWD Road and Bridge Foremen~~ will be responsible for pullout construction, notification of box holders and scheduling of construction. This does not apply to subdivision mailboxes within the subdivision **which are the responsibility of the individual private landowners.** **—Mailbox banks or clusters shall be consolidated to the extent possible and practical.**

~~(3)~~A mailbox damaged by the impact of plowed snow or ice shall be replaced or repaired by the mailbox owner and at the owner's expense. When a mailbox is hit by a plow and damaged, it will be replaced by the County.

5.9.7 Utilities

~~f.~~ **—Utility company facilities:** Park County recognizes the need and importance of public utility facilities to be placed within the County rights-of-way. These facilities may create hazards or be detrimental to the purpose and function of the County road system. It is therefore the policy of Park County to locate these facilities in accordance with the following standards.

~~(1)~~ **—Standards:** All utilities shall be installed in accordance with the plans and specifications approved by the utility owner and the County Engineer. Where applicable, the plans for installation must bear the name, seal and signature of a registered Wyoming Professional Engineer responsible for their preparations. The alignment of all utilities within County rights-of-way and major floodways is subject to approval by the County Engineer. All alignments of utilities within special flood hazard areas must have a floodplain permit.

5.9.7.1 Underground Utilities

~~(a)~~ **—Underground utilities:** All accesses to underground utilities from the road surface (e.g. manholes, vaults) shall be of heavy-duty construction made of cast iron, capable of safely supporting anticipated maintenance equipment and vehicular traffic and a minimum AASHTO HS-20 loading. No aluminum castings will be allowed. Concrete collars in conformance with the WPWSS standard drawings are required.

All valves, manholes, vaults or other appurtenant structures located within the rights-of-way shall not interfere with the County's use of the rights-of-way and shall be buried a minimum of twelve (12) inches, except in paved areas. In paved areas, these appurtenant structures shall conform to the finished grade of the road.

Seep plugs shall be installed in trenches used for underground utilities at no less than five hundred (500) foot intervals if the possibility exists that the surrounding water table will be lowered and this will have an adverse effect on surrounding wells and vegetation dependent on the water table elevation.

All non-metallic facilities shall have tracer wire installed with the facility. Minimum wire shall be #12AWG, solid copper. Park County accepts no responsibility for facilities not adequately marked or buried.

5.9.7.2 Aboveground Utilities

~~(b) — Aboveground utilities:~~ All aboveground utilities shall be located and installed so as not to cause unnecessary obstruction to pedestrian and vehicular traffic or damage to the utility itself. No pole or structure above ground shall be placed within a pedestrian walkway nor set closer than twelve (12) feet to the shoulder of any County road. A lesser distance, however, may be allowed if insufficiently cleared right-of-way is available to meet this minimum distance. In no case will a pole, guy and/or anchor be permitted within twelve (12) feet of the shoulder of a County road except light and traffic control poles with breakaway bases. Overhead lines shall be permitted to sag under worst anticipated conditions no lower than eighteen (18) feet above the roadway and shall preferably be no closer than twenty-three (23) feet.

~~(c) — Utilities in major floodways:~~ All utilities within or adjacent to major floodways shall comply with the *Park County Floodplain Regulations* and other applicable floodplain regulations and shall be located and installed in a manner that will prevent objectionable damage such as land erosion, water pollution or flood diversions.

5.9.7.3 Utilities Relocation

~~d) — Changes affecting utilities:~~ Future changes to County roads may require the relocation or removal of utility installations. For minor changes, the affected utility company shall complete the relocation or removal within thirty (30) days after notification by the County Engineer. For major utility relocation projects involving extensive design and securing of contracts or material orders, the affected utility company shall complete the relocation or removal within ninety (90) days, or a time frame as mutually agreed upon, with approval from the County Engineer of the final design. ~~To avoid the necessity for such changes, u~~Utility companies are encouraged to locate their facilities consistent with future plans for County roadways **to avoid the necessity of such changes**. Any removal and/or relocation of utilities within the County rights-of-way shall be the expense of the Utility.

5.9.7.4 Road Cuts vs. Borings

Road cuts are discouraged by PCPWD on paved roadways. All paved roadways shall be bored unless a bore creates an unsafe or hazardous condition. If, after the contractor attempts a bore, and the bore is shown to be physically impossible using standard boring techniques, the County Engineer may allow an open cut. Open cuts will only be allowed when the County Engineer determines that a bore is impossible based on a field inspection of the bore location.

If an open cut is allowed, the contractor, at his own expense, shall meet the following minimum criteria:

- i. Submit a Traffic Control Plan for approval **by PCPWD (See 5.7.5);***

- ii. Install and maintain all traffic control devices prior to the start of and during construction;
- iii. Work continuously to complete the work as expeditiously as possible;
- iv. Trench backfill shall be ~~completed~~ ~~made~~ with ~~WyDOT WPAWSS~~ Grading "W" crushed base compacted to ninety-five (95) percent of ~~the soil maximum dry density based on a Modified Proctor Standard Proctor~~ at $\pm$ two (2) percent of optimum moisture in accordance with ~~ASTM D-698 D-1557~~ or with an approved lean concrete slurry. Compaction by flooding the trench shall not be allowed; and
- v. Any other requirements deemed necessary by the County Engineer and/or the Road and Bridge Foremen.

~~To maintain a reasonable uniform road surface appearance,~~ The County Engineer may require a chip seal over the area of disturbance plus ten (10) feet on either side if deemed necessary to maintain a uniform road surface. Should this be required and the contractor is unable to chip the area due to weather or other causes, the contractor may bond the work, or a cost shall be determined by the County Engineer and invoiced to the contractor upon completion of the chip surfacing by the County.

5.9.8 Irrigation Systems

~~e. — Irrigation facility policy:~~ Park County recognizes the need and importance of irrigation facilities. However, these facilities may create hazards or be detrimental to the purpose and function of the County road system. It is therefore the policy of Park County to locate irrigation facilities outside of County rights-of-way whenever possible.

~~(1) — Irrigation facilities within County road rights-of-way:~~ Irrigation facilities which must be within the County rights-of-way shall be constructed to reduce impacts to the road system and eliminate hazards to the traveling public. At a minimum, the following practices shall not be allowed within the County rights-of-way:

- i. New supply and waste ditches;
- ii. New diversion structures; or
- iii. Open or unlined channels.

Park County will work cooperatively with irrigators to resolve conflicts with new and existing facilities.

~~(2) — Irrigation facilities within subdivisions:~~ Subdivision irrigation improvements shall conform to the requirements of the State Engineer's Office and any applicable irrigation district or ditch company. Construction Plans for all irrigation facilities required for the development shall be submitted with all other construction plans.

5.9.9 Signage

Site signage for development shall conform to the minimum requirements outlined in the DSR. Signage pertaining to public traffic control including driveways and roadways shall conform to **MUTCD Standards**, latest edition (See also 5.7.4.1).

5.9.10 Above Ground Obstructions

In developing a comprehensive site plan, above-ground obstructions should be identified and evaluated to determine if they adversely impact site use or maintenance. If necessary, obstructions that hinder site use, maintenance, or visibility (site distance) should be relocated or removed. Possible above ground obstructions include but are not limited to utility poles, trees, fences, and structures.

During initial site preparation, dead trees, old foundations, and failed structures should be removed. If utility relocation is necessary to move phone pedestals, transformer boxes, or power poles, work shall be coordinated with the appropriate utility service provider. Walkways shall be clear and free of protruding obstructions which could represent a tripping hazard.

5.9.11 Sidewalks

Sidewalks are generally not used or required for rural road or site development in unincorporated areas of the County. However, sidewalks may be considered in transition zones to urban or suburban areas. In addition, depending on the size and scope of the development, sidewalks may be incorporated into the site plan at the developer's discretion. When sidewalks are incorporated in the design of a roadway or commercial/residential development that will be accessed by the public, the following design standards shall be incorporated in the design.

Commercial and industrial sidewalk specifications for development accessible to the public requires a minimum clear width of 5 to 8 feet, a concrete thickness of 4 inches standard, a maximum 2% cross slope (1/4" per foot), and ADA accessibility compliance. The width should be increased to 6 to 8 feet near larger parking areas or adjacent to multi-tenant complexes. The thickness of the concrete should be increased to at least 6 inches and incorporate welded wire fabric in areas where service vehicles or light trucks cross the walking path.

ADA ramps along concrete sidewalks shall have a maximum slope of 12:1 (8.33%) and shall have landing strips or transitions that are at least 5' x 5'. ADA accessibility requires a minimum 36-inch-wide continuous unobstructed path from the handicap vehicle parking spot to the building entrance. ADA compatible curb ramps and crossings are required for transitions from parking lot grades to elevated pathways.

In areas where there is considerable traffic or heavy truck traffic, it is recommended that a buffer strip be incorporated into site design to protect pedestrians. When buffer strips are used, the minimum distance from the back of the curb (when used) or roadway surface to the sidewalk shall be 4 feet. Although not required, buffer zones should be landscaped when possible and practical. Industrial site plans must isolate pedestrian paths from heavy truck routes, loading docks, and large vehicle turning aprons.

When sidewalks are constructed, transverse contraction joints shall be incorporated at regular intervals equal to the sidewalk width (typically every 5 to 6 feet) to control cracking.

5.9.12 Curb and Gutter

Curb and gutters are not required for any of the road functional classifications in the CRBS as it relates to unincorporated Park County. However, in certain situations, curb and gutters may be considered in transition zones to urban or suburban areas. In addition, depending on the size and scope of the development, curb and gutter may be incorporated into the site plan at the developer's discretion. When curb and gutter are incorporated in the design of a roadway or commercial/residential development that will be accessed by the public, the following design standards shall be adhered to.

Standard curb and gutter design specifications for site development require a monolithic concrete mix with a minimum compressive strength of 4,000 psi. A standard vertical or barrier curb features a 6-inch top width, a 7-inch reveal above the flow line, and an 18 to 24-inch gutter pan width. The curb/gutter shall be placed over a 4- to 6-inch-thick compacted aggregate base. The gravel base shall be compacted to at least 95% of the material maximum dry density based on a modified proctor. The gutter shall slope to the nearest drainage inlet at a rate of 1/8 to 1/4 inch per foot. Gutter and pavement interface must maintain a minimum 0.5% to 2% slope to prevent standing water or ponding.

Concrete shall be air entrained (5-8% air content) for maximum freeze-thaw resistance. The curb and gutter must be cast simultaneously as a single unit without a cleavage plan. Expansion joints shall be spaced at 10 to 30 feet intervals and contraction/control joints shall be placed every 10 feet. Expansion joints must align directly with adjacent sidewalk or pavement joints when applicable.

Rolled/mountable curbs shall be used in residential subdivisions (when curbs are used) to allow smooth low-speed vehicle transitions into driveways without specialized curb cuts. Barrier or vertical curbs should be reserved for commercial/industrial sites in situations where it is necessary to restrict vehicular movement and protect pedestrians.

5.9.13 Landscaping & Seeding

Landscaping: Whenever roadway or bridge construction or site development results in earth disturbance of 1 acre or more, revegetation and reforestation is required and shall be completed during the first planting season after construction. Native or similar horticultural materials shall be used. All areas disturbed by construction operations not otherwise covered by structures or pavement must be seeded, fertilized, mulched, planted and otherwise treated to provide an established stand of vegetation in accordance with WPWSS Standards. Cut and fill slopes must be treated to prevent erosion. Areas not disturbed by construction shall be left in their present vegetative state, except the thinning of trees may be required.

~~h. —Landscaping installations:~~ Earth-cuts, embankment slopes and all other areas where the ground cover has been disturbed during the course of ~~road~~ construction shall be ~~well~~ revegetated and reforested equal to or better than conditions existing prior to construction. Landscaping material shall be installed in accordance with plans approved as part of any ROW permit and/or approved construction plans, and shall be fertilized, mulched, watered and otherwise treated to provide an established stand of vegetation by the end of the first full growing season after completion of construction. All

Landscaping shall conform to these Standards and the WPWSS requirements for Landscaping.

5.9.14 Erosion Control

Road construction and/or site development resulting in a disturbance of 1 acre or more requires that a site-specific drainage and erosion control plan be prepared (see 5.6.1). The plan should incorporate Best Management Practices (BMP) including physical controls to minimize the potential for soil loss from the construction site.

Structural BMPs like silt fences, sediment basins, and stabilized construction entrances shall be installed before heavy grading begins. In order to minimize the potential for soil loss during construction, phasing should be considered when possible and practical. This ensures only active work zones are cleared to minimize the amount of area disturbed (and exposed to the elements) at any given time.

Depending on the size and scope of the development, BMP to minimize the potential for erosion should consider one or more of the following:

- *Perimeter Controls: Install silt fences or filter barriers along downgradient boundaries to intercept sheet flow.*
- *Access Stabilization: Build gravel-stabilized construction entrances to prevent mud tracking onto public roads.*
- *Inlet Protection: Place fabric or manufactured barriers around catch basins and storm drains.*
- *Runoff Diversion: Construct temporary earth dikes or swales to divert clean upslope water around disturbed areas.*
- *Temporary Cover: Apply straw mulch, wood chips, or erosion control blankets on idle slopes within required local timeframes (often 7 to 14 days).*
- *Permanent Vegetation: Establish permanent turf or native ground cover reaching over 70% density to achieve final site stabilization.*

BMPs should be inspected weekly and after every significant rain event, removing trapped sediment and repairing damaged barriers. The Water Division of WDEQ shall be consulted regarding sample or template storm water pollution prevention plans (SWPPP) and guidelines for installing and maintaining BMP.

5.9.15 Guardrails

Guardrail systems, when required shall follow rural highway guidelines developed by the FHWA and AASHTO. WYDOT design and construction details and standard specifications shall be incorporated into site specific construction plans when guardrail systems are used (See **APPENDIX X**).

In most cases involving public roads including dedicated County Roads the Midwest Guardrail System shall be used. The Midwest Guardrail system incorporates a standard 31-inch rail height, 12-gauge or 10-gauge W-beam or box steel railing (AASHTO M180 compliant rail elements), wood or steel posts embedded in compacted soil, and

crashworthy end terminals to safely redirect vehicles. The 31-inch rail height allows for future overlays, allowing a minor tolerance down to 28 inches.

Guardrail (when required) shall be positioned horizontally roughly 2 feet beyond the outer shoulder edge, provided the roadside slope is stable (flatter than 6:1). Typical guardrail installations run a minimum of 175 to 200 feet (including end treatments) to develop proper tension and anchorage with at least 50 feet extending before the hazard.

Non-interstate rural roads shall be designed based on Test Level 2 (TL-2) or Test Level 3 (TL-3) systems, rated for passenger cars and pickup trucks traveling at designated rural design speeds. End terminals shall be energy absorbing or flared breakaway terminals at both the approach and trailing ends to prevent spearing or vaulting during head-on impacts.

5.9.16 Lighting

Site development lighting requirements (refer to DSR) dictate the minimum illumination, fixture shielding, and maximum power density needed for safety, visibility, and compliance. Standard requirements emphasize using full-cutoff or fully shielded fixtures to prevent light trespass onto neighboring properties, alongside specific foot-candle levels for walkways, parking lots, and active work zones.

Unless otherwise specified in the DSR, the following general lighting guidelines shall be incorporated into the site design:

- *General Work/Construction Areas: Minimum of 5 foot-candles (fc) at ground or floor level.*
- *Walkways, Stairs, and Passageways: Minimum of 2 to 5 foot-candles to ensure safe navigation and exit path visibility.*
- *Loading and Storage Platforms: Minimum of 3 to 5 foot-candles depending on active material handling.*

Luminaires shall be used designed to direct light downward, ensuring zero upward transmission of light into the night sky. Fixtures must be fully shielded so the bare bulb or light source is not visible from adjacent properties or public rights-of-way. Illumination spilling onto adjacent parcels typically cannot exceed 0.1 to 1.0 foot-candle at the property boundary unless otherwise specified in the DSR.

5.9.17 Multi-use and Recreational Pathways

Multi-use and recreational pathways require planning for width, grade, and accessibility. Standard AASHTO guidelines recommend a minimum 10-foot width for two-way traffic, scaling up to 12–14 feet for heavy-use corridors, with maximum sustained slopes generally kept under 5% to ensure safety and ADA compliance. 10 feet is the baseline minimum for two-way multi-use paths; 12 to 14 feet is preferred for high-volume commuter or recreational areas. An 8-foot width may be used in restricted spatial conditions on a case-by-case basis.

Multi-use and recreational pathways shall have a minimum vertical clearance of 8 feet for pedestrians and bicyclists, increasing to 10 feet under overpasses or tunnels for

emergency and maintenance vehicle access. When possible and practical, a 2-foot clear shoulder or graded zone on each side of the path should be maintained.

When possible and practical, pathways shall have a continuous grade of less than 5%. Steeper pitches (up to 8.33% or 1:12) are permissible only for short, regulated distance ramps up to 200 feet. Maximum cross slope of 1% to 2% shall be considered to allow proper surface drainage without destabilizing wheelchairs or cyclists. When possible pathways shall be separated from roadways by at least 5 feet. Road crossings for pathways including striping, signage and alert systems shall comply with MUTCD Guidelines, latest edition.

Surfacing for multi-use and recreational pathways can vary depending on the planned use. Asphalt and concrete surfacing are preferred for heavy commuter and all-weather use. Stabilized crushed aggregate is suitable for permeable, eco-sensitive environments.

5.10 BRIDGES AND LARGE CULVERTS

5.10.1 Bridge and Large Culvert Design

~~d. —Bridge design:~~

~~(1) —Design standards for bridges:~~ Design of bridges ~~Bridges~~ shall conform to AASHTO *Standard Specifications for Highway Bridges, latest edition*. Plans and a design report shall be prepared by a Wyoming licensed structural engineer and shall be submitted to the County Engineer for review and approval prior to construction. Clear deck width, at a minimum, must accommodate the full width of the travel lanes of approach roads., ~~as indicated in Table 5-3.~~

5.10.1.1 Design Loading

For standard bridges accessible to the public, the required design live loading is the HL-93 model specified by AASHTO (American Association of State Highway and Transportation Officials). It combines a 72,000-lb design truck (or 50,000-lb design tandem) with a continuous 640-lb/ft design lane load. ~~The design loading requirements shall conform, at a minimum, to AASHTO HS20-44 specifications.~~

Bridge design requires accounting for multiple distinct categories of loads to ensure safety and structural integrity over a long service life. The design engineer shall consider various loading combinations using Load and Resistance Factor Design (LRFD) methods to test the structure against worst-case scenarios. These loadings should include the following at a minimum:

- *Dead Load: The permanent weight of the bridge structure itself, including girders, decks, barriers, and future wearing surfaces.*
- *Live Load (HL-93): Moving vehicular loads, which include a combination of design trucks/tandems and uniform lane loads placed to create maximum stress.*
- *Dynamic Load Allowance (Impact): An added percentage increase applied to live loads to account for the bouncing or vibration of vehicles moving over the deck.*

- *Environmental Loads: Forces from wind pressure, thermal expansion and contraction due to temperature changes, and seismic activity where applicable.*
- *Special Forces: Longitudinal braking forces, centrifugal forces on curved bridges, and hydraulic pressures from water currents or ice loads for substructures in rivers.*

5.10.1.2 Hydraulic Design

The waterway area shall accommodate a one hundred (100) year frequency storm, unless otherwise specified by the County Engineer. Where flood studies from the U.S. Army Corps of Engineers or the Federal Emergency Management Agency are available, bridges shall be designed to accommodate the "Standard Project Flood". A minimum of two (2) foot of freeboard is required. Additional freeboard shall be required when debris laden flows are anticipated.

5.10.1.3 Scour Protection

Bridge scour protection guidelines and specifications shall be incorporated to mitigate the potential for loss of streambed material around bridge foundations caused by water. Calculations for scour potential must consider "total scour" which includes long-term degradation, contraction, and local scour. Design calculations must evaluate the scour potential associated with a 100-year flood for standard risk scenarios and the 500-year flood (when possible) as a check to determine maximum potential scour depth.

Deep foundation elements like shafts and piles must extend significantly below the calculated total scour depth. Countermeasures must be sized so that the local flow velocity cannot dislodge the armor material.

Standard rock riprap sizing and gradation requirements (such as Class III or larger) must be placed with a proper underlying geotextile filter fabric to prevent fine soil migration. Articulated Concrete Blocks (ACB) consisting of concrete block mats can be used to offer a high-performance, interlocking smooth surface alternative to loose riprap. Gabions and Reno Mattresses consisting of wire mesh containers filled with graded stone can be used to reinforce revetments on sloping banks or stream channels.

5.10.1.4 Bridge Deck

Bridge decks shall be designed using AASHTO LRFD Bridge Design Specifications based on HL-93 design vehicular live loading. Concrete bridge decking shall have a 28-day compressive strength of 4000 psi and a typical slab thickness of 8 inches (including 0.5 inch wearing surface). Minimum clear cover shall be at least 2.5 inches for top reinforcement and 1.0 inch for bottom reinforcement to prevent steel corrosion. Steel reinforcement shall be epoxy-coated or galvanized.

Bridge deck clear widths shall be at least 16 feet wide for single lane bridges and 20 feet wide for two-lane bridges.

5.10.1.5 Bridge Railing

Bridge railing design specifications are governed primarily by AASHTO LRFD Bridge Design Specifications. Bridge guardrails require safety performance crash testing (such as TL-3 or TL-4 via MASH or NCHRP Report 350). Hardware must successfully transfer impact forces into the deck reinforcement without causing vehicle rollover, snagging on vertical posts, or deck punching shear failure.

The following minimum bridge guardrail heights shall be incorporated into the design of the bridge:

- *Vehicular Traffic Rails: Typically range from 27 to 32 inches minimum height for standard passenger cars, while heavy truck-rated barriers (TL-4 and higher) require a minimum height of 42 inches.*
- *Pedestrian Railings: Minimum height is 42 inches from the top of the walkway. Opening limits dictate that a 6-inch sphere cannot pass through the lower 27 inches.*
- *Bicycle / Equestrian Railings: Minimum height increases to 54 inches, with tighter upper-region spacing (an 8-inch sphere maximum opening rule).*

5.10.2 Bridge and Large Culvert Materials

Bridge material specifications cover the technical requirements for structural components like concrete, steel, and timber used in construction. Governed by agencies like AASHTO and the Federal Highway Administration, standards dictate material strength, composition, and testing methods to ensure public safety and longevity. Acceptable materials for bridge construction include the following:

- **Concrete:** *High-strength mixes, low water-cement ratios, and special admixtures for pre-stressed or reinforced superstructures.*
- **Structural Steel:** *Grades with specified minimum yield strengths (such as 36,000 psi or higher) used for girders, trusses, and plates.*
- **Reinforcing Steel:** *Deformed carbon-steel bars satisfying strict elongation and cold-bend criteria.*
- **Timber & Composites:** *Processed wood or lightweight fiber-reinforced polymers utilized for specific load or pedestrian applications.*

Bridge hardware and fasteners shall be designed and sized to provide a rigid fixed connection between bridge members and shall be corrosion resistant.

5.10.3 Bridge Construction Cost Responsibility

~~(2) — Payment of bridge construction costs:~~ If the design of the roadway serving a new development requires construction of new bridges or upgrading of existing bridges, the developer shall be required to pay the cost of such construction. Where construction of a bridge benefits future developments or cures a safety hazard affecting more than the proposed development, the Board of County Commissioners may establish a plan of compensation to the original developer whereby other beneficiaries pay a fair share for use of the bridge.

5.11 GENERAL CONSTRUCTION REQUIREMENTS

~~SECTION 3: — Road and Bridge Construction Specifications~~

~~a. Purpose and intent:~~ This section sets forth specific standards for roadway and bridge construction in Park County, and is intended for use by developers, property owners, contractors, utilities and others engaging in construction of new roads, upgrading of existing roads, building of bridges and other construction activities within the County road rights-of-way.

5.11.1 Construction Plans and Technical Specifications

Construction plans and technical specifications for new roads or major improvements to existing roads for which these design and construction standards are applicable (See Section 2.1) shall be prepared under the direct supervision of a Wyoming licensed professional engineer. Plans shall include the minimum information provided in **APPENDIX H** and shall be reviewed and approved by the County Engineer before initiation of construction.

5.11.2 Construction Contract Administration

When construction projects involve a dedicated public road right-of-way or a dedicated County Road, the engineer of record shall administer the procurement or a qualified contractor through a competitive bid process. The engineer shall ensure that the construction contract is administered and all requirements in the bid and contract documents are met. This includes but is not limited to:

- Advertisement for request for bids;
- Review of bids;
- Contract award;
- Review and administration of construction contract;
- Review and approval of contractor submittals;
- Review and ensure bonding requirements (if any) are met;
- Review and approval of contractor pay estimates;
- Address Contractor Requests for Information (RFI);
- Review and approve contract change orders;
- Administer Final Project Closeout
 - Certificate of Substantial Completion
 - Certificate of Final Completion
- Ensure warranties are in place (if applicable);

5.11.3 Surveying and Construction Staking

Site development construction staking is necessary to translate approved engineering plans into physical three-dimensional points on the ground. This can involve initial boundary and site control setup, setting rough and fine grading stakes, precise building or foundation layouts, and underground utility alignment. For road systems construction staking involves establishing horizontal and vertical alignment, edge of road surface, location and elevations for drainage culverts, and edge of right-of-way.

When substantial site grading and earthwork is involved construction staking (placement of physical stakes into the ground) is often replaced at least partially by GPS equipment combined with machine control technology. This technology utilizes a digital terrain model that is generated from the design and translated to earthmoving equipment. The equipment operator can then tell the horizontal/vertical location of the working grade in relation to the 3D model.

Surveying necessary to identify, locate, or establish real property boundaries shall be conducted under the direct supervision of a licensed professional land surveyor. Surveying and/or construction staking for earthwork, infrastructure, and/or utilities shall be completed under the supervision of the design engineer or a professional land surveyor.

5.11.4 Construction Quality Assurance

~~Construction testing:~~ Quality control supervision of the construction shall be done by the developer's engineer at no expense to the County. The County Engineer shall be permitted access to the construction site at all times to make spot checks of quality control. Any additional testing or corrective work deemed necessary shall be done within the time determined by the County Engineer at no expense to Park County.

~~(b) Periodic inspection during construction:~~ The County Engineer may conduct periodic inspections during construction to assure compliance with approved construction plans. The County Engineer may establish specific checkpoints when inspections must be conducted, and approvals granted before construction is continued. The contractor or developer shall contact the County Engineer twenty-four (24) hours in advance of any required inspections.

~~(a) Sampling of materials:~~ Samples for preliminary approval or production control may be submitted by the producer to the developer's engineer. The developer's engineer shall use an appropriate ASTM or AASHTO procedure to determine the acceptability or rejection of the sample.

~~(c) Final inspection:~~ Upon completion of construction and prior to County approval of the completed work, copies of the "As-Built" plans, concrete cylinder test reports, compaction test reports and other test data shall be delivered to the County Engineer. The County Engineer may conduct an inspection to determine if the construction meets County Standards. If the inspection discloses any work, in whole or in part, as unsatisfactory, the County Engineer shall give the developer's engineer the necessary instructions for correction, and the contractor shall comply with and execute such

instructions. At the discretion of the County Engineer, the County may withhold the granting of future Building/Zoning Permits or ROW Permits until such time corrective work is completed.

5.11.5 As-built Drawings

~~As-built record drawings:~~ Prior to granting construction acceptance, as-built drawings are required on all construction. As-built drawings shall include revision blocks, as necessary, and shall be signed and sealed by a Wyoming **licensed** Professional Engineer. The as-built drawings will be ~~field~~ checked by the County Engineer **for comparison to these RBS and the approved construction plans**. Should the County Engineer determine the as-built drawings do not correctly represent constructed field data, the as-built drawings will be returned for verification and correction.

As-built drawings shall include, but not be limited to, the following information:

- (1) Plan:
 - (a) *Property and/or rights-of-way lines, easement and/or tracts. Type and dimension of easement or tract clearly labeled: property lines and rights-of-way lines dimensioned;*
 - (b) *Lots and blocks shown and numbered;*
 - (c) *All drainage facilities, including culverts, detention, retention areas, drainage channels and erosion controls to be shown;*
 - (d) *Survey lines and stations, based on centerline stationing. Stationing equated to flow-line at bubbles, cul-de-sacs, horizontal curves and other departures from normal road cross-section, and two hundred (200) feet from all intersections;*
 - (e) *Road and road names, including road width;*
 - (f) *Match-lines referring to next sheets of design;*
 - (g) *Station and elevation (flow-line) of all curb returns, horizontal P.C.s, P.T.s, etc., also the high or low point on all vertical curves;*
 - (h) *Directional flow arrows on each side of the road;*
 - (i) *Curb return radius, existing and proposed;*
 - (j) *Complete horizontal curve data, radius, delta, length, tan, cord, cord bearing.*
 - (k) *Centerline stations of all intersecting roads;*
 - (l) *Survey line ties to section corners and quarter corners;*
 - (m) *Handicap and mid-block ramp locations;*
 - (n) *All storm sewer laterals, mains or trunk lines shall be tied to perpendicular off the centerline;*
 - (o) *Storm sewer manholes, numbered and stationed;*
 - (p) *Inlets numbered and stationed;*
 - (q) *Size, type of pipe, slope and footage noted between all manholes, appurtenances and inlets;*
 - (r) *Benchmark description with elevations;*
 - (s) *Mailbox locations;*
 - (t) *Revegetation Plan; and*
 - (u) *Traffic control and sign plan.*

- (2) Profile:
- (a) Existing ground profile (dotted or dashed) and labeled;
 - (b) All as-built elevations shall be centerline, flow-line or invert of pipe; top pipe is acceptable for existing utilities;
 - (c) Centerline or flow-line stationing continuous for entire length of road project with centerline station of all intersecting roads;
 - (d) Existing improvements in the profile shall include, but not be limited to, sidewalks, curbs and gutters, with certified as-built grades and elevations;
 - (e) Existing and proposed utilities including, but not limited to, water, gas, telephone, storm sewer, sanitary sewer, irrigation ditches, electric, structures, cable, conduits and under-drains where crossed with grades and elevations;
 - (f) Invert elevations at all stubs;
 - (g) Station and elevation of all horizontal P.C., P.T., etc.;
 - (h) Station and elevation of grade breaks;
 - (i) As-built construction, including stations and elevations (vertical curves, with VPI, VPC and VP high and low point, not the middle ordinate;
 - (j) Curb return profiles;
 - (k) Storm sewer manholes numbered, stationed, rim elevations and invert elevations (E., W., N., and S.) and dimensioned offset from centerline;
 - (l) Size, type of pipe, footage and slope (in Percent) of storm sewer between manholes and inlets; and
 - (m) Match-lines indicating references to next sheets.
- (3) Details: Details of special conditions and construction **shall be included** in the as-built **drawings**.

5.11.6 Certification

Upon completion of the project, the engineer of record shall prepare and submit a certification statement to the County Engineer certifying that ~~shall be given by the developer's engineer that~~ construction has been completed in conformance with the approved construction plans, ~~and standards and~~ specifications, **and these RBS**.

6.0 MAINTENANCE

6.1 MAINTENANCE CLASSIFICATIONS

The ownership and type of road (County, County maintained public road, non-county maintained public road, or private road), determines the County's role if any in maintaining specific roadways that comprise the PCRBS. The following sections outline the various levels of maintenance and the applicability to the types of roads.

6.1.1 Full Maintenance

All County Roads that have been improved within dedicated County Road rights-of-way shall receive full maintenance by the PCPWD. Full maintenance includes the following at a minimum:

- Routine road grading to re-establish the horizontal and vertical limits of the road surface (gravel roads);
- Chip sealing (paved roads);
- Crack sealing (paved roads);
- Road sweeping (paved roads);
- Mowing of road shoulder;
- Repair and replace road delineators and signage;
- Road striping (depending on classification);
- Cattleguard cleaning, repair, and replacement;
- Culvert cleaning, repair, and replacement;
- Cleaning and re-establishing roadside ditches;
- Snow plowing and sanding;

Routine road grading will typically be conducted twice per year (once in the spring and once in the fall) for all County gravel roads. Additional road grading may be conducted as needed depending on weather conditions, road conditions, and the availability of resources. Additional maintenance measures and surface treatment (i.e., fog sealing or application of magnesium chloride) will be considered by the County Engineer and R&B Foremen on a case-by-case maintenance when deemed necessary to improve safety and to preserve the road surface while maximizing maintenance efficiency.

6.1.2 Limited Maintenance

Most dedicated County Roads receive full maintenance (6.1.1). However, in some cases the BOCC have authorized PCPWD staff to provide limited maintenance for certain public roads that are not dedicated County roads. Unless otherwise specified in individual agreements approved and signed by the BOCC, limited maintenance shall consist of road grading (2 times per year – typically in the Spring and Fall) and snow plowing. Road grading and snow plowing of the dedicated County Roads shall be completed by PCPWD before initiation of limited maintenance on non-County roads. In no case shall limited maintenance involve replacing culverts, replacing cattleguards, or adding road materials without prior approval by the BOCC.

6.1.3 Provisional Maintenance

Provisional maintenance for roads refers to temporary or stopgap repair work done to keep a roadway safe and passable before a permanent, long-term engineering solution can be funded, scheduled, or completed. Examples include cold-patching potholes, placing temporary gravel over a sinkhole, or erecting temporary traffic barriers after a washout. Provisional maintenance shall focus entirely on halting rapid deterioration or mitigating sudden hazards to prevent vehicle damage or accidents. These measures are intended to last weeks or months, and not years.

The R&B Foremen are generally responsible for identifying and implementing provisional maintenance until a more permanent solution can be implemented. Provisional maintenance measures will be prioritized based on severity and safety. Provisional maintenance needs that represent an immediate or emergency safety

concern shall take precedence followed by measures necessary to preserve the long term integrity of the road, bridge, or associated component.

6.1.4 No Maintenance

Roads that receive no maintenance by the PCPWD include all private roads and public roads that have not been accepted for County maintenance by the Board of County Commissioners. Roads that receive no County maintenance are to be maintained by the adjacent landowners or through cooperative agreements with the HOA or HMID through a shared maintenance and improvement agreement. All costs to maintain the roadways that are not maintained by the County shall be the responsibility of the adjacent landowners.

6.2 COUNTY ACCEPTANCE OF MAINTENANCE RESPONSIBILITY

~~a. Road Establishment:~~ There are several subdivision roads and existing roads within Park County which are not County roads. In *State Statue WS §24-3-101*, et seq, there is a process to be followed to petition the County to establish a non-County road as a County road (See Figure 3). The Board of County Commissioners may, at their sole discretion, grant or deny establishment and maintenance by the County of these roads. Any road petitioned shall be brought up to the RBS ~~current County Standards~~ prior to requesting the road be established as a County Road.

~~b. Acceptance procedure for roads constructed by developers:~~ The Park County ~~DSR Development Standards and Regulations~~ require developers to construct roads necessary to serve approved subdivisions and developments. These roads shall be built to County Standards if applicable (See Section 2.1.2). The developer may propose to build either public or private roads, and the County may require that roads be offered for dedication to the public.

The County may accept other rights-of-way and roads; however, that does not constitute an acceptance of maintenance responsibilities. Acceptance of maintenance responsibilities is at the sole discretion of the Board of County Commissioners. **Even if a road is accepted for limited maintenance by the BOCC, the road is not considered a County Road unless the statutory process outlined in FIGURE 3 is followed.**

6.3 COUNTY PRIORITIES

The County Engineer and R&B Foremen shall establish maintenance priorities for the CRBS. Maintenance priorities shall focus on preserving public health and safety. Road or bridges that represent an immediate or urgent safety risk shall receive top priority in all cases.

For non-emergency related maintenance, priorities for maintenance shall be based on use. Specifically, roads that are classified as arterials, or collector roads will receive the top maintenance priority as they serve the greatest amount of traffic (ADT). These roads are the most critically important in maintaining connectivity during an emergency and are of principal importance in ensuring efficient flow of people, goods, and materials.

Once the collector and arterial roads are adequately maintained, the next level of priority is the roads classified as Local Access Roads and then Rural Roads (See Section 3.3.2).

Roads and routes that are designated school bus routes will receive precedence regarding maintenance priorities when compared to roads that are not associated with bus routes. Low Volume Rural Roads will receive maintenance once the other higher volume roads are maintained.

All maintenance including the degree and frequency is subject to availability of resources (labor, equipment, and materials) which is tied to the operating budget. The operating budget is approved by the BOCC in July of each year for the given fiscal year (July 1st through June 30th). The level and frequency of maintenance will be scaled as necessary to meet budget limits with priority given to the higher volume roads as described herein.

6.4 PROVISIONS FOR PRIVATE ROAD MAINTENANCE

Private roads should be maintained by the adjacent landowners as needed. The level of maintenance should be sufficient to satisfy the needs of the people that own and use the road. However, in all cases, private roads shall be maintained to a level that allows access by emergency vehicles.

Dedicated public roads that are not maintained by the County shall be maintained by the adjacent landowners, preferably administered through a shared maintenance agreement and/or HOA. The adjacent landowners have the option of creating a RIMD to establish a legal mechanism for administering and funding improvements and maintenance (See Section 2.4).

6.5 PUBLIC SERVICE REQUESTS

Park County residents may request service, maintenance, or improvements as it relates to public roads (including dedicated County Roads) that are maintained by the Road and Bridge Division of the PCPWD. The R&B Division will not in any case provide improvements or maintenance for public roads that are not dedicated County Roads without prior approval by the BOCC. Service requests may be submitted by calling (307) 527-8520 or via email to pubworks@parkcounty-wy.gov.

Service requests are logged and tracked by PCPWD staff and are distributed to the R&B Foreman, Sign Manager, Project Manager, or County Engineer for review and possible action depending on the nature, scope, and severity of the request. For maintenance related requests, priority is given to emergency issues where access and/or public safety may be in jeopardy. Resources including County labor, equipment, and materials will be allocated depending on availability based on the importance and severity of the service request. For non-emergency requests, priority will be given to the roads that have the highest level of use (ADT).

6.6 SNOW REMOVAL POLICY

a. Snow removal policy: It is the County's policy to plow County maintained roads after an accumulation of at least four (4) inches of snow. Snow removal operations will start early enough to allow for the necessary snow removal on the main school bus routes prior to the scheduled bus runs.

During major snowstorms, snow removal operations will concentrate on keeping the main traveled roads open during normal school/working hours. Driveway approaches and mailbox pull-outs may be plowed only after the road system has been plowed and clean-up operations can commence. Driveway entrances are generally the responsibility of the property owner with possible exception to emergency situations.

Snow removal operations will not be conducted between the hours of 9:00 p.m. and 4:00 a.m. for any reason except a life-threatening emergency. Snow removal operations utilize the following priority system (see **TABLE 10**) :

TABLE 10- COUNTY SNOW REMOVAL OPERATION PRIORITY SCHEDULE

ROAD CLASSIFICATION	PRIORITY
Primary/Arterial Roads	1
Major Collector Roads	2
Minor Secondary /Collector Roads	2 3
Local Access Roads	4
Rural Road	5
Low Volume Rural Road	6
Residential-Major*	3 7
Residential-Minor*	8
Recreational Roads	5 9

**PCPWD generally is not responsible for maintenance or snow plowing for subdivision roads unless the BOCC have accepted limited maintenance responsibility in prior agreements. The BOCC have not accepted any subdivision roads for maintenance in more than 25 years due to cost and limited availability of County resources.*

Exceptions to the ~~this~~ priority system shown in **TABLE 10** include school bus routes and emergency access requirements. It is the County's policy to provide snow removal operation in a consistent and economical manner on County **maintained** roads.

The County does not provide snow removal operations on private roads, except in emergency conditions.

The County may move a vehicle that causes an obstruction to a County road so the vehicle does not cause such obstruction. The County is not responsible for any damage caused to the vehicle.

6.7 BRIDGE INSPECTION & MAINTENANCE

Bridge inspection and maintenance guidelines are regulated by the Federal Highway Administration under the National Bridge Inspection Standards. Publicly owned highway bridges longer than 20 feet must undergo routine safety inspections at least once every 24 months, carried out or supervised by certified team leaders. Assessments of structural elements below the water surface and above the mudline are typically conducted at 60-month intervals for public bridges.

Bridge inspections are intended to check and evaluate fracture-critical members (steel or concrete components whose failure could cause collapse) and damage inspections following severe weather, floods, or vehicle impacts. All County bridges in Wyoming (longer than 20 feet) are inspected by WYDOT with reports provided to the County Engineer. The County

Engineer is responsible for reviewing inspection reports to identify root causes of distress, such as concrete spalling, structural steel corrosion, rotating or failing abutments, or timber rotting. Load ratings shall be incorporated and posted when deemed necessary based on analyses by a qualified structural engineer in circumstances where deterioration is noted or traffic weight demands changes.

PCPWD shall be responsible for inspecting and maintaining bridges that are not part of the National Bridge Inventory System (less than 20 feet). These structures shall be inspected at least every 5 years and shall be repaired or replaced when deemed to be structurally deficient.

Privately owned and maintained bridges are the responsibility of the adjacent landowners and/or the person or entity that maintains the bridge. Although private bridges do not have to be designed, constructed, or maintained to County standards, they should be maintained in a condition that allows safe and reliable access by emergency service vehicles (ie, fire trucks). It is recommended (but not required) that private bridge owners provide load rating information to the County fire chief who maintains a database. This information is used by the Fire Chief in allocating and assigning proper equipment during a fire emergency.

6.8 SIGN INSPECTION & MAINTENANCE

Road sign inspection and maintenance involves scheduled daytime and nighttime evaluations, inventory tracking, and routine servicing to comply with Federal Highway Administration (FHWA) and MUTCD standards. Procedures ensure structural stability, clear legibility, and mandatory nighttime reflectivity for driver safety.

Daytime checks shall evaluate structural integrity, plumbness of posts, clearance height, lateral placement, and surface damage (cracking, fading, or delamination). Nighttime evaluations shall assess retroreflectivity using calibrated measuring equipment, mobile retroreflectometers, or structured visual estimation by trained observers. During the inspections hardware shall be adjusted if necessary including tightening of bolts, securing loose brackets, and straightening or replacing bent posts. Severely faded, vandalized, or structurally compromised signs shall be replaced. Typically replacement schedules based on sheeting time can range from 7 to 15 years.

Logs shall be kept of installation dates, material sheeting types, and condition updates. The information shall be maintained into a centralized database maintained by PCPWD.

Road signs on private roads shall be maintained by the private landowners and replaced as necessary based on condition and visibility. Signage for all roads accessible by the public shall be installed and maintained in accordance with MUTCD guidelines. All costs for private road signage are the responsibility of the adjacent landowners.

6.9 CULVERT INSPECTION & MAINTENANCE

Road culvert inspection and maintenance requires routine visual checks annually and after major storms. Assessments shall be focused on clearing inlet/outlet blockages, inspecting barrel structural integrity (corrosion, cracking, or joint separation), and preventing embankment erosion or scour.

Culverts shall be inspected bi-annually and immediately after major heavy rain or flood events. High-risk or damaged sites require more frequent monitoring. Inspectors shall identify and address accumulated debris, trash, sediment, and vegetation blocking water flow. Scour, bank erosion, or undermining of headwalls shall be identified, logged, and addressed as soon as available resources will allow. For large walk-through culverts, adhere to confined space entry rules (OSHA 29 CFR 1910.146) only during low, safe water flows.

When inspecting metal pipes, check for rust, coating loss, seam cracking, and shape deformation. For metal pipes, look for wall cracking, spalling, joint separation, and settlement. For plastic pipes, check for buckling, stress cracks, and excessive deflection.

Debris that could inhibit or restrict flow shall be cleared and removed from the culvert openings (upgrading and downgradient). Leaves, branches, silt, and garbage shall be cleared away from the opening and the channel immediately upgradient and downgradient of the culvert. Eroded areas shall be stabilized with riprap, rock armor, or geotextiles at inlets and outlets. Culverts that are severely corroded or damaged beyond repair shall be replaced as soon as possible and practical as resources allow.

6.10 CATTLEGUARD INSPECTION & MAINTENANCE

Cattle guard maintenance requires clearing trapped debris from the grates and inspecting the structure for rust, bent rails, or a sinking foundation. Proper upkeep keeps livestock safely contained and ensures the unit can handle heavy vehicle loads without failing.

During the inspection, cracked, bent, or twisted steel rails, and damaged welds shall be identified. The pit below the grating shall be deep enough and clear so livestock cannot touch the bottom and cross. Inspector shall verify that water drains properly and does not pool underneath, which can cause rust or shift the foundation. Wing walls shall be inspected to make sure there are no sharp or protruding members and to ensure animals cannot bypass the edge of the cattleguard. Cattleguards that are severely damaged and beyond repair shall be replaced.

6.11 TREE TRIMMING

Rural road tree trimming requires maintaining a vertical clearance of 14 feet over travel lanes and clear sight lines at curves and intersections. Maintenance involves clearing the right-of-way of woody brush, removing dead or leaning hazard trees, and preventing utility line conflicts. Brush and trees shall be cut or trimmed back from the shoulder and ditch line to protect drainage flow and road foundations.

County R&B crews (or contracted services) are responsible removing trees or thick brush within the designated County Road ROW when the growth inhibits line of sight or restricts maintenance activities. Dead, diseased, or structurally compromised trees outside the immediate shoulder that could strike the roadway or utility wires when falling shall also be removed.

Trees or brush that are located outside the County road right-of-way that represent safety or visibility concerns shall be removed by the landowner at the landowner's expense. For the purpose of these RBS, a tree or brush is considered outside the County Road ROW when the majority of the primary tree or shrub base is located outside the physical limits of the

ROW. In situations where private trees or shrubs are encroaching on the County Road ROW, the landowner will be notified. The landowner will have 6 months to correct the encroachment. If no action is taken by the landowner within the specified time, PCPWD shall trim the trees, limbs, branches or brush up to the limits of the ROW line at the Owner's expense. Professional tree trimming or pruning practices shall be followed to the extent possible and practical to preserve tree health when maintaining trees and shrubs that are to remain.

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